Resource limitation rewires chromosome instability and ploidy evolution across in vitro and in vivo cancer models

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Abstract

Chromosome instability (CIN) generates karyotypic diversity, but whether altered progeny persist may depend on resource availability and ploidy. We combined lineage-matched near-diploid and near-tetraploid SUM159 culture and tumor datasets with a chromosome-number-structured mechanistic model linking oxygen/resource stress to proliferation, death, whole-genome doubling, chromosome missegregation, and post-missegregation survival. *In vitro*, oxygen deprivation reduced passage growth and remodeled low- and high-chromosome components. The model reproduced the emergence and expansion of high-chromosome states and complementary ploidy shifts from 2N and 4N starting states. Across the ten lowest-objective fits, the fitted model attributed less than 1.9% of total 4N lineage burden to a compartment representing death associated with resource stress (Supplementary Table 6), although the model did not recover the distinct terminal mixture proportions of parallel oxygen-deprived 2N lineages (Supplementary Table 5). Separate *in vivo* fits captured cohort-scale tumor growth and ploidy remodeling but admitted multiple near-equivalent parameterizations. Across three primary-family joint-fit ensembles, joint fits consistently estimated a lower maximal division rate *in vivo* and higher complete effective per-chromosome missegregation probabilities at 0% and 1% O_2 (Supplementary Tables 8 and 14). Complete post-missegregation survival functions revealed a two-part context contrast: altered daughters had higher absolute fitted survival *in vivo* at both 44 and 88 chromosomes, whereas the survival increase between these states was larger *in vitro* (Supplementary Table 14). Fixed-O_2 attractor analyses retained multiple compatible response regimes, indicating that the global oxygen–CIN–ploidy landscape was not uniquely resolved (Supplementary Tables 9 and 10). Together, these results support a model in which resource context changes both the production of chromosome-number variation and the ploidy-dependent filtering of viable descendants. These conclusions remain conditional on the model, data streams, parameter bounds, regularization, and sampled optimizer landscapes.

Introduction

Whole-genome doubling (WGD) and aneuploidy are common features of cancer evolution, yet their prevalence varies substantially across biological contexts.[1–3] A central unresolved question is why high-ploidy states are tolerated in some environments but selected against in others.[4,5] One clue comes from the striking difference between cultured cancer cell lines and primary tumors: across organ systems, cell lines exhibit markedly higher ploidy than primary tumors. This observation suggests that standard in vitro culture conditions, which provide abundant nutrients and simplified environmental constraints, may relax selection against high-ploidy states, whereas in vivo tumor environments impose stronger constraints on cells carrying excess genomic material.[4,5]

A plausible basis for this context dependence is the energetic cost of increased chromosome content.[4–6] WGD increases the biosynthetic burden associated with DNA replication, RNA production, and especially protein synthesis.[4–6] Aneuploidy further compounds this cost by disrupting gene dosage balance, producing proteotoxic, osmotic, oxidative, replication, and endoplasmic-reticulum stress.[7–9] These stresses require active homeostatic responses, including protein quality control, degradation of excess proteins, DNA repair, and metabolic adaptation.[5,7–9] Thus, high-ploidy cells are expected to carry a resource-dependent fitness cost, particularly when oxygen, glucose, or other metabolic substrates become limiting.[4,5]

At the same time, resource scarcity can also increase genomic instability.[4,5] Environmental stresses such as serum starvation, hypoxia, folate deficiency, and energy restriction have been linked to karyotypic instability, chromosome mis-segregation, polyploidization, and increased cellular heterogeneity. The evolutionary consequences of these events depend strongly on ploidy. Chromosome mis-segregation can trigger p53-mediated arrest, senescence, or death, particularly when chromosome loss creates severe dosage imbalance.[8,10–13] WGD and high-ploidy states can buffer these effects by reducing the relative dosage impact of individual chromosome gains or losses, allowing some mis-segregations to be converted into heritable karyotype variation rather than lethal events.[7,14,15] Consistent with this view, ALFA-K, a computational method that infers chromosome-level karyotype fitness from longitudinal single-cell karyotype data, showed that mis-segregations impose stronger fitness costs near diploidy than near tetraploidy.[16] Thus, ploidy is a quantitative determinant of whether CIN generates nonviable progeny or heritable karyotypic diversity.

Oxygen provides a tractable first-order proxy for this resource landscape.[4–6] Across tissue sites, WGD prevalence correlates with peripheral oxygen partial pressure, consistent with the idea that well-oxygenated tissues are more permissive for WGD-positive cancers, whereas poorly oxygenated tissues are less permissive.[5,6] However, oxygen is an imperfect surrogate for total energy availability. Tissue resource states also depend on amino acid and glucose availability, local metabolic consumption, and metabolic specialization. For example, tissues that rely heavily on glycolysis or have unusually high glucose availability may not experience low oxygen as equivalent to global energy scarcity.

Together, these observations identify a central tension in cancer ploidy evolution. Resource scarcity should select against WGD and other high-ploidy states because additional chromosome content increases biosynthetic, proteostatic, and homeostatic demands.[4–6] Yet adverse environments can also promote CIN, chromosome mis-segregation, and polyploidization, generating the karyotypic variation on which selection acts.[4,5] Whether this variation is eliminated or retained depends in part on ploidy: near-diploid cells are more vulnerable to lethal dosage imbalance after chromosome loss, whereas WGD or high-ploidy cells can better buffer individual segregation errors and convert them into heritable karyotype variation.[7,14–16] Thus, resource limitation may simultaneously oppose and promote WGD, depending on how strongly it

suppresses proliferation, induces CIN, and filters the survival of altered progeny. A key question is therefore which environments select for WGD, which select against it, and through which mechanisms. Answering this requires a quantitative framework that can jointly represent resource-dependent proliferation, stress-induced chromosome instability, and ploidy-dependent survival of resulting karyotypic states.

The central analytical target of this study is therefore not a single fitted parameter, but the context-dependent mapping from resource stress to chromosome-number variation and from altered daughters to viable diversity. By fitting one mechanistic structure to lineage-matched SUM159 culture and tumor data, we asked which complete response functions showed concordant context contrasts, which observations the model failed to reproduce, and which mechanistic interpretations remained unresolved.

Methods

Study design and data streams. We analyzed near-diploid and near-tetraploid SUM159 lineages across two experimental contexts. *In vivo*, orthotopic tumors initiated from the near-diploid and near-tetraploid lineages were followed with longitudinal tumor-burden measurements and harvested for terminal single-cell chromosome-number profiling; where matched terminal histology was available, necrosis was quantified at harvest. The *in vivo* calibration used the untreated subset of eight tumors, four initiated from each lineage. *In vitro*, the corresponding lineage pair was analyzed in passage-structured culture experiments with measured passage growth, karyotype/chromosome-number observations, and G0/G1 flow-density profiles under controlled oxygen conditions (Supplementary Table 1). The shared lineage origins placed the two datasets on a common modeling axis, but the experiments were not biological replicates of one another and did not constitute a randomized intervention comparing culture with tumors.

Model state and oxygen context. The model represents viable cells as a distribution over total autosomal chromosome-number states. It also tracks non-viable biomass in separate dead compartments, so simulated bulk tumor burden includes both live cells and uncleared dead material. Terminal single-cell chromosome-number profiles sample only the viable population at harvest, whereas tumor burden and necrotic fraction depend on live plus uncleared dead compartments. This state representation lets one simulation be compared simultaneously with longitudinal tumor volume, terminal necrosis, and single-cell chromosome-number distributions, while preserving the model's focus on karyotype evolution.

Resource context is represented differently in the two settings. *In vivo*, oxygen is modeled as an effective mean-field level that declines with chromosome-weighted tumor demand and relaxes toward a supply-demand target. *In vitro*, each passage segment is simulated at its assigned fixed oxygen level.

Mechanistic modules. The mechanistic core of the model has three modules corresponding to the routes through which resource context could select for or against WGD. First, oxygen/resource stress suppresses proliferation through a chromosome-number-dependent growth penalty, so high-chromosome states pay a larger growth cost when resources are limiting. Stress also induces a chromosome-number-dependent live-to-dead hazard, and the model assigns the resulting burden to a dedicated compartment representing death associated with resource stress. This partition is an internal model attribution rather than a direct observation of the biological cause of death. Second, stress can increase chromosome missegregation, while WGD is modeled as a constant-probability division-associated conversion branch that provides a route into high-ploidy states, generating the karyotypic variation on which selection can act. Third, daughters produced after missegregation pass through a ploidy-dependent survival filter: daughters of lower-ploidy mothers are more vulnerable to lethal dosage imbalance, whereas higher-ploidy mothers can better buffer chromosome gains or losses and convert some CIN events into heritable karyotype variation. These modules map directly onto the Introduction's mechanistic alternatives by asking whether resource-poor environments suppress high-ploidy proliferation, increase CIN, alter the selective consequences of a constant-probability WGD route, or determine whether altered progeny die or persist.

Observation models. The observation models link each measured data stream to a specific biological component of this mechanism, so the fitted trajectory is constrained by both growth dynamics and chromosome-number evolution:

Data stream	**Biological component constrained**
Longitudinal tumor burden	Net growth, death, clearance, and resource-dependent proliferation
Terminal single-cell chromosome-number distributions	Karyotype evolution, WGD, missegregation, and survival of altered progeny
Terminal necrotic fraction	Uncleared dead biomass relative to total tumor burden
In vitro passage growth rates	Proliferation under controlled culture and oxygen conditions
In vitro karyotype/chromosome-number and flow-density profiles	Chromosome-number evolution in culture

Joint fitting. We used the same cellular response model for the separate and joint *in vivo*/*in vitro* fits: chromosome-number transitions, hypoxia/resource-stress effects on proliferation and death, stress-induced missegregation, WGD generation, and post-missegregation survival were represented by the same mechanistic structure. The contexts differed in how oxygen trajectories were supplied and in their observation models. In the reported joint analysis, all 14 cross-context mechanistic parameters were soft-coupled. Each received separate *in vivo* and *in vitro* values reconstructed around a transformed-scale center, and a bounded Welsch penalty regularized their separation (Supplementary Tables 8 and 7). The joint fit therefore tested which parameter contrasts were repeatedly recovered across optimizer starts and landscape-informed warm-start pairs. These contrasts are conditional on the model, data streams, regularization, and shared *in vitro* anchor; they identify model-supported hypotheses rather than direct measurements of environmental causes.

Interpretive hierarchy. We distinguished three evidentiary levels throughout the analysis. Experimental burden, chromosome-number, flow-density, and necrosis records are direct observations. Fitted trajectories, latent *in vivo* O_2, and context-specific response functions are conditional model outputs constrained by those observations. Fixed-O_2 dominant-eigenvector curves are post-fit asymptotic diagnostics under standardized conditions and are therefore more extrapolative than fitted finite-time trajectories. Numerical optimizer seeds and warm-start pairs assess search dependence; they do not create additional biological replication or sampling-based confidence intervals.

Together, this framework allowed us to ask whether fitted differences between tumors and culture were associated with WGD-favoring or WGD-opposing regimes through resource-dependent proliferation, stress-induced chromosome missegregation, WGD generation, or the ploidy-dependent survival of karyotypically altered progeny.

Results

Lineage-matched culture and tumor trajectories motivate a resource-stress model of ploidy evolution

The motivating observations differed by experimental setting. In culture, longitudinal chromosome-number measurements showed complementary remodeling under oxygen deprivation: the 2N-starting cohort acquired a high-chromosome component, whereas the 4N-starting cohort shifted downward (Figure 1B). In tumors, longitudinal burden measurements were paired with chromosome-number and histology/necrosis measurements at harvest; the chromosome-number data therefore compare the starting cell-line references with distinct terminal tumor samples rather than tracking individual tumors longitudinally (Figure 1D). The near-diploid and near-tetraploid SUM159 lineage pairs nevertheless placed the *in vitro* and *in vivo* data on a common interpretive axis (Figure 1A; Supplementary Table 1), allowing fitted context contrasts to be compared without treating the two experiments as direct biological replicates.

These observations motivated a model in which resource limitation can both oppose and promote high-ploidy states. In the model overview, resource stress imposes growth and death costs on chromosome-rich states while increasing stress-linked chromosome missegregation; a constant per-division WGD branch provides a separate route into high-ploidy states, and high ploidy can then buffer some missegregation events that would be lethal or strongly deleterious near diploidy (Figure 2). Thus, the predicted ploidy outcome depends on the balance among growth suppression, stress-associated death, CIN generation, the constant-probability WGD route, and post-missegregation survival filtering.

In vitro ploidy evolution under hypoxia occurs with little death attributed by the model to resource stress

The culture measurements underlying these comparisons are catalogued in Supplementary Table 1 and its machine-readable companions.

Separate *in vitro* fits showed that oxygen stress can reshape ploidy without requiring a large fitted contribution from death attributed by the model to resource stress. Passage-growth observations shifted to a lower regime under oxygen deprivation in both starting-ploidy cohorts, and the fitted trajectories reproduced this cohort-level shift while not matching every passage-level observation (Figure 3A). Observed flow-density profiles contained low- and high-DNA-content components under both control and deprivation conditions, with pronounced changes in their relative weights during oxygen deprivation; the fitted model qualitatively reproduced this distributional remodeling across sampled passages (Figure 3B). The corresponding inferred chromosome-state distribution developed a broad high-chromosome mode while retaining a lower-chromosome mode; overlaying the direct single-cell karyotypes and fitted mean trajectory shows where measurements anchor this otherwise model-inferred passage sequence (Figure 3C). To evaluate whether the shared fitted state also captured lineage-specific mixture weights, we examined the final direct karyotype samples from two parallel oxygen-deprivation lines, O1 and O2, which had been propagated separately from the same 2N precursor through the matched nominal oxygen schedule (Methods; Supplementary Table 5). O1 remained bimodal, with 12 of 20 cells at $N = 46$–49 and eight at $N = 90$–100, whereas O2 was dominated by the high-chromosome component, with 16 of 20 cells at $N = 93$–99. The shared model distribution retained low- and high-chromosome modes but did not reproduce their lineage-specific proportions. The supported result is therefore a shared process capable of generating and expanding high-chromosome states, rather than deterministic prediction of each lineage-specific endpoint.

We next examined whether the fitted remodeling was attributed primarily to death assigned by the model to resource stress or to chromosome-number transitions among descendants. The post-missegregation survival function increased with chromosome number, allowing a subset of chromosome-loss daughters from high-ploidy mothers to remain viable (Figure 3D–E). Consistent with this mechanism, the paired severe-deprivation trajectories moved in complementary directions: the 2N lineage generated higher-chromosome descendants, whereas the 4N lineage shifted downward, and in both cases the burden assigned by the model to death associated with resource stress remained small relative to nonviable chromosome-transition products (Figure 3F). In the lowest-objective separate fit, the 4N mean declined from 84.3 to 80.8 chromosomes while the model assigned less than 1.8% of total burden to this compartment. Across the ten lowest-objective *in vitro* fits, the corresponding 4N decline was 3.54–4.17 chromosomes and the maximum fitted contribution of this compartment over the daily 4N trajectory remained below 1.9% (Supplementary Table 6). Seed 10 was the lowest retained objective among 500 numerical starts, its local refinement was accepted with convergence code 1, and five of 20 active parameters were exactly at configured bounds; the differential-evolution convergence flag was absent for all 500 starts (Figure 3G; Supplementary Table 6). These diagnostics characterize the sampled numerical search and do not establish posterior uncertainty or a proven global optimum. Thus, the fitted result supports bidirectional ploidy redistribution through viable descendants rather than loss of 4N cells attributed by the model primarily to resource stress.

In vivo fits capture cohort-scale behavior while retaining multiple compatible parameterizations

The ten lowest-objective separate *in vivo* fits had objectives of 14.119–14.175. Across these fits, log-burden RMSE was 0.663–0.680 across positive observations and 0.628–0.644 after equal weighting of tumor-specific mean squared errors. Terminal mean chromosome-number MAE was 2.41–3.71 chromosomes; terminal-distribution Wasserstein-1 distance was 4.18–5.24 chromosomes and total-variation distance was 0.681–0.714. Terminal necrosis-fraction MAE was 0.064–0.103 (Supplementary Table 2). The fits therefore captured broad cohort-level growth and endpoint scales but did not provide uniformly high-fidelity individual-tumor trajectories or terminal distribution shapes. The dynamic parameters were shared within lineage cohorts rather than fitted as mouse-specific latent effects.

The lowest-objective separate fit (seed 25) reproduced the cohort-scale increase in tumor burden while resolving markedly different ploidy trajectories for the two starting states (Figure 4A). The 2N-initiated cohort remained close to diploid throughout tumor growth, whereas the 4N-initiated cohort underwent a pronounced downward shift from near-tetraploid toward a state only slightly above diploid. In both cohorts, this remodeling occurred as fitted effective O_2 declined with increasing burden and modeled dead material accumulated alongside the viable compartment. CIN-associated dead burden constituted the largest modeled dead component by the end of both cohort trajectories, while the fitted contribution assigned to death associated with resource stress was greater in the 4N-initiated cohort. At harvest, the predicted distributions preserved the distinct central ploidy of the observed 2N and 4N cohorts, but were broader than the observed distributions. Thus, the representative fit captured the direction and cohort-level endpoint of ploidy remodeling together with tumor growth, while its broader predicted terminal distributions and model-derived intermediate states preclude interpretation as precise mouse-level trajectory recovery (Supplementary Table 2).

Near-equivalent objectives nevertheless arose from substantially different parameter combinations. To preserve the quantitative ploidy outcome rather than dichotomizing it at ploidy 2, we evaluated all 500 separate-fit parameter sets without refitting at 201 fixed O_2 concentrations from 0% to 5% and calculated, at each concentration, the Spearman association between each of 18 fitted parameters and continuous dominant mean ploidy (Figure 4B; Methods). Separating parameters by the sign of the association at their maximum absolute correlation revealed two fitted-landscape directions. The strongest positive peaks were baseline chromosome-missegregation probability $p_{\text{mis,base}}$ ($\rho = +0.649$ at 4.05% O_2) and maximum division rate $\lambda_{\max}$ ($\rho = +0.467$ at 0.025%), whereas the strongest negative peaks were stress-associated death scale μ_{hp} ($\rho = -0.691$ at 0%), post-missegregation viability-loss strength β_{buf} ($\rho = -0.447$ at 1.90%), and the ploidy exponent of stress-associated death γ_μ ($\rho = -0.439$ at 0.975%; Supplementary Table 3). The signed heatmap distinguishes locally strong from persistent associations: $p_{\text{mis,base}}$ was positively associated with dominant mean ploidy at all 201 concentrations, whereas μ_{hp} attained its largest magnitude at the anoxic boundary even though its correlation was positive at 83.6% of grid values. Maximum $|\rho|$ therefore identifies the strongest oxygen-specific fitted-landscape relationship rather than a grid-averaged or globally stable parameter importance. These are within-ensemble associations, not perturbational effects or independent predictive importance.

The aligned endpoint-and-range summaries in Figure 4B place these associations in their numerical-search context. For each parameter, the distribution of the 500 retained optimizer endpoints is shown above its configured fitting interval and initial value, with seed 25 marked within the endpoint distribution. The endpoint densities occupy markedly different portions of their allowed intervals, and several approach a configured boundary; the ensemble therefore does not collapse to a single resolved parameter vector. The separate t-SNE view in Figure 4C retained three exploratory groups of 43, 412, and 45 solutions (average silhouette 0.655), while cluster-stratified distributions for all 18 parameters are shown in Supplementary Figure 1. In an exploratory three-group comparison, 14 of 18 parameters passed the Benjamini–Hochberg threshold of $q < 0.05$. Ranking those parameters by Kruskal–Wallis epsilon-squared selected μ_{hp}, κ_O, $\lambda_{\max}$, n_{exp}, $p_{\text{mis,base}}$, and α_{O_2}, with effect sizes of 0.154, 0.151, 0.143, 0.115, 0.101, and 0.049, respectively (Figure 4D; Supplementary Table 4). Because the groups were constructed from the same fitted parameters being compared, these statistics quantify internal separation of the fitted landscape rather than independent validation or biological significance. Complete separate-*in vivo* numerical-search diagnostics are shown in Supplementary Figure 2 and Supplementary Table 4. These are optimizer-derived endpoint distributions and search diagnostics, not posterior samples, credible intervals, biological replicates, or causal "drivers." Figure 4 therefore separates what the representative fit reproduces (panel A), the continuous but non-unique parameter–ploidy structure among compatible fits (panel B), the exploratory geometry of the fitted endpoints (panel C), and the parameters that most strongly distinguish those internally defined groups (panel D). Several parameters also spanned wide ranges among the ten lowest-objective fits, and the fitted burden-error scale reached its upper bound in all ten. The resulting boundary dependence, oxygen-dependent association changes, and multiple fitted-solution clusters are consistent with practical non-identifiability rather than a uniquely resolved biological parameter vector.

Joint fitting yields concordant function-level context contrasts across warm-start-conditioned search ensembles

Joint soft-coupled fits were used to ask which fitted mechanisms remained context-specific when the SUM159 lineage pairs were compared in culture and tumors (Figure 5A). The principal analytical targets were the complete effective missegregation and post-missegregation survival functions; individual scalar parameters were treated as intermediate components of those functions. One objective-minimum separate *in vivo* representative was retained from each of the three primary regions C01–C03 in the pooled 14-parameter warm-start landscape (seeds 366, 25, and 311, respectively). Each was paired with the same lowest-objective separate *in vitro* fit (seed 10) and optimized from 500 numerical seeds, giving 1,500 joint searches. To distribute these warm starts, transformed, standardized parameter samples and final solutions from the separate *in vivo* and *in vitro* searches were pooled into that fitted-landscape embedding (Figure 5C; Methods; Supplementary Methods). This pooled warm-start embedding is distinct from the *in vivo*-only 18-parameter landscape in Figure 4B. The marked regions are inputs to joint optimization, not optimized joint-fit endpoints, and the two color scales retain the separate component objectives. All 14 cross-context mechanistic parameters were represented by context-specific values centered on a common transformed-scale value. A bounded Welsch penalty regularized context separation: it is near-quadratic close to zero and saturates for large offsets. Family-level summaries first aggregated the 500 optimizer starts within each retained pair and then compared directions across C01–C03. Optimizer starts quantify numerical search variation and are not biological replicates; moreover, all three pairs share the same *in vitro* anchor. The best retained objective differed among the three pairs, ranging from 18.852 to 19.414, so family-to-family variation was retained rather than collapsed into a claimed global optimum (Supplementary Table 7).

The warm-start audit supported coverage of the frozen embedding but not a unique biological partition (Supplementary Figure 6A–B; Supplementary Table 7). On the saved t-SNE coordinates, $k = 3$ and $k = 4$ differed in average silhouette by only 0.000957; $k = 4$ created a singleton region, so the smaller non-singleton $k = 3$ solution was retained. Across 100 80%-endpoint subsamples drawn without replacement, re-optimizing the region count selected $k = 3$ 58 times and $k = 4$ 42 times, although the fixed-$k = 3$ assignment agreed closely with the saved labels (median adjusted Rand index 1.00). A separate $k = 3$ partition in standardized 14-parameter endpoint space agreed poorly with the t-SNE regions (adjusted Rand index 0.052). We therefore interpret C01–C03 only as embedding-dependent numerical-search regions used to diversify joint initialization, not as biological subtypes.

Across the three selected pair winners, the direct in-sample summaries retained the broad burden and terminal mean-chromosome-number scales *in vivo* and the low- versus high-growth regimes *in vitro*, but visible residual scatter and underprediction of several high mean-chromosome-number culture endpoints remained (Figure 5B). Circles and triangles distinguish 2N- and 4N-initiated observations without changing the fixed blue/pink context encoding. These fits therefore provided a basis for comparing fitted functions without establishing uniformly accurate prediction of every observation (Supplementary Tables 1 and 7).

Across the 1,500 feasible, unprojected endpoints from the three retained pairs, the C01–C03 family—rather than an individual optimizer start—was the comparison unit (Supplementary Table 7). This prevents repeated numerical searches from being treated as independent biological or inferential evidence while retaining variation among the three primary warm-start regions.

Figure 5D resolves the context-specific natural-scale parameter distributions for the one-pair-per-family endpoint analysis against the pair-specific differential-evolution (DE) initial populations that generated those searches. The mirrored display places the *in vitro* marginal above the baseline and the *in vivo* marginal below it, so movement and separation of the two contexts can be read directly without using a ratio as the horizontal coordinate. In the accompanying paired-ratio audit, 12 of the 14 parameter contrasts lay on the same side of equality in all three selected families. The concordant lower-*in vivo* contrasts were $\lambda_{\max}$, $O_{2,c}$, $s_{\max}$, and β_{buf}; the concordant higher-*in vivo* contrasts were $p_{\mathrm{mis,base}}$, p_{misseg}, $k_{o,\mathrm{mis}}$, α_{O_2}, γ_{growth}, μ_{hp}, γ_μ, and $n_{\exp}$. The exceptions were p_{wgd} and n_O, whose directions differed among families. For all 14 paired-ratio summaries, every selected-family endpoint 5th–95th percentile width was narrower than the corresponding pair-specific DE-initial width; 13 were narrower than half that width in all three families, while p_{wgd} reached 0.690 of the initial width in one family. For 11 parameters the widest family-specific endpoint ratio span was exactly zero because at least 90% of numerical starts returned the same ratio. This concentration demonstrates repeated optimizer convergence relative to the actual starting populations under the tested search design; it is not posterior contraction or proof of structural identifiability. Six parameters—α_{O_2}, γ_{growth}, μ_{hp}, γ_μ, $s_{\max}$, and $n_{\exp}$—had active-bound occupancy of at least 10% in one or more selected families, further limiting identifiability claims (Supplementary Table 8).

The pair-median *in vivo*/*in vitro* ratio for the maximal division rate, $\lambda_{\max}$, ranged from 0.177 to 0.222 across C01–C03, supporting a lower fitted proliferative ceiling *in vivo*. The maximal stress-induced missegregation increment, p_{misseg}, was higher *in vivo* in all three families, with pair-median ratios from 11.1 to 17.7. The oxygen threshold $O_{2,c}$ and post-missegregation viability-loss severity parameter β_{buf} were lower *in vivo* in all three, whereas μ_{hp}, γ_μ, $k_{o,\mathrm{mis}}$, and γ_{growth} were higher (Supplementary Table 8). Lower β_{buf} increases survival in the fitted function; it is not, by itself, a direct measure of biological buffering strength. Several of these axes were boundary-limited, so the ratios are descriptive fitted-solution contrasts rather than precise effect sizes.

Because the complete effective per-chromosome missegregation probability includes both $p_{\mathrm{mis,base}}$ and the clipped stress-linked increment determined jointly by p_{misseg}, $k_{o,\mathrm{mis}}$, and the death-hazard terms, we evaluated that full function rather than assigning its direction from one parameter (Figure 5E). At 0% O_2, the three pair-level *in vivo*/*in vitro* ratios ranged from 11.1 to 20.7 at 44 chromosomes and from 11.2 to 20.6 at 88 chromosomes. At 1% O_2, the corresponding ranges were 7.42–17.0 and 10.4–19.6. Thus, the complete effective probability was higher *in vivo* at both reference chromosome states in all three families under severe and intermediate hypoxia. At 5% O_2, all three ratios remained above one at 88 chromosomes, whereas the 44-chromosome ratio was below one in C01 and above one in C02 and C03 (Supplementary Table 14).

The three post-missegregation survival parameters individually showed mixed directions, but their complete nonlinear function gave a consistent and biologically more informative comparison (Figure 5F). Pair-level per-copy survival ranged from 0.728 to 0.819 at 44 chromosomes and from 0.904 to 0.914 at 88 chromosomes *in vivo*, compared with 0.204 and 0.837, respectively, for the common *in vitro* anchor. Absolute survival was therefore higher *in vivo* at both reference states in all three selected winners. Conversely, the fitted 44-to-88 chromosome survival increase was 0.0849–0.186 *in vivo* versus 0.633 *in vitro*, and was larger *in vitro* in every selected family (Supplementary Table 14). The supported result is therefore two-part: altered daughters have higher absolute fitted survival in tumors, while culture has the stronger fitted ploidy-dependent survival gradient. These are fitted-function comparisons, not biological confidence intervals.

The bounded Welsch penalty limits how strongly large context separations are regularized, and every endpoint in the three selected 500-start ensembles had at least one active parameter bound. All selected differential-evolution runs stopped under the configured relative-tolerance/step-tolerance rule before the 500-iteration target, and all attempted local refinements were rejected (Supplementary Table 7). The observed directional concordance is therefore limited to these three primary-family search ensembles; it does not resolve the global optimum or behavior in unsampled regions of the objective landscape and does not identify unique context effect sizes.

Fixed-O_2 analyses expose an uncertainty architecture rather than a unique response regime

The separate *in vivo* calibration left multiple parameterizations that could explain the observed ploidy endpoints. We characterized this uncertainty with fixed-O_2 dominant-eigenvector response curves: for each candidate fit, the analytical attractor was evaluated across an oxygen grid and grouped by the regression-smoothed shape of the predicted O_2–ploidy relationship (Methods; Supplementary Methods). Among the 500 separate *in vivo* fits, 316 were complex nonmonotone, 50 inverted-U-shaped, 50 monotone increasing, 47 U-shaped, 26 approximately flat, five increase-then-plateau, four decrease-then-plateau, and two monotone decreasing (Supplementary Figure 5A; Supplementary Table 9). Spectral-gap diagnostics classified only 141 fits as reliable, 168 for caution, and 191 as unreliable. In the broader six-candidate fixed-O_2 sensitivity archive, three warm-start representatives were classified for caution and three as unreliable; none of the three primary-family representatives displayed in the main analysis was classified as reliable (Supplementary Table 9).

Because these curves are asymptotic operator diagnostics rather than fitted finite-time observations, small spectral gaps weaken their biological interpretation over experimental time horizons. We therefore replaced the single-winner comparison with an objective-defined joint-fit ensemble (Figure 6A). All 3,000 endpoints passed the finite-objective, parameter-completeness, feasibility, no-projection, and configuration checks, and all 600 endpoints in the widest retained set passed the post-fit operator checks. Within each warm-start pair, the primary analysis retained the 50 lowest-objective endpoints; the pair-specific decile cutoffs lay 0.00359–0.0716 objective units above the corresponding pair minimum (Supplementary Figure 6C; Supplementary Table 7). The resulting 300 complete 201×60 response surfaces were summarized pointwise by their median, while the unmodified fitted trajectories were summarized by medians and 10th–90th percentile bands (Supplementary Table 10). Figure 6A displays the C01Sc01, C02Sc01, and C03Sc02 surfaces as C01, C02, and C03, respectively. Complete numerical summaries for all six warm-start pairs are provided in Supplementary Tables 7 and 10, while Supplementary Figure 6C–D shows the corresponding objective-eligibility and qualitative robustness audit.

The ensemble resolved one conditional response direction but not one complete response topology. When effective per-chromosome missegregation was held fixed, every endpoint in every primary 50-seed ensemble predicted higher dominant mean ploidy at 5% than at 0% O_2 at the low, intermediate, and high diagnostic missegregation levels. Across the six pairs, the corresponding median 5%-minus-0% ploidy differences were 0.466–1.678, 0.434–1.693, and 0.953–4.888, respectively. A denser line representation evaluated 496 fixed missegregation probabilities from 0.005 through 0.500 in increments of 0.001 for each displayed pair. Movement along one colored curve therefore isolates the oxygen response at one constant $p_{\mathrm{miss,eff}}$, whereas movement between curves changes the imposed chromosome-error probability (Figure 6B; Supplementary Table 10). The unmodified trajectories in Figure 6A ask a different question because the fitted missegregation function is restored and allowed to change with oxygen. Those coupled trajectories decreased in four pairs and increased in two, with pair-median ploidy changes ranging from -1.172 to $+1.175$. Among the three displayed pairs, median population-average effective per-chromosome missegregation along the unmodified trajectory fell from 0.115 to 0.00110 in C01, from 0.178 to 0.00236 in C02, and from 0.212 to 0.0475 in C03 between 0% and 5% O_2. The presence of both ploidy regimes at 0%, 1%, or 5% O_2 and the prevalence of small spectral gaps also differed among pairs. All 48 pair-by-diagnostic modal results were unchanged in the nested 5%, 10%, and 20% objective sets, with 100% within-set support, and all 144 seed-weighted versus unique-parameter-endpoint comparisons retained the same modal result. In the primary complete surfaces, parameter-endpoint deduplication preserved the high- versus low-ploidy classification at every grid point; the most duplicated pair, C02Sc01, represented 50 seeds by 15 unique *in vivo* response-parameter endpoints (Supplementary Table 10). The six pair-level surfaces nevertheless did not collapse to one common arrangement of low- and high-ploidy regions, weak-spectral-gap boundaries, or model-implied trajectories. Thus, the fixed-missegregation oxygen response was numerically stable across the sampled endpoints, whereas the emergent coupled oxygen–CIN–ploidy response remained warm-start-pair dependent. These standardized surfaces and trajectories are post-fit model diagnostics, not independent CIN measurements, causal oxygen-by-CIN interventions, or biological uncertainty intervals.

We next asked whether the weak-gap regions in Figure 6A represented disagreement about the qualitative ploidy regime or primarily slow separation of the dominant asymptotic mode. We classified dominant mean ploidy as low at values no greater than 2, intermediate between 2 and 4, and high at values of at least 4. Among the 3,287, 2,838, and 2,118 sampled weak-gap grid cells for C01, C02, and C03, respectively, every cell retained at least 90% agreement among the 50 accepted optimizer-seed endpoints on this three-level classification. Stable-high cells accounted for 57.8%, 55.8%, and 13.6% of these regions; stable-intermediate cells accounted for 36.3%, 37.2%, and 69.0%; and stable-low cells accounted for 5.9%, 7.0%, and 17.4%, respectively. None met the prespecified mixed-regime definition (Supplementary Figure 7A; Supplementary Table 11). A change among the low, intermediate, and high classes occurred at an immediately adjacent O_2 or fixed-missegregation grid cell for at least half of the endpoints at 8.67%, 8.17%, and 4.86% of weak-gap cells (Supplementary Figure 7B). Exact dominant-mean-ploidy values were tightly aligned across endpoints over most of these regions: the median 90th–10th percentile spread was 0.000035, 0.000317, and 0.00672 ploidy units, and its 90th percentile was 0.000733, 0.0237, and 0.0107, respectively (Supplementary Figure 7C). Local response magnitude was less uniform at narrow boundaries: the median, across endpoints, of the largest change to an immediate grid neighbor exceeded one ploidy unit at 9.13%, 11.31%, and 4.39% of weak-gap cells (Supplementary Figure 7D; Supplementary Table 11). Thus, the small-gap mask primarily identifies conditions where asymptotic mode separation is slow; it does not generally erase the ensemble-supported low/intermediate/high dominant-mean-ploidy conclusion. The local-neighbor calculation is a grid-sensitivity diagnostic, however, and neither its grid-cell fractions nor optimizer-endpoint agreement are biological switching probabilities.

Finally, response class and the complete fitted *in vivo* MAP objective were overlaid on the pooled 14-parameter t-SNE coordinate system used for joint warm-start selection. Several common response classes overlapped in parameter space and objective value, and no single response class uniquely contained the lowest-objective solutions. These complementary diagnostics therefore retained competing regimes rather than selecting one response shape by assumption (Supplementary Figure 5B–C; Supplementary Tables 9 and 4).

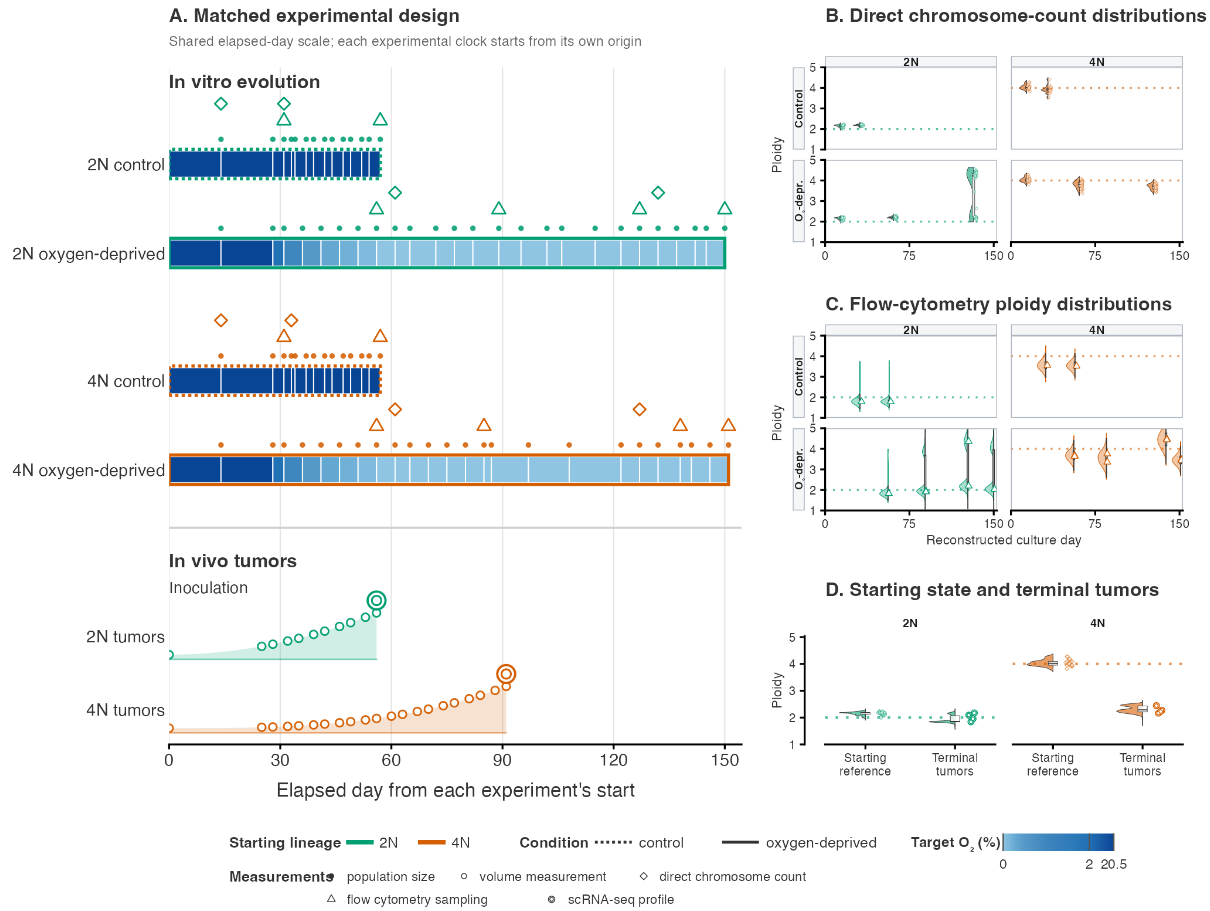


Figure 1: **Matched experimental systems and ploidy readouts.** Green and orange identify the 2N- and 4N-initiated cohorts throughout; blue is reserved for target O_2. **(A)** The culture and tumor designs are placed on the same 0–150-day elapsed-time axis, although each experiment begins at its own origin and the two clocks do not denote synchronized biological states. Each O_2-colored culture block represents one reconstructed passage; the display scale expands 0–2% O_2 and compresses 2–20.5% O_2 so that the low-O_2 steps remain distinguishable, while the labels report the actual target concentrations. Dotted and solid cohort-colored outlines distinguish control and oxygen-deprived trajectories. Filled circles mark population-size measurements at every displayed passage endpoint, with shared pre-branch measurements repeated on the two aligned eventual branches only for visual comparison. Open triangles and diamonds mark flow-cytometry and direct chromosome-count sampling, respectively. In tumors, open circles mark volume-measurement days on schematic nonlinear growth wedges; wedge height is not a measured burden value. Double concentric circles mark single-cell RNA-sequencing-derived chromosome profiles; only the final 4N profile is displayed. Symbol height offsets in the culture tracks separate coincident assays and do not encode measurement magnitude. **(B)** Direct chromosome-count measurements are shown as an explicit condition-by-starting-ploidy grid, with control above oxygen-deprived cultures and 2N beside 4N. Counts were converted to ploidy by division by 22. At each sampled day, a left half-violin shows the distribution of directly counted cells, the central box shows the median and interquartile range, and small open diamonds to the right show the individual cell measurements. Horizontal offsets separate the three display layers and do not change sampling time. The shared directly measured starting distribution is repeated at the beginning of both condition rows for alignment and is not counted as an independent observation. **(C)** Observed flow-cytometry profiles use the same condition-by-starting-ploidy grid and reconstructed culture-day axis as panel B. At each sampled day, the cohort-, condition-, and day-specific G0/G1 density is shown as a left half-violin with a density-weighted box summary; parallel O1 and O2 profiles sampled on the same day are pooled with equal sample weight. Open triangles, matching the flow-cytometry sampling symbol in panel A, show the global density peak of each individual profile. Panels B and C use a

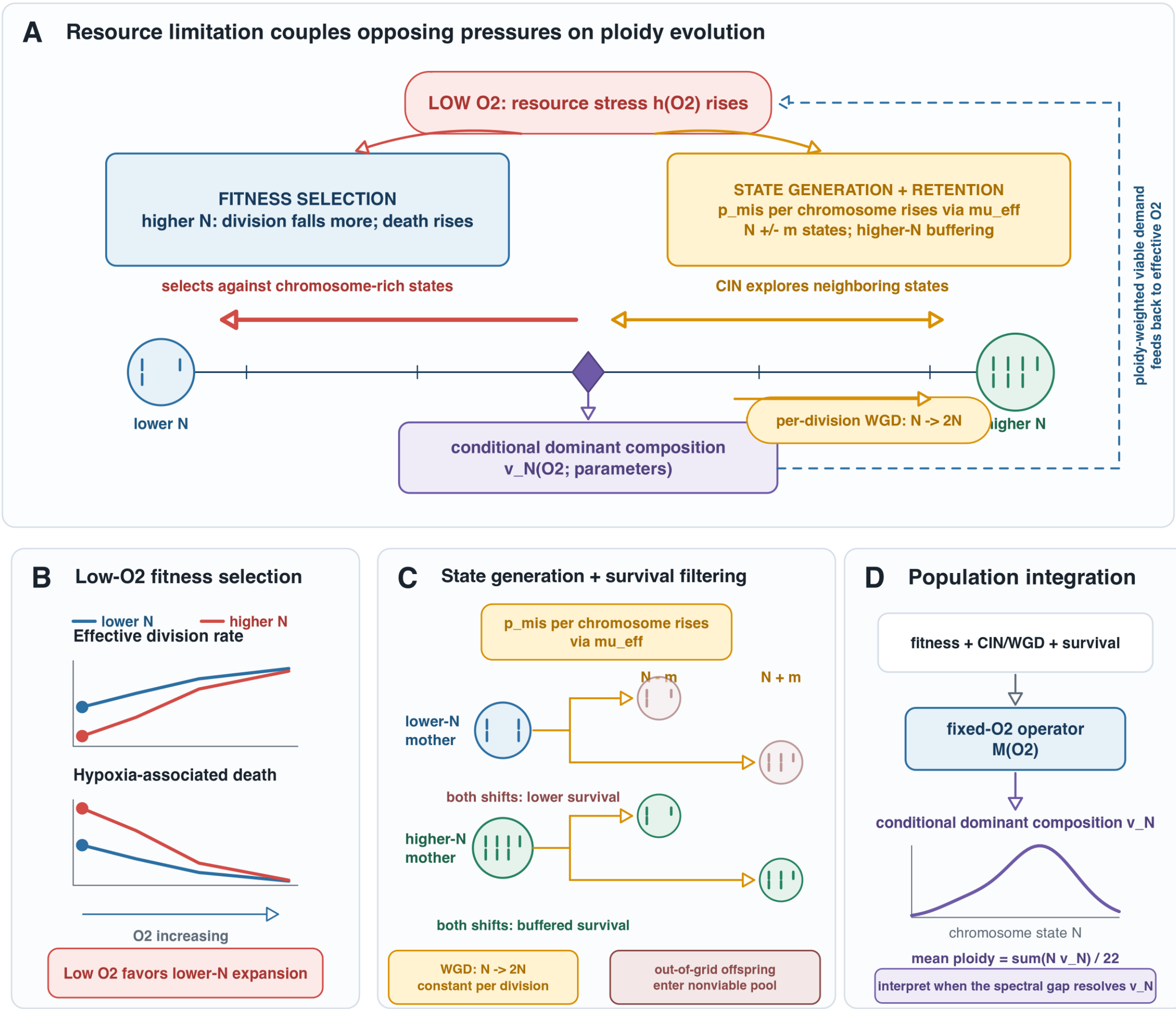


Figure 2: **Model overview: resource limitation can impose opposing pressures on ploidy evolution. (A)** Low oxygen raises bounded resource stress and simultaneously strengthens selection against chromosome-rich states while increasing the per-chromosome missegregation probability that supplies chromosome-state variation. In the dynamic model, viable burden and ploidy-weighted oxygen demand close the feedback loop; the resulting composition depends jointly on oxygen and the fitted model parameters. **(B)** Under the ploidy-related stress mode, low oxygen suppresses effective division and increases the modeled death hazard associated with resource stress more strongly in higher-chromosome-count states, creating a relative expansion advantage for lower-N states. **(C)** The per-chromosome missegregation probability rises through the stress-linked death hazard and generates symmetric $N - m$ and $N + m$ daughter branches. For a given $|m|$, altered daughters from higher-N mothers have greater post-missegregation survival. Constant per-division WGD provides a separate tracked-lineage transition $N \to 2N$, and out-of-grid offspring enter the CIN-associated nonviable pool. **(D)** Growth, death, CIN/WGD transitions, and post-missegregation survival define the fixed-O_2 population operator. When its dominant mode is sufficiently separated, the normalized dominant eigenvector gives the conditional asymptotic chromosome-state composition and mean ploidy.

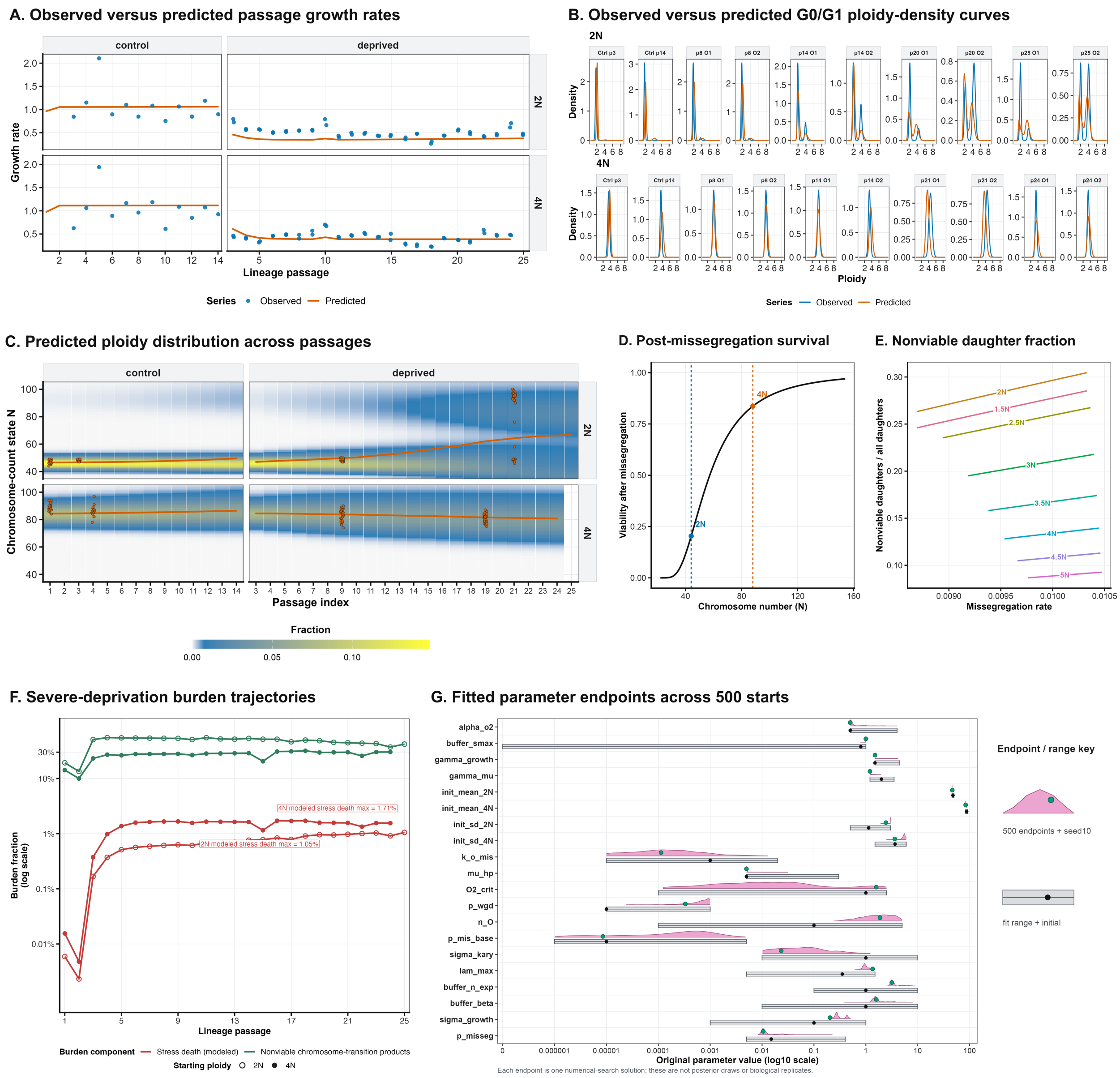


Figure 3: **In vitro ploidy reshaping occurs with little death attributed by the model to resource stress.** **(A)** Observed (blue points) and fitted (orange line) passage growth rates from the lowest-objective retained separate *in vitro* fit (seed 10), arranged as 2N/4N rows and control/deprived columns. The fitted trajectories reproduce the lower growth-rate regime under oxygen deprivation while individual passage measurements remain variable. The underlying culture observations are catalogued in Supplementary Table 1 and its machine-readable companions. **(B)** Observed (blue) and fitted (orange) G0/G1 DNA-ploidy density curves from the same fit. Parallel O1 and O2 measurements at the same deprived passage are displayed in separate sample-level subpanels rather than connected or overlaid as one observed curve; the corresponding shared prediction is repeated in each subpanel. Both sources are drawn as thin solid lines, with the fitted curve layered above the observed curve where they overlap. High-DNA-content components are present under both control and deprivation conditions, while their relative peak weights change prominently during deprivation; the shared fitted state captures the qualitative remodeling but not every lineage-specific peak proportion or width. **(C)** The predicted chromosome-state distribution across passages shows the oxygen-deprived 2N lineage developing a high-chromosome component while retaining a lower-chromosome mode. The orange line gives the fitted mean chromosome number and the orange points with dark outlines give direct single-cell karyotypes; these measurements occur only at sampled passages, whereas the intervening heatmap is model inferred. The displayed chromosome range is derived from the pooled weighted predicted and observed 1st–99th percentiles

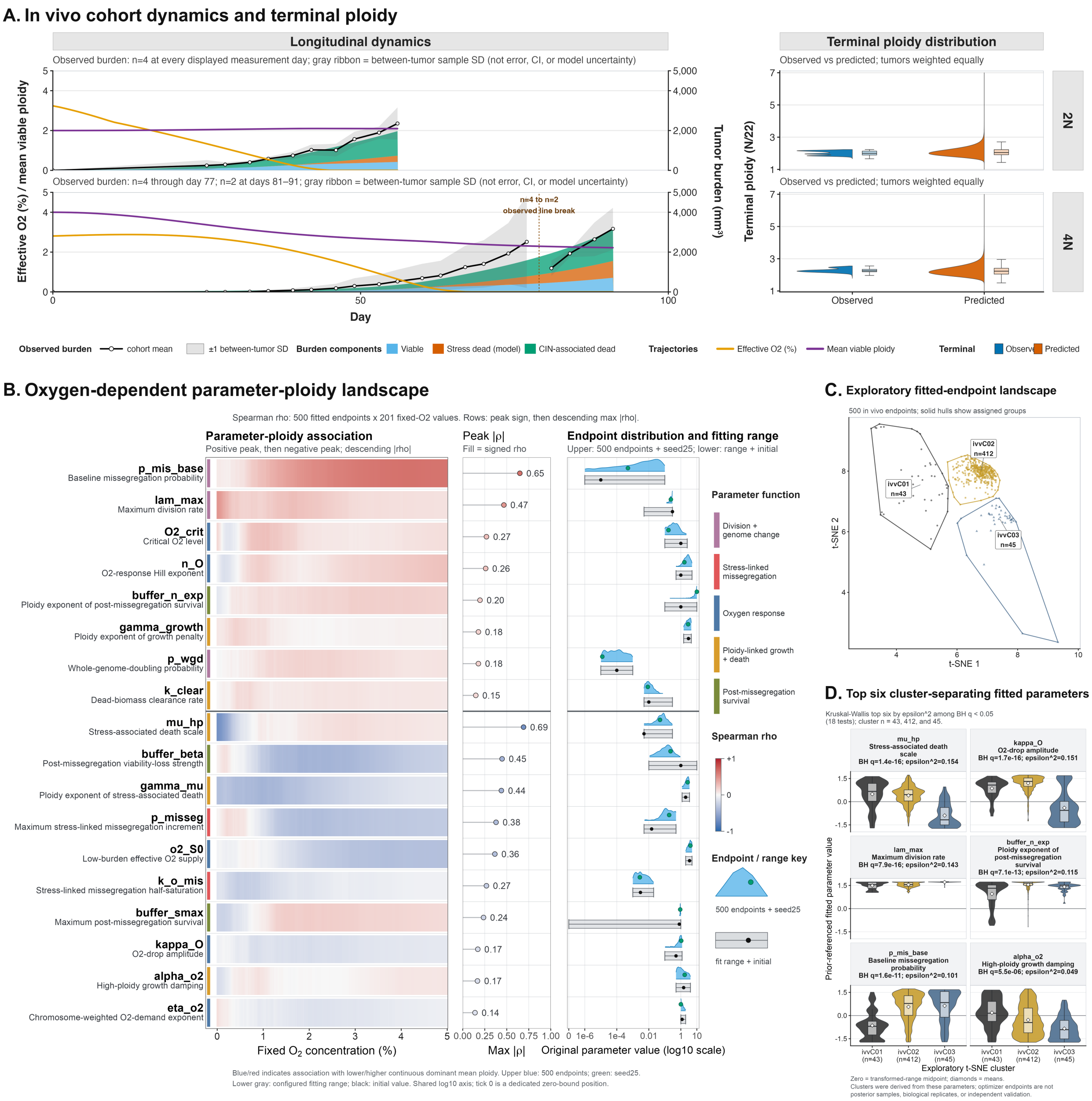


Figure 4: **Separate in vivo fits capture burden, resource, and ploidy dynamics while retaining multiple resource-response parameterizations.** **(A)** Cohort-level summary of the lowest-objective retained separate fit (seed 25). The left column shows 2N-initiated and 4N-initiated longitudinal dynamics on a common 0–100-day axis. Stacked areas, averaged over the tumor trajectories represented at each day, are read from the right axis and show predicted viable burden, burden assigned by the model to death associated with resource stress, and CIN-associated dead burden. Black points and curves show observed cohort means at recorded measurement days. The gray ribbon is the cohort mean plus or minus one between-tumor sample SD; it is not a measurement-error bound, confidence interval, or model-uncertainty interval. Orange and purple curves show cohort-mean fitted effective O_2 and viable-cell mean ploidy, respectively, on the shared numerical 0–5 left axis. Mean ploidy is viable-cell weighted mean chromosome number divided by $N_{\text{unit}} = 22$; sharing this numerical axis does not equate the physical units of O_2 and ploidy. The right column reconstructs the report's terminal observed-versus-predicted chromosome-number comparison after conversion to ploidy, using blue for observed and orange for predicted distributions. In each half-violin/box summary, the left half shows the weighted distributional shape and the box-and-whisker at right shows the weighted median and quartiles. Each tumor receives equal total weight within its cohort and source. Effective O_2, burden compartments, and intervening ploidy paths are model predictions; chromosome-number profiles directly constrain only

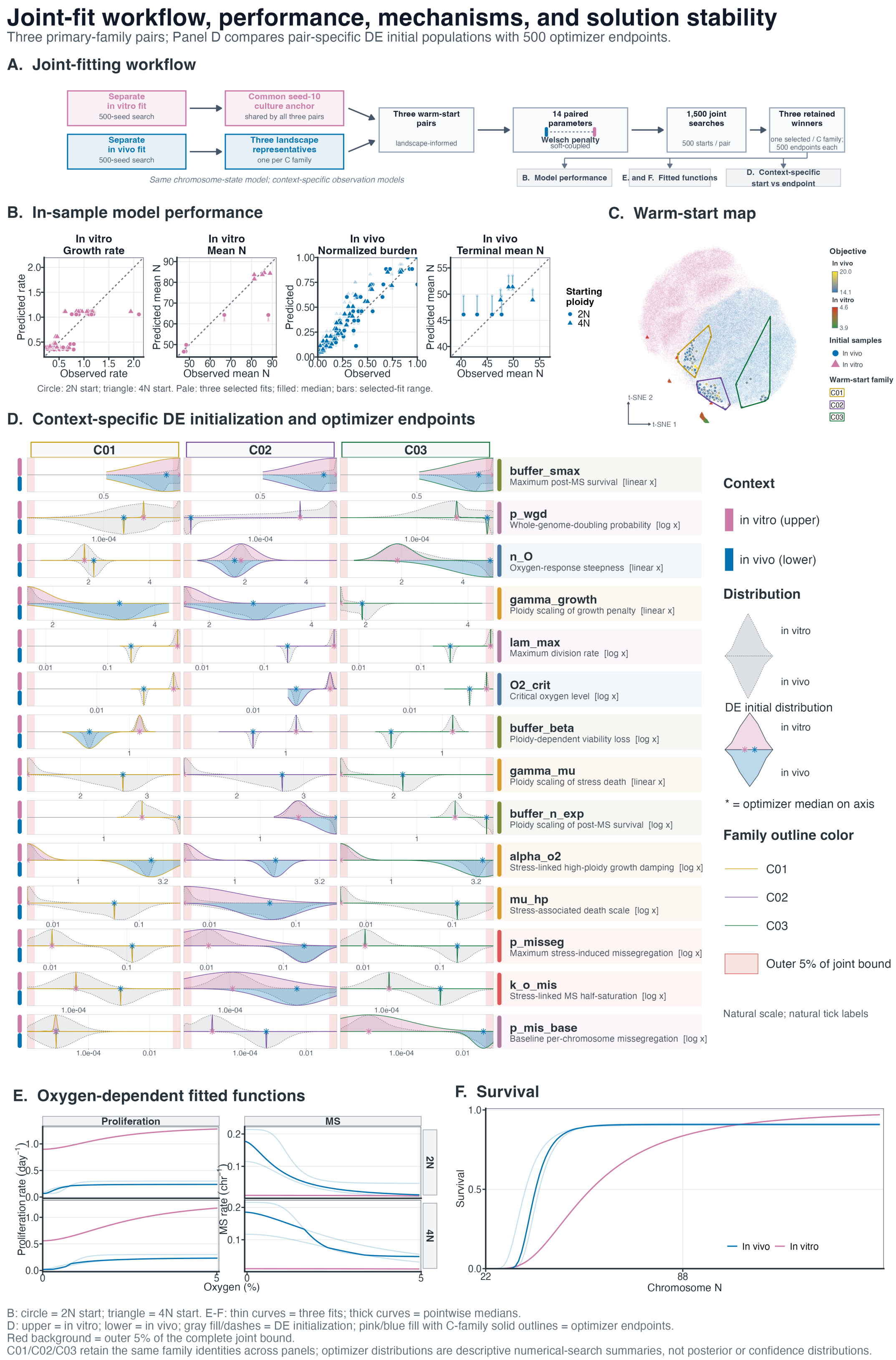


Figure 5: **Joint-fit workflow, model performance, fitted mechanisms, and optimizer-solution stability. (A)** Joint-fitting workflow. Separate 500-seed *in vitro* and *in vivo* searches form two input lanes. One objective-minimum *in vivo* representative from each primary C01–C03 region is paired with the common seed-10 *in vitro* anchor. Each of these three pairs enters the same chromosome-state model with 14 paired context-specific parameters linked by the bounded Welsch soft-coupling penalty. Five hundred numerical starts per pair give 1,500 joint searches, and the lowest retained joint objective within each pair supplies the three winners summarized in panels B, E, and F. Panel D uses all 500 feasible, unprojected endpoints from each retained pair and its

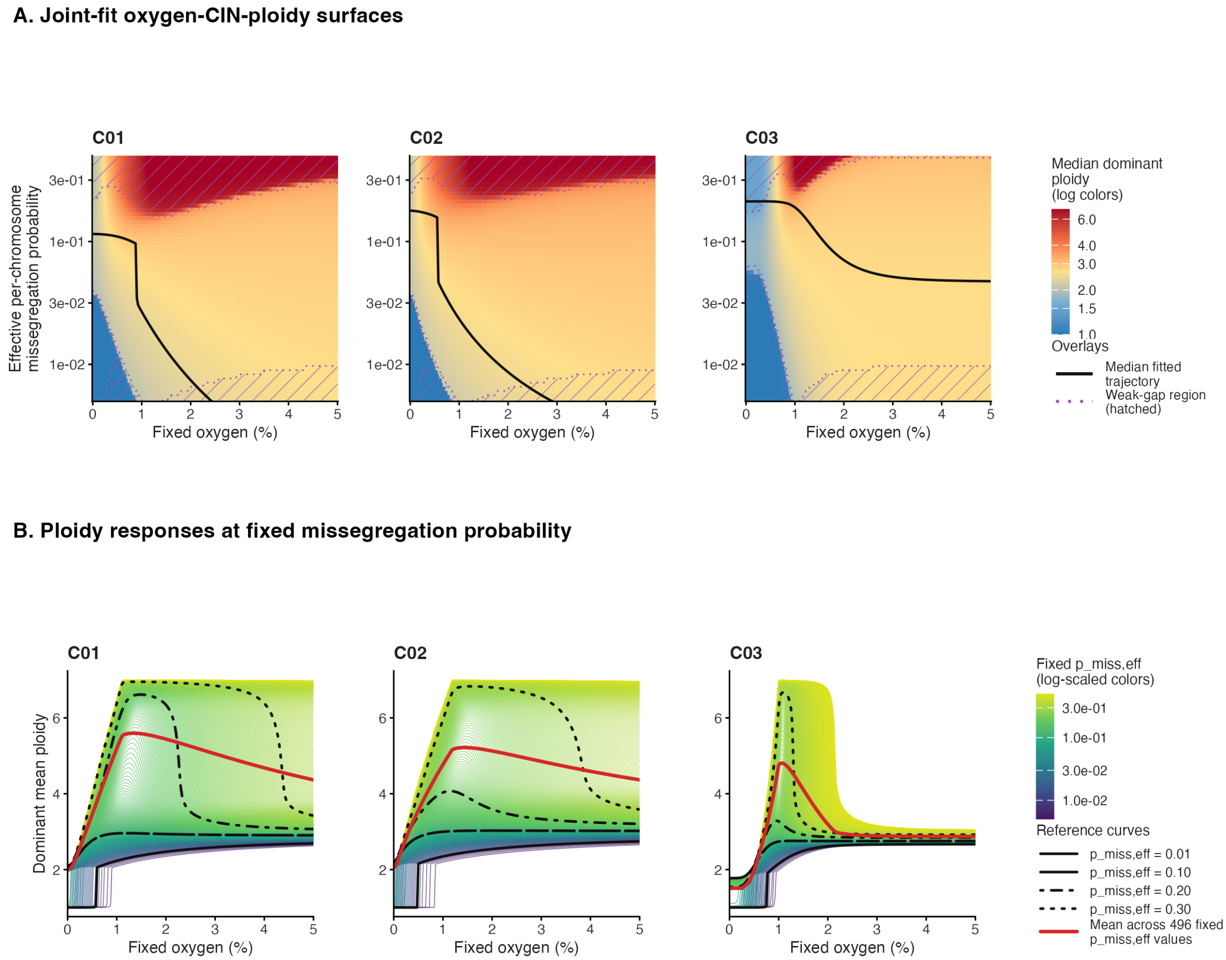


Figure 6: **Joint-fit fixed-O_2 diagnostics retain pair-dependent oxygen–CIN–ploidy responses. (A) Joint-fit oxygen–CIN–ploidy surfaces.** Three pair-specific post-fit response ensembles are arranged in one row: C01Sc01, C02Sc01, and C03Sc02 are labeled C01, C02, and C03, respectively. Each plotting region is square. Within each displayed warm-start pair, the complete grid of 201 oxygen concentrations and 60 forced effective per-chromosome missegregation probabilities was evaluated for the 50 endpoints in the pair-specific lowest-objective decile. Fill reports median dominant mean ploidy in its original units, with logarithmically scaled color spacing. Black curves show the pointwise median model-implied population-average missegregation probability along the unmodified fixed-O_2 trajectories, and gray ribbons show their 10th–90th percentile intervals. Purple diagonal hatching marks positions where at least half of eligible endpoints have spectral gap < 0.005, and purple dotted contours delimit those weak-gap regions; white crosses mark sampled positions where fewer than 80% agree on whether dominant mean ploidy is at least 2. The overlay legend identifies the black median trajectory and hatched weak-gap region. Exact duplicate 14-parameter endpoints were additionally collapsed to one analysis unit in the sensitivity analysis summarized in Supplementary Figure 6D. Complete summaries for all six warm-start pairs, together with their objective eligibility and qualitative robustness, are reported in Supplementary Tables 7 and 10. These surfaces summarize numerical fit multiplicity and are not independently fitted CIN measurements, causal oxygen-by-CIN interventions, or biological confidence regions. **(B) Ploidy responses at fixed missegregation probability.** A denser line representation uses the same three displayed pairs and C01–C03 order. Each of the 496 colored curves holds $p_{\mathrm{miss,eff}}$ constant at one value from 0.005 through 0.500 in increments of 0.001 and shows the median dominant mean ploidy across the 50 eligible numerical endpoints at all 201 oxygen concentrations. These curves were obtained by direct operator evaluation rather than interpolation. Curve color identifies the fixed probability; colorbar labels remain in the original probability units while color spacing is

Discussion

Here, we integrated lineage-matched *in vitro* and *in vivo* ploidy-evolution datasets into a common mechanistic framework. The main result is not a uniquely identified parameter vector, but a small set of function-level contrasts that remained directional across three primary-family search ensembles. Within the fitted model, tumors had a lower proliferative ceiling and higher complete effective per-chromosome missegregation probabilities at 0% and 1% O_2. At the same time, altered daughters had higher absolute fitted per-copy survival *in vivo* at both 44 and 88 chromosomes, whereas culture had the stronger fitted survival increase between those states (Supplementary Tables 8 and 14). Absolute altered-daughter survival and its ploidy-dependent gradient are therefore separable axes: a context can retain altered daughters more readily overall without imposing the stronger ploidy dependence on that retention.

This separation sharpens the biological significance of the model. Resource context may change not only how much chromosome-number variation is generated, but also how chromosome state filters whether that variation persists. In that sense, the model links CIN production to the retention of viable karyotypic diversity rather than treating CIN, WGD, and survival as interchangeable proxies. The complete-function comparison was essential: directions of individual survival parameters were mixed, whereas the evaluated nonlinear survival function yielded the two-part context contrast in all three selected pair-level winners.

The multi-endpoint oxygen analysis further separates a conditional response from the coupled model trajectory. At a standardized fixed effective missegregation probability, higher oxygen consistently shifted the asymptotic dominant mode toward higher ploidy across the sampled endpoints. Once each endpoint's oxygen-dependent missegregation function was restored, however, the net 0%–5% trajectory direction differed among warm-start pairs (Supplementary Table 10). These findings are not contradictory: the response surface conditions on one transition input, whereas the unmodified trajectory allows oxygen-linked chromosome-error production and state-specific growth and survival to change together. The distinction identifies where a robust conditional model response coexists with unresolved coupled dynamics. Within the weak-gap regions of the three displayed pair ensembles, every sampled grid cell retained at least 90% optimizer-endpoint agreement on the low/intermediate/high dominant-ploidy classification, although exact ploidy magnitude remained locally sensitive at narrow boundaries (Supplementary Figure 7; Supplementary Table 11). This qualitative agreement does not remove the finite-time limitation implied by slow dominant-mode separation. Because both the response surfaces and unmodified trajectories are dominant-eigenvector post-fit diagnostics rather than randomized perturbations or longitudinal CIN measurements, neither establishes a causal oxygen effect.

These results remain conditional. Soft coupling permits context-specific values while regularizing their separation on the transformed parameter scale, but the reported Welsch penalty is bounded, so sufficiently large differences contribute a saturated penalty rather than an indefinitely increasing cost. Repeated separation across warm-start pairs therefore identifies model-supported hypotheses conditional on the likelihoods, bounds, regularization, and common *in vitro* anchor; it is not an equality test and does not establish that environmental context caused the fitted difference. Likewise, fitted effective O_2 and stress-linked parameter contrasts are latent model quantities, not direct measurements of tissue oxygen, CIN, or death mechanisms.

The explicit *in vitro* model limitation is also informative. Observed flow-density and chromosome-count distributions support the presence and remodeling of low- and high-ploidy components, while the late O1 and O2 samples differ in the relative occupancy of those components (Supplementary Table 5). The model generated a shared mixed distribution but did not reproduce the lineage-specific late composition of both lines. This localizes the limit of the shared modeled process: lineage-specific latent histories, stochastic diversification, measurement heterogeneity, or other unmodeled processes would be needed to explain the divergent compositions. The supported claim is therefore a common process capable of generating and filtering a mixed distribution, rather than complete reconstruction of lineage-specific evolution.

Several observations limit a purely resource-dependent interpretation of ploidy evolution. First, the present model does not include a separate oxygen-independent cost of abnormal chromosome content. Aneuploid states can incur proteotoxic, replication, mitotic, and gene-dosage stresses even outside the specific oxygen/resource term represented here.[7–9] Consequently, the fitted resource-dependent penalty should not be interpreted as an exhaustive account of high-ploidy or aneuploid fitness costs.

Second, oxygen is only one component of the broader resource environment experienced by tumor cells. The analyzed datasets did not directly measure nutrient availability, perfusion, local consumption, or spatial oxygen heterogeneity. The fitted *in vivo* O_2 trajectory is therefore a latent model variable, and nominal *in vitro* oxygen and fitted *in vivo* oxygen should not be treated as experimentally equivalent energetic states. Oxygen is informative within the model but remains an incomplete proxy for resource stress.

Third, the model omits other mechanisms that could alter the generation or filtering of high-ploidy states, including context-dependent WGD generation, tissue-specific polyploidization programs, and immune or stromal selection. The present SUM159 analysis cannot distinguish these unmodeled possibilities from resource-mediated effects. Its conclusions should therefore remain restricted to the fitted mechanisms and matched culture/tumor datasets rather than generalized to all tissue-specific ploidy differences.

The fitted contrasts remain limited by practical identifiability and search coverage. Near-equivalent separate *in vivo* objectives were obtained from boundary-dependent parameter combinations, and every endpoint in the three retained joint ensembles had at least one active bound. The three primary-family pairs sampled three embedding-dependent *in vivo* warm-start regions but only one *in vitro* warm start; their best objectives differed by more than the within-pair numerical spread. The cluster audit supports these regions as a coverage device, not as stable biological groups, and the archived bundle does not permit t-SNE seed or perplexity sensitivity to be re-estimated. Within each joint pair, qualitative oxygen-response results were unchanged across the 5%, 10%, and 20% objective sets and after exact response-parameter endpoint deduplication; across pairs, however, the unmodified trajectory direction, ploidy-regime occupancy, and spectral-gap reliability remained different (Supplementary Tables 7 and 10). The bounded Welsch penalty was close to saturation for many large context splits, so it did not continue to impose strong shrinkage once those splits became large. Fixed-O_2 interpretation was also limited by small spectral gaps: only 28.2% of separate fits were classified as reliable, and none of the three displayed separate-*in vivo* warm-start representatives met that criterion (Supplementary Table 9). The directional concordance reported here is limited to these three primary-family search ensembles; a unique effect size, global optimum, and behavior in unsampled objective-landscape regions remain unresolved.

XX Before submission, repeat the joint fitting with multiple independently selected *in vitro* anchors and a balanced set of *in vivo* and *in vitro* warm start combinations. The sensitivity analysis should vary the coupling scale and saturation constant and should compare the bounded Welsch penalty with a nonsaturating quadratic coupling penalty. Report whether the direction and magnitude of the complete function level contrasts remain stable under these alternatives.

One implementation limitation requires separate caution. Out-of-range chromosome-transition mass is routed to the CIN-associated dead compartment, so that compartment combines biological nonviability with finite-grid boundary loss.

These findings also suggest hypotheses for therapies that induce chromosome-segregation errors. Many anticancer agents elevate chromosome mis-segregation either directly, by disrupting mitosis, or indirectly, by producing replication and DNA-damage lesions that persist into mitosis.[17–23] Within the fitted ploidy-dependent survival framework, such agents are predicted not to have uniform effects across chromosome-number states: near-diploid cells are predicted to be more vulnerable to missegregation-induced dosage imbalance, whereas WGD or high-ploidy cells may better buffer individual chromosome gains or losses and convert some errors into heritable karyotype variation. This offers a model-based hypothesis for prior drug-response analyses in which multiple mitotic spindle poisons, including paclitaxel, vinblastine, and vinorelbine, were among the strongest ploidy-associated agents.[24] It also suggests a possible explanation for why gemcitabine, despite being classically viewed as an S-phase antimetabolite, may show a strong association with ploidy: independent chromosome-loss assays have identified gemcitabine as a potent inducer of chromosome instability.[17] Thus, replication-stress-induced chromosome mis-segregation is a candidate ploidy-dependent therapeutic vulnerability generated by the model, not a response tested in the present SUM159 experiments.

1 Supplementary Information

1.1 *In vivo* experiments

The *in vivo* analysis used near-diploid and near-tetraploid variants of the SUM159 triple-negative breast cancer cell line, denoted `SUM159_NLS_2N_A6M` and `SUM159_NLS_4N_A4M`, respectively. For each tumor, 1×10^6 cells were injected orthotopically into mice. The subset analyzed here comprised eight tumors: four initiated from the near-diploid 2N line and four initiated from the near-tetraploid 4N line (Supplementary Table 1).

- **2N cohort:** four mice injected with `SUM159_NLS_2N_A6M`.
- **4N cohort:** four mice injected with `SUM159_NLS_4N_A4M`.

Tumors were harvested at predefined endpoints, and single-cell RNA sequencing was performed on all eight tumors. Chromosome-number distributions used for model calibration were derived from these endpoint single-cell profiles and analyzed together with the corresponding tumor-burden trajectories. Only this untreated subset was used for the modeling analyses presented here. Processed data and associated experimental metadata were organized and retrieved through CLONEID, a framework for structuring clone-specific genotypic and phenotypic measurements.[25]

XX Before submission, complete the animal and ethics reporting by specifying the mouse strain and genetic background, sex, age at implantation, source, housing and husbandry conditions, allocation and randomization procedures, blinding procedures, injection matrix and volume, tumor measurement schedule, predefined experimental and humane endpoints, exclusion criteria, numbers excluded with reasons, euthanasia method, approving institutional committee, protocol number, and compliance with applicable animal welfare guidelines. Also report the tissue collection and processing procedures used for necrosis assessment and single cell profiling, together with the chromosome inference method, filtering criteria, quality control thresholds, uncertainty assessment, and number of cells retained for each tumor.

Supplementary Table 1: Experimental data inventory for the analyzed SUM159 culture and tumor datasets. Counts distinguish biological specimens, measurement records, and display summaries. Individual burden measurements, chromosome estimates, culture counts, passage records, and flow-density values are provided in the accompanying machine-readable supplementary tables.

Setting	Data stream or design quantity	Biological or measurement units	Count or value	Scope and interpretation
In vivo	Orthotopic inoculum	Cells per tumor	1×10^6	Experimental input for each analyzed tumor
In vivo	Untreated tumor cohort	Tumors	8	Four near-diploid initiated and four near-tetraploid initiated tumors
In vivo	Longitudinal burden	Tumor measurements	120	Measurements from the eight tumors used for joint burden and terminal chromosome-number calibration
In vivo	Terminal chromosome-number profiles	Single-cell estimates	6,502	3,728 estimates from four near-diploid initiated tumors and 2,774 from four near-tetraploid initiated tumors
In vivo	Matched terminal necrosis	Tumor-level fractions	6	Three tumors per starting-ploidy cohort had matched necrosis measurements used by the observation model
In vitro	Passage-structured fit records	Passage entries	131	All entries retained assigned oxygen, timing, and linked population measurements
In vitro	Passage growth	Finite estimates	114	Entries with a finite segment-level growth estimate
In vitro	Direct chromosome counts	Samples; cells	12; 220	Passage-linked chromosome-number samples and their total single-cell observations
In vitro	G0/G1 DNA-content profiles	Profiles; grid points per profile	20; 200	Normalized profiles used in the flow-density comparison
In vitro	Oxygen schedule	Target O_2 values (%)	20.5, 2, 1, 0.5, 0.3, 0.2, 0.1, 0	Control and serial deprivation levels; a post-anoxia branch was evaluated at 1%
Figure 1 display	Culture population markers	Displayed endpoints	77	Representative cohort, condition, and reconstructed-day summaries
Figure 1 display	Flow summaries	Profiles; grouped time points	20; 12	Individual profile peaks and equal-weight grouped density summaries
Figure 1 display	Ploidy conversion	Autosomal chromosomes per ploidy unit	22	Used only to express chromosome counts on the plotted ploidy scale
Figure 1 display	Flow plotting viewport	Ploidy range; minimum retained mass	1–5; 85.43%	Full normalized profiles were used for calculation; only the display viewport was cropped
Figure 1 display	Starting chromosome counts	Directly counted cells	40	Twenty cells from each starting-ploidy cohort

Note: Counts summarize the analyzed data and do not imply equal replication across data streams. Display counts can exceed the number of unique biological samples when a shared starting distribution is repeated for alignment. The machine-readable companion tables preserve the individual observations and identifiers needed to reconstruct these totals.

Supplementary Table 2: Absolute fit-quality metrics for the ten lowest-objective separate *in vivo* fits. Burden errors are calculated on the log-burden scale. Terminal chromosome-number metrics and necrosis-fraction MAE are averaged equally across tumors or harvests.

Rank	Fit	Objective	Burden log-RMSE	Tumor-balanced log-RMSE	Terminal mean-N MAE	Terminal Wasserstein-1	Terminal total variation	Necrosis-fraction MAE
1	`seed25`	14.1193	0.6660	0.6317	2.41	4.51	0.697	0.091
2	`seed366`	14.1340	0.6680	0.6341	2.54	4.75	0.708	0.078
3	`seed292`	14.1372	0.6724	0.6384	2.79	4.59	0.698	0.064
4	`seed392`	14.1406	0.6784	0.6426	3.33	4.92	0.689	0.093
5	`seed90`	14.1524	0.6799	0.6439	3.65	4.86	0.684	0.092
6	`seed391`	14.1553	0.6797	0.6439	3.38	4.75	0.682	0.091
7	`seed264`	14.1558	0.6735	0.6400	2.56	4.18	0.682	0.079
8	`seed109`	14.1724	0.6629	0.6284	2.85	4.30	0.681	0.103
9	`seed322`	14.1724	0.6677	0.6331	2.52	4.86	0.714	0.078
10	`seed26`	14.1748	0.6784	0.6425	3.71	5.24	0.698	0.094

Note: Positive-observation burden RMSE weights every retained measurement equally; tumor-balanced RMSE first averages squared residuals within each tumor and then weights tumors equally. Wasserstein-1 and mean-N MAE are reported in chromosomes. The ten optimizer-selected fits are not biological replicates or confidence-interval samples.

Supplementary Table 3: Peak continuous fixed-O_2 fitted-parameter associations underlying Figure 4B. Parameters are grouped by the sign of the Spearman correlation at their maximum absolute association, with the positive group shown before the negative group, and ordered by decreasing maximum $|\rho|$ within each group. Signed peak ρ and peak O_2 report the first oxygen value attaining that maximum. Sign stability is the percentage of oxygen values sharing the predominant correlation sign and is not used for ordering.

Within-group rank	Parameter	Max $\|\rho\|$	Signed peak ρ	Peak O_2 (%)	Sign stability (%)
Positive peak ρ					
1	`p_mis_base`	0.649	+0.649	4.050	100.0
2	`lam_max`	0.467	+0.467	0.025	100.0
3	`O2_crit`	0.269	+0.269	1.100	93.0
4	`n_0`	0.261	+0.261	4.975	91.0
5	`buffer_n_exp`	0.196	+0.196	2.925	95.0
6	`gamma_growth`	0.178	+0.178	0.400	100.0
7	`p_wgd`	0.178	+0.178	1.900	97.5
8	`k_clear`	0.148	+0.148	0.700	100.0
Negative peak ρ					
1	`mu_hp`	0.691	-0.691	0.000	83.6
2	`buffer_beta`	0.447	-0.447	1.900	92.5
3	`gamma_mu`	0.439	-0.439	0.975	100.0
4	`p_misseg`	0.378	-0.378	2.075	85.6
5	`o2_S0`	0.365	-0.365	3.700	91.5
6	`k_o_mis`	0.269	-0.269	0.450	100.0
7	`buffer_smax`	0.236	-0.236	0.000	84.6
8	`kappa_0`	0.171	-0.171	2.050	100.0
9	`alpha_o2`	0.167	-0.167	0.025	78.1
10	`eta_o2`	0.139	-0.139	2.150	93.0

Interpretation: The display group uses the sign at maximum $|\rho|$, which can differ from the predominant grid sign when a strong association is localized to a narrow oxygen range. For example, `mu_hp` has a negative peak at 0% O_2 although 83.6% of its grid correlations are positive. The full signed heatmap, rather than the grouped peak order alone, should therefore be used to assess oxygen dependence. The gray fitting ranges, black initial values, endpoint densities, and seed25 markers in Figure 4B are optimizer-derived quantities, not posterior samples, credible intervals, measurement bounds, or biological confidence intervals.

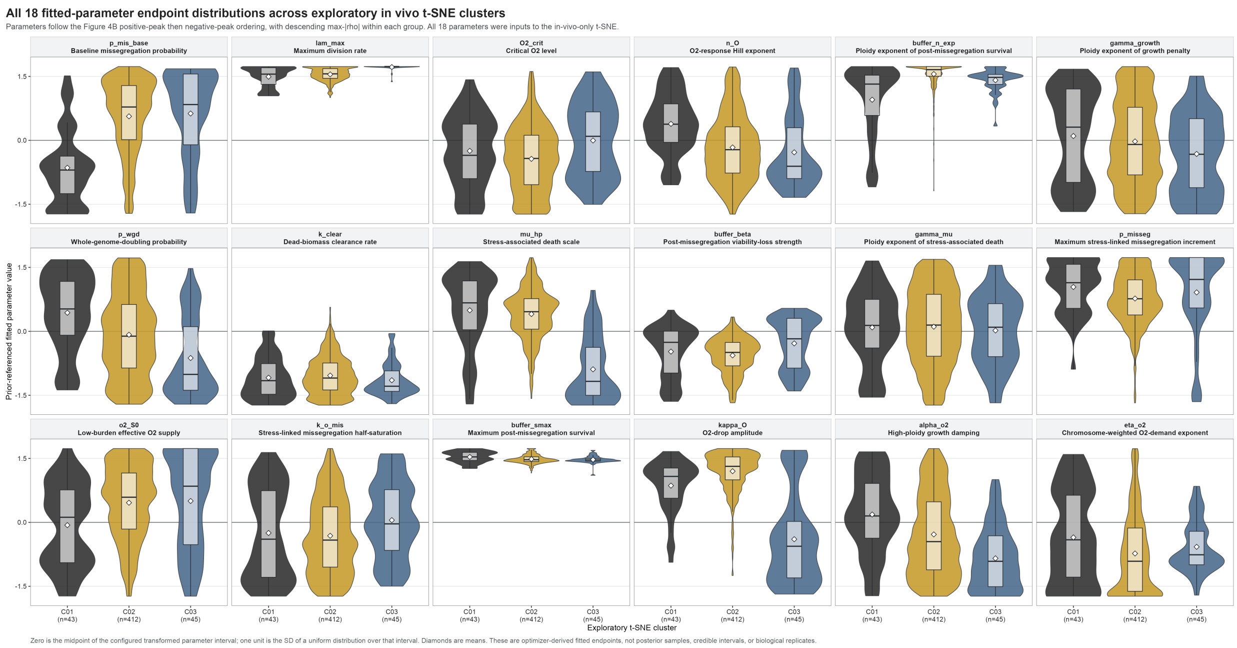
Supplementary Table 4: Separate *in vivo* fitted-landscape and numerical-search summary. Cluster comparisons are descriptive because the groups were constructed from the same fitted parameters subsequently compared. Optimizer endpoints are numerical solutions, not biological replicates or posterior samples.

Analysis	Quantity	Value	Definition or interpretation
Fixed-O_2 association audit	Evaluation dimensions	$500 \times 18 \times 201$	Fitted solutions, parameters, and oxygen concentrations from 0% to 5% in 0.025-percentage-point increments
Embedding input	Numerical rows	$127{,}500 + 500$	Retained initial-population rows plus fitted endpoints; 18 fitted parameters were embedded
Exploratory clustering	C01, C02, C03 sizes	43, 412, 45	Endpoint counts in the selected three-group solution
Exploratory clustering	Selected k; average silhouette	3; 0.655	Descriptive separation on the saved *in vivo*-only embedding
Omnibus comparison	BH-significant parameters	14 of 18	Kruskal–Wallis tests with Benjamini–Hochberg $q < 0.05$
Top cluster contrast	Stress-associated death scale, μ_{hp}	0.154	Kruskal–Wallis epsilon-squared
Top cluster contrast	Oxygen-drop amplitude, κ_O	0.151	Kruskal–Wallis epsilon-squared
Top cluster contrast	Maximum division rate, $\lambda_{\max}$	0.143	Kruskal–Wallis epsilon-squared
Top cluster contrast	Survival ploidy exponent, $n_{\exp}$	0.115	Kruskal–Wallis epsilon-squared
Top cluster contrast	Baseline missegregation probability, $p_{\mathrm{mis,base}}$	0.101	Kruskal–Wallis epsilon-squared
Top cluster contrast	Resource-dependent growth damping, α_{O_2}	0.049	Kruskal–Wallis epsilon-squared
Objective proximity	Starts within 1%, 5%, and 10%	29, 241, 405 of 500	Relative excess above the lowest retained objective
Optimizer record	DE termination flag	500 of 500	Implementation-specific differential-evolution stopping flag
Optimizer record	Local refinement attempted; accepted	500; 460	Recorded local-refinement outcomes
Optimizer record	Accepted local code 0	276	Accepted local refinements carrying convergence code 0
Selected solution	Active configured bounds	2 of 20	Exact active-bound count for the lowest-objective separate fit, seed 25

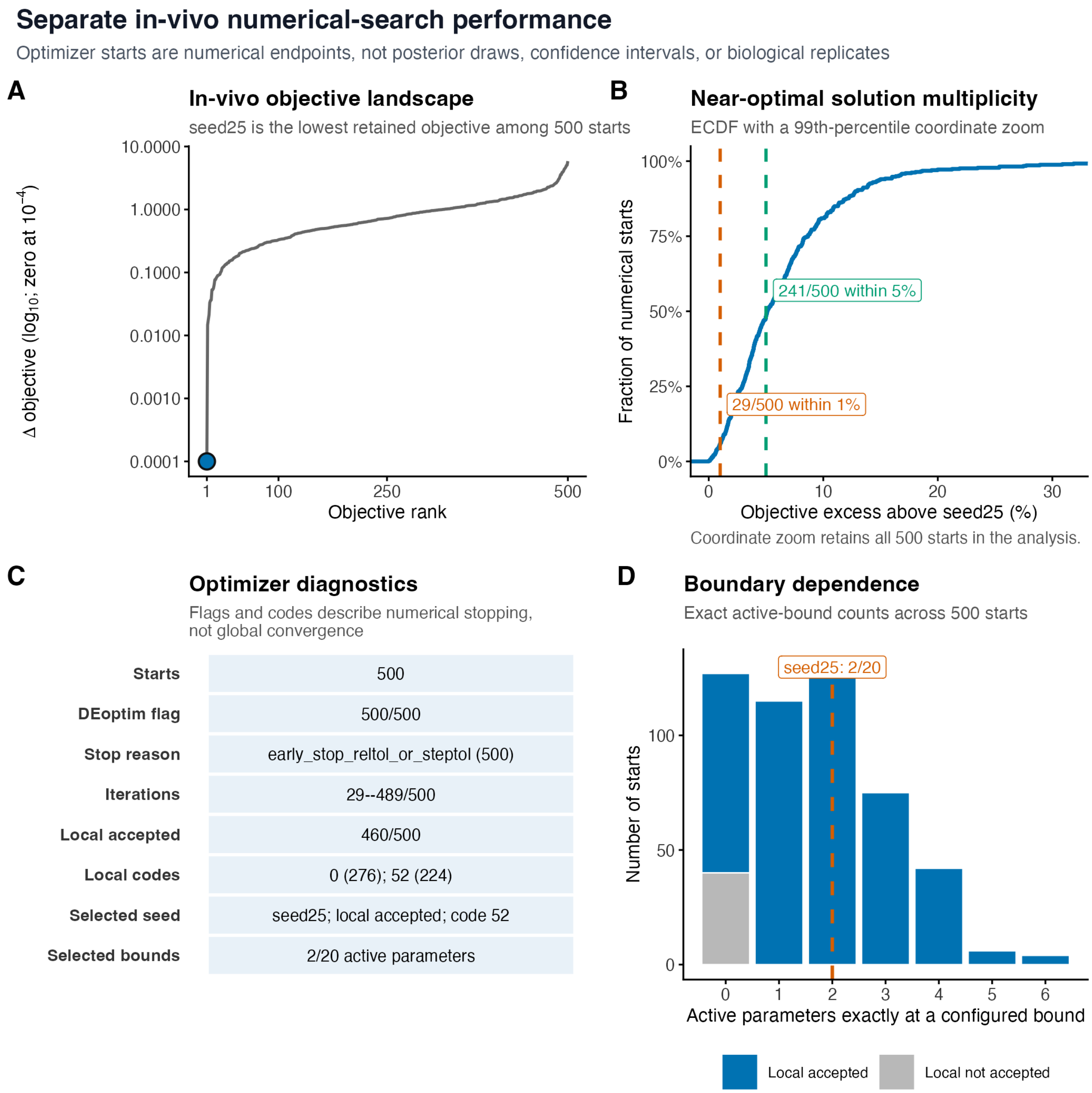
1.2 *In vitro* experiments

The *in vitro* analysis used matched near-diploid and near-tetraploid SUM159 lineages. For each starting ploidy, a control branch was maintained at 20.5% O_2, and two parallel oxygen-deprivation lineages (O1 and O2) were stepped over successive passages through target levels of 2%, 1%, 0.5%, 0.3%, 0.2%, 0.1%, and 0% O_2. The main deprivation branch was then maintained at target 0% O_2; an additional branch after the first 0% passage was evaluated at target 1% O_2. We describe O1 and O2 as parallel lineages because the available records do not establish whether they should be classified as biological replicates (Supplementary Table 1).

For calibration, each fit entry retained the assigned oxygen level, passage duration, initial and final corrected cell counts, and a segment-level growth-rate estimate. Chromosome-number measurements and G0/G1 DNA-content density profiles were linked to the corresponding passage when



Supplementary Figure 1: **All 18 fitted-parameter endpoint distributions across the exploratory separate-*in vivo* t-SNE clusters.** The 18 facets are arranged in three rows and six columns in the same positive-peak then negative-peak order as Figure 4B, with decreasing maximum $|\rho|$ within each sign group. Each facet shows the prior-referenced fitted-value distribution for clusters C01, C02, and C03; boxplots show medians and interquartile ranges and diamonds show means. Figure 4D displays the six-parameter subset selected from these same values by the pre-specified omnibus-test and effect-size rule. All displayed parameters contributed to the *in vivo*-only 18-parameter t-SNE and endpoint clustering. Zero is the midpoint of the configured transformed parameter interval and one unit is the standard deviation of a uniform distribution over that interval. Cluster sizes, tests, and effect sizes are reported in Supplementary Table 4. These are distributions of optimizer-derived fitted endpoints, not posterior samples, credible intervals, independent validation, or biological replication.



Supplementary Figure 2: **Complete numerical-search diagnostics for the separate *in vivo* fit ensemble. (A)** Objective difference by rank across 500 starts, with the lowest retained fit, seed 25, highlighted. **(B)** Empirical cumulative distribution of relative objective excess above seed 25. Vertical markers show that 29 starts were within 1% and 241 were within 5%; the displayed x-range ends at the 99th percentile by coordinate zoom without removing starts from the analysis. **(C)** Recorded differential-evolution termination, iteration, local-refinement, selected-seed, and boundary diagnostics. Flags and codes describe implementation-specific numerical stopping and do not establish global convergence. **(D)** Distribution of exact active-parameter boundary counts across all 500 retained endpoints, separated by local-refinement acceptance status; the dashed line marks seed 25, for which two of 20 active parameters were exactly at configured bounds. The corresponding counts are reported in Supplementary Table 4. Optimizer starts are numerical endpoints, not posterior draws, biological replicates, or inferential uncertainty.

available. The current fit objects contain 131 entries, including 114 with finite growth estimates, 12 with chromosome-number measurements (220 single-cell values in total), and 20 with matched G0/G1 profiles. Each G0/G1 profile was represented on a 200-point ploidy grid and normalized before comparison with the simulated viable chromosome-number distribution (Supplementary Table 1).

For the Figure 1 timeline, population-size measurement availability was audited against the raw passage snapshot: all 131 passage IDs had a positive seeding count and at least one positive linked harvest count. The displayed representative tracks collapse matching records to cohort, condition, and reconstructed day; common pre-branch endpoints are repeated on both aligned eventual branches. For panel B, direct autosomal chromosome counts were converted to ploidy by division by $N_{\text{unit}} = 22$. For panel C, each observed flow profile was normalized to unit density mass. Profiles from parallel O1 and O2 lines at the same cohort, condition, and reconstructed day were averaged with equal sample weight for the half-violin and density-weighted box summary, while the global density maximum of each individual profile supplied its open-triangle marker. The complete normalized profiles were used to calculate peaks, quantiles, and whiskers before applying the ploidy 1–5 plotting viewport. This viewport retained at least 85.43% of normalized density mass in every sample; density above ploidy 5 was cropped without within-window renormalization (Supplementary Table 1). These display summaries did not alter the flow-density likelihood.

Within the passage-structured model, oxygen was held fixed at the assigned value for each segment. For segments represented by parallel records, passage duration and endpoint cell counts were summarized by the median, while all available growth, chromosome-number, and flow-density observations contributed to their corresponding likelihood terms.

XX Before submission, complete the cell culture and assay reporting by specifying cell line provenance and authentication, mycoplasma testing, passage range, culture medium and supplements, incubation conditions, oxygen control equipment, independent verification of oxygen exposure, equilibration procedures, seeding and passaging protocols, replicate structure, allocation procedures, cell counting method, sampling schedule, chromosome counting method, flow cytometry preparation and gating, assay specific quality control, and all prespecified exclusion criteria.

Supplementary Table 5: Final karyotype distributions of the parallel 2N oxygen-deprivation lines and their shared fitted state. O1 and O2 were propagated as separate lines from the same 2N precursor through the matched nominal oxygen schedule. The fitted row is the single model distribution evaluated against both observed lines at the final matched chromosome-count segment.

Distribution	n	Mean N	Median N	$N \geq 80$	Component structure
O1 observed	20	66.85	49	8/20 (40.0%)	12 cells at $N = 46$–49; 8 cells at $N =$ 90–100
O2 observed	20	88.05	95	16/20 (80.0%)	4 cells at $N = 49$–76; 16 cells at $N =$ 93–99
Shared fitted state	—	64.27	60	32.5% probability	dominant local modes at $N = 40$ and $N = 83$

Note: $N \geq 80$ is a descriptive bin chosen within the unoccupied interval between the low- and high-chromosome components of the final observed samples; it is not a fitted threshold or inferential cutoff. Observed entries are single-cell counts. The fitted entry is a probability distribution and therefore has no observed sample size. Values are descriptive and do not constitute an additional statistical test.

Supplementary Table 6: Modeled 4N chromosome-number decline and the fitted contribution of the compartment representing death associated with resource stress across the ten lowest-objective separate *in vitro* fits. Initial and terminal mean chromosome numbers refer to the first and last modeled states of the selected 4N lineage; its terminal portion was maintained at target 0% O_2. The last column is the maximum daily burden assigned by the model to death associated with resource stress, expressed as a percentage of total burden over the full modeled 4N trajectory.

Rank	Fit	Objective	Initial mean N	Terminal mean N	Decline in mean N	Maximum modeled stress death burden (%)
1	`seed10`	3.8525	84.34	80.76	3.58	1.76
2	`seed132`	3.8533	84.21	80.66	3.55	1.76
3	`seed81`	3.8541	84.22	80.67	3.54	1.77
4	`seed294`	3.8594	84.71	80.89	3.82	1.74
5	`seed337`	3.8598	84.66	80.76	3.91	1.74
6	`seed106`	3.8605	84.45	80.53	3.91	1.82
7	`seed317`	3.8610	84.30	80.13	4.17	1.76
8	`seed140`	3.8610	84.21	80.54	3.67	1.74
9	`seed285`	3.8618	84.24	80.20	4.04	1.77
10	`seed464`	3.8623	84.68	80.73	3.96	1.77

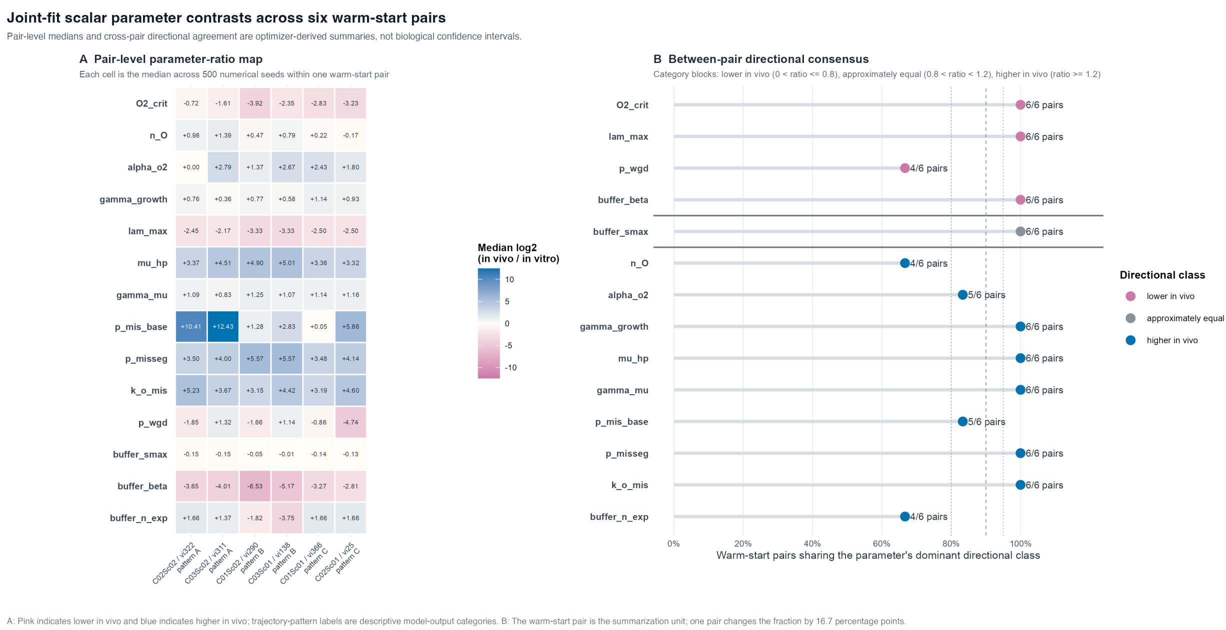
Seed-10 audit quantity	Value	Scope
Maximum passage-end stress-associated dead-burden fraction	2N: 1.05%; 4N: 1.71%	Severe-deprivation passage-end summaries shown in Figure 3F
Numerical starts and fitted parameters	500; 20	Separate *in vitro* calibration
Differential-evolution convergence flags	0 of 500	Recorded implementation-specific flag
Selected local refinement	accepted, code 1	Lowest-objective retained seed 10
Selected active configured bounds	5 of 20	Exact active-bound count

Note: Fits are ordered by the retained objective value. The decline is the initial minus terminal predicted mean chromosome number. The dead burden percentage is the maximum, over all daily 4N states, of the burden assigned by the model to death associated with resource stress as a percentage of total burden. It is therefore a trajectory maximum rather than an endpoint-only value. These optimizer-selected solutions are not biological replicates or a confidence interval.

Supplementary Table 7: Warm-start selection, joint numerical-search diagnostics, and objective eligibility. Every pair used the same separate *in vitro* seed-10 anchor and 500 numerical starts. The three primary pairs are marked in the second column. The lowest-objective decile retained 50 endpoints per pair. Numerical endpoints and cluster assignments are search diagnostics rather than biological replicates, confidence sets, or biological groups.

Pair	Primary	Separate *in vivo* start	Joint winner	Minimum objective	Iterations of 500	DE flags of 500	Local refinement	Winner bounds of 20	Decile cutoff above minimum
C01Sc01	yes	366	472	18.8523	26–82	500	rejected, code 52	2	0.038700
C01Sc02	no	290	155	19.7913	26–90	500	rejected, code 52	4	0.061291
C02Sc01	yes	25	497	18.9705	26–46	500	rejected, code 52	2	0.003593
C02Sc02	no	322	54	18.8901	26–76	500	rejected, code 52	2	0.061055
C03Sc01	no	138	122	19.9782	26–116	500	rejected, code 52	3	0.071578
C03Sc02	yes	311	18	19.4145	26–73	500	rejected, code 52	2	0.061229

Audit quantity	Value	Interpretation
Complete endpoints passing hard configuration and feasibility checks	3,000 of 3,000	Six pairs, 500 endpoints per pair
Widest retained objective set passing post-fit operator checks	600 of 600	Lowest-objective 20%, 100 endpoints per pair
Primary complete response surfaces	300	Lowest-objective 10%, 50 endpoints per pair across six pairs
Parameter-specific context-ratio records passing feasibility checks	42,000 of 42,000	Six pairs, 500 endpoints, and 14 paired parameters
Unique exact 28-value context-specific parameter signatures	947	Lowest-objective representative retained within each pair
Pooled warm-start embedding	228,000 vectors	127,500 *in vivo* population points, 99,500 *in vitro* population points, and 500 final solutions per context
Pooled embedding configuration	seed 123; perplexity 30; $\theta = 0.5$; 1,000 iterations	Saved t-SNE coordinates used for warm-start selection
Saved-embedding silhouette, $k = 3$; $k = 4$; difference	0.815694; 0.816651; 0.000957	$k = 4$ was numerically near-tied and produced a singleton region
Selected k across 100 subsamples retaining 80% of endpoints	$k = 3$: 58; $k = 4$: 42	Region-count resampling on the saved embedding
Fixed-$k = 3$ label agreement under the same subsampling	Median ARI 1.00	Agreement with saved labels
Standardized 14-parameter-space $k = 3$ agreement	ARI 0.052	Poor agreement with the saved-embedding regions



Supplementary Figure 3: **Scalar joint-fit parameter contrasts across the six warm-start pairs.** **(A)** Pair-by-parameter map of the median $\log_2(\textit{in vivo}/\textit{in vitro})$ ratio across the 500 numerical seeds within each warm-start pair. Pink indicates lower fitted values *in vivo*, blue indicates higher fitted values *in vivo*, and the printed signed values give the exact pair medians. Pairs are grouped by their final descriptive 0–1,000-day modeled ploidy-trajectory pattern; these are model-output patterns rather than biological groups, and all six pairs share the same seed-10 *in vitro* anchor. **(B)** Directional consensus across warm-start pairs under the prespecified reclassification used here: lower *in vivo* for $0 < \textit{in vivo}/\textit{in vitro} \leq 0.8$, approximately equal for $0.8 < \textit{in vivo}/\textit{in vitro} < 1.2$, and higher *in vivo* for ratios ≥ 1.2. Parameters are arranged in these three directional blocks rather than sorted by consensus magnitude. Horizontal position and labels show the fraction and exact number of the six pairs sharing the parameter's dominant class. The pair, not the 3,000 numerical runs or biological specimens, is the summarization unit. The numerical search inventory is reported in Supplementary Table 7.

Supplementary Table 8: Context-specific differential-evolution initial-population and optimizer-endpoint distributions for the 14 paired parameters. Each C-family cell reports the in vivo and in vitro natural-scale parameter distributions separately. The DE-initial distribution comprises 200,000 members reconstructed from the 500 runs (400 members per run); the endpoint distribution comprises the corresponding 500 feasible final solutions. Entries report median [5th, 95th percentile], followed by context-specific active-bound occupancy. Cross-family direction is an auxiliary audit computed from the paired endpoint ratio. These numerical-search distributions are not posterior or confidence intervals.

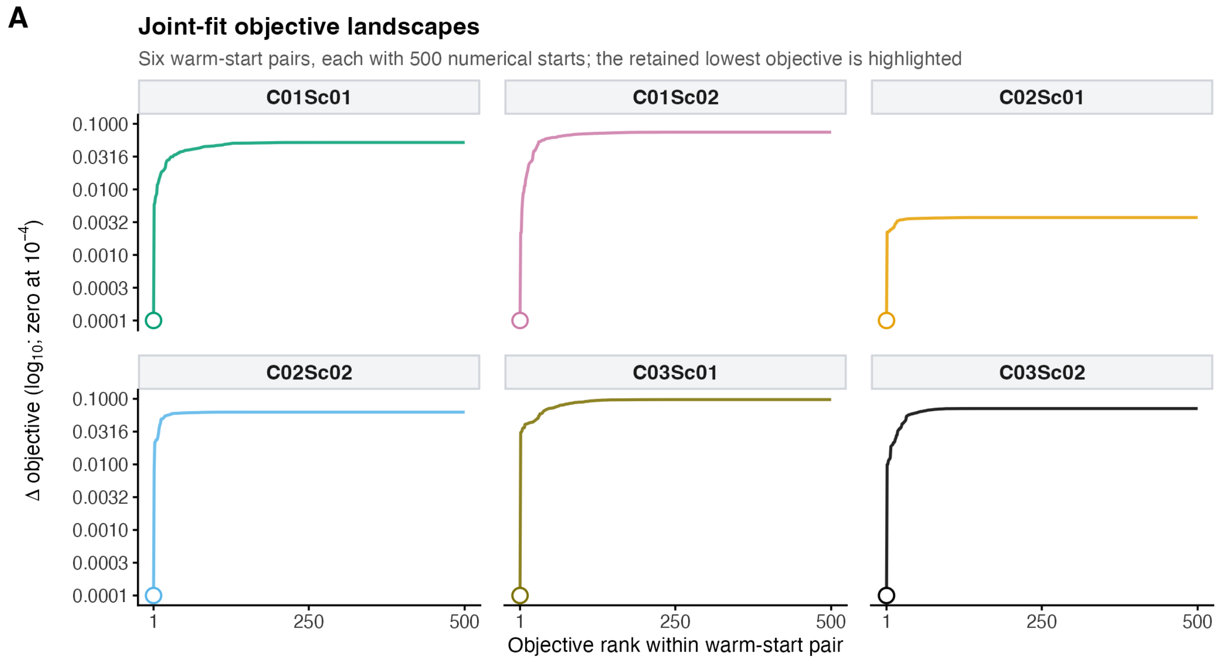
Parameter	C01: in vivo; in vitro	C02: in vivo; in vitro	C03: in vivo; in vitro	Directional/search audit
`lam_max`	in vivo: DE init 0.247 [0.201, 0.306]; endpoint 0.239 [0.239, 0.239]; bound=0%; in vitro: DE init 1.32 [1.09, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	in vivo: DE init 0.247 [0.202, 0.306]; endpoint 0.239 [0.239, 0.239]; bound=0%; in vitro: DE init 1.32 [1.09, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	in vivo: DE init 0.308 [0.259, 0.366]; endpoint 0.300 [0.300, 0.300]; bound=0%; in vitro: DE init 1.33 [1.13, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	lower in vivo; max paired W_E/W_I=0; max bound=0%
`p_mis_base`	in vivo: DE init 0.00000988 [0.00000155, 0.0000954]; endpoint 0.00000882 [0.00000337, 0.00000882]; bound=0%; in vitro: DE init 0.00000958 [0.00000151, 0.0000925]; endpoint 0.00000854 [0.00000329, 0.00000856]; bound=0.2%	in vivo: DE init 0.000495 [0.0000634, 0.00356]; endpoint 0.000495 [0.000495, 0.000495]; bound=0%; in vitro: DE init 0.00000854 [0.00000133, 0.0000615]; endpoint 0.00000854 [0.00000854, 0.00000854]; bound=0%	in vivo: DE init 0.0428 [0.00795, 0.0918]; endpoint 0.0471 [0.0471, 0.0471]; bound=0%; in vitro: DE init 0.00000982 [0.00000172, 0.000123]; endpoint 0.00000854 [0.00000854, 0.00000854]; bound=0%	higher in vivo; max paired W_E/W_I=0.372; max bound=0.2%
`p_wgd`	in vivo: DE init 0.000164 [0.0000334, 0.000617]; endpoint 0.000182 [0.000182, 0.000202]; bound=0%; in vitro: DE init 0.000299 [0.0000607, 0.000964]; endpoint 0.000331 [0.000298, 0.000331]; bound=0%	in vivo: DE init 0.0000144 [0.0000103, 0.000197]; endpoint 0.0000124 [0.0000124, 0.0000124]; bound=0%; in vitro: DE init 0.000333 [0.0000181, 0.000947]; endpoint 0.000331 [0.000331, 0.000331]; bound=0%	in vivo: DE init 0.000626 [0.000163, 0.000988]; endpoint 0.000830 [0.000657, 0.000830]; bound=0%; in vitro: DE init 0.000250 [0.0000648, 0.000841]; endpoint 0.000331 [0.000331, 0.000405]; bound=0%	family-dependent or overlaps equality; max paired W_E/W_I=0.690; max bound=0%
`p_misseg`	in vivo: DE init 0.117 [0.0542, 0.239]; endpoint 0.117 [0.117, 0.117]; bound=0%; in vitro: DE init 0.0106 [0.00562, 0.0215]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	in vivo: DE init 0.186 [0.0887, 0.368]; endpoint 0.187 [0.187, 0.187]; bound=0%; in vitro: DE init 0.0106 [0.00572, 0.0213]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	in vivo: DE init 0.169 [0.0799, 0.337]; endpoint 0.169 [0.169, 0.169]; bound=0%; in vitro: DE init 0.0105 [0.00570, 0.0211]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	higher in vivo; max paired W_E/W_I=0; max bound=0%
`k_o_mis`	in vivo: DE init 0.00103 [0.000205, 0.00513]; endpoint 0.00103 [0.00103, 0.00103]; bound=0%; in vitro: DE init 0.000113 [0.0000225, 0.000563]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	in vivo: DE init 0.00274 [0.000591, 0.0126]; endpoint 0.00274 [0.00274, 0.00274]; bound=0%; in vitro: DE init 0.000113 [0.0000243, 0.000525]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	in vivo: DE init 0.00144 [0.000296, 0.00698]; endpoint 0.00144 [0.00144, 0.00144]; bound=0%; in vitro: DE init 0.000113 [0.0000231, 0.000548]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	higher in vivo; max paired W_E/W_I=0; max bound=0%
`O2_crit`	in vivo: DE init 0.223 [0.178, 0.278]; endpoint 0.223 [0.223, 0.223]; bound=0%; in vitro: DE init 1.59 [1.27, 1.98]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.169 [0.130, 0.219]; endpoint 0.169 [0.169, 0.169]; bound=0%; in vitro: DE init 1.59 [1.22, 2.05]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.518 [0.463, 0.580]; endpoint 0.518 [0.518, 0.518]; bound=0%; in vitro: DE init 1.59 [1.42, 1.78]; endpoint 1.59 [1.59, 1.59]; bound=0%	lower in vivo; max paired W_E/W_I=0; max bound=0%

Parameter	C01: in vivo; in vitro	C02: in vivo; in vitro	C03: in vivo; in vitro	Directional/search audit
`n_0`	in vivo: DE init 2.17 [1.77, 2.58]; endpoint 2.17 [2.17, 2.17]; bound=0%; in vitro: DE init 1.87 [1.46, 2.27]; endpoint 1.87 [1.87, 1.87]; bound=0%	in vivo: DE init 1.66 [1.31, 2.02]; endpoint 1.66 [1.66, 1.66]; bound=0%; in vitro: DE init 1.87 [1.52, 2.22]; endpoint 1.87 [1.87, 1.87]; bound=0%	in vivo: DE init 4.77 [4.15, 4.98]; endpoint 4.89 [4.89, 4.89]; bound=0%; in vitro: DE init 1.97 [1.13, 3.27]; endpoint 1.87 [1.87, 1.87]; bound=0%	family-dependent or overlaps equality; max paired W_E/W_I=0; max bound=0%
`alpha_o2`	in vivo: DE init 2.50 [2.17, 2.74]; endpoint 2.69 [2.69, 2.69]; bound=0%; in vitro: DE init 0.537 [0.503, 0.614]; endpoint 0.500 [0.500, 0.500]; bound=100%	in vivo: DE init 1.65 [1.49, 1.75]; endpoint 1.74 [1.74, 1.74]; bound=0%; in vitro: DE init 0.527 [0.502, 0.582]; endpoint 0.500 [0.500, 0.500]; bound=100%	in vivo: DE init 3.15 [2.64, 3.60]; endpoint 3.45 [3.45, 3.45]; bound=0%; in vitro: DE init 0.544 [0.504, 0.636]; endpoint 0.500 [0.500, 0.500]; bound=100%	higher in vivo; max paired W_E/W_I=0; max bound=100%
`gamma_growth`	in vivo: DE init 3.21 [2.26, 3.82]; endpoint 3.31 [3.31, 3.31]; bound=0%; in vitro: DE init 1.62 [1.51, 2.01]; endpoint 1.50 [1.50, 1.50]; bound=100%	in vivo: DE init 2.79 [1.95, 3.32]; endpoint 2.86 [2.86, 2.86]; bound=0%; in vitro: DE init 1.60 [1.51, 1.96]; endpoint 1.50 [1.50, 1.50]; bound=100%	in vivo: DE init 1.96 [1.55, 2.29]; endpoint 1.93 [1.93, 1.93]; bound=0%; in vitro: DE init 1.55 [1.50, 1.86]; endpoint 1.50 [1.50, 1.50]; bound=99.8%	higher in vivo; max paired W_E/W_I=0; max bound=100%
`mu_hp`	in vivo: DE init 0.0459 [0.00957, 0.121]; endpoint 0.0513 [0.0513, 0.0513]; bound=0%; in vitro: DE init 0.00592 [0.00507, 0.0118]; endpoint 0.00500 [0.00500, 0.00500]; bound=99.8%	in vivo: DE init 0.0447 [0.00935, 0.118]; endpoint 0.0499 [0.0499, 0.0499]; bound=0%; in vitro: DE init 0.00591 [0.00507, 0.0118]; endpoint 0.00500 [0.00500, 0.00500]; bound=100%	in vivo: DE init 0.0964 [0.0219, 0.244]; endpoint 0.114 [0.114, 0.114]; bound=0%; in vitro: DE init 0.00612 [0.00509, 0.0117]; endpoint 0.00500 [0.00500, 0.00500]; bound=99.8%	higher in vivo; max paired W_E/W_I=0; max bound=100%
`gamma_mu`	in vivo: DE init 2.56 [1.80, 3.04]; endpoint 2.64 [2.63, 2.64]; bound=0%; in vitro: DE init 1.30 [1.21, 1.61]; endpoint 1.20 [1.20, 1.21]; bound=94%	in vivo: DE init 2.60 [1.83, 3.09]; endpoint 2.68 [2.68, 2.68]; bound=0%; in vitro: DE init 1.30 [1.21, 1.61]; endpoint 1.20 [1.20, 1.20]; bound=98.6%	in vivo: DE init 2.09 [1.46, 2.49]; endpoint 2.14 [2.14, 2.14]; bound=0%; in vitro: DE init 1.27 [1.21, 1.55]; endpoint 1.20 [1.20, 1.20]; bound=98.4%	higher in vivo; max paired W_E/W_I=0.0191; max bound=98.6%
`buffer_smax`	in vivo: DE init 0.857 [0.695, 0.984]; endpoint 0.909 [0.909, 0.909]; bound=0%; in vitro: DE init 0.949 [0.786, 0.999]; endpoint 1.00 [1.00, 1.00]; bound=100%	in vivo: DE init 0.861 [0.699, 0.984]; endpoint 0.914 [0.914, 0.914]; bound=0%; in vitro: DE init 0.947 [0.785, 0.999]; endpoint 1.00 [1.00, 1.00]; bound=100%	in vivo: DE init 0.854 [0.691, 0.984]; endpoint 0.904 [0.904, 0.904]; bound=0%; in vitro: DE init 0.950 [0.787, 0.999]; endpoint 1.00 [1.00, 1.00]; bound=100%	lower in vivo; max paired W_E/W_I=0; max bound=100%
`buffer_beta`	in vivo: DE init 0.165 [0.127, 0.215]; endpoint 0.165 [0.165, 0.165]; bound=0%; in vitro: DE init 1.59 [1.22, 2.07]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.227 [0.183, 0.283]; endpoint 0.227 [0.227, 0.227]; bound=0%; in vitro: DE init 1.59 [1.28, 1.98]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.0990 [0.0710, 0.138]; endpoint 0.0990 [0.0990, 0.0990]; bound=0%; in vitro: DE init 1.59 [1.14, 2.22]; endpoint 1.59 [1.59, 1.59]; bound=0%	lower in vivo; max paired W_E/W_I=0; max bound=0%
`buffer_n_exp`	in vivo: DE init 9.24 [6.95, 9.93]; endpoint 10.0 [10.0, 10.0]; bound=99.8%; in vitro: DE init 3.35 [2.19, 6.61]; endpoint 3.16 [3.16, 3.16]; bound=0%	in vivo: DE init 9.24 [6.95, 9.93]; endpoint 10.0 [10.0, 10.0]; bound=100%; in vitro: DE init 3.35 [2.19, 6.60]; endpoint 3.16 [3.16, 3.16]; bound=0%	in vivo: DE init 8.13 [5.80, 9.84]; endpoint 8.13 [8.13, 8.13]; bound=0%; in vitro: DE init 3.16 [2.25, 5.15]; endpoint 3.16 [3.16, 3.16]; bound=0%	higher in vivo; max paired W_E/W_I=0; max bound=100%

1.3 A Mechanistic Model of Chromosome-Number Evolution

We model tumor population dynamics as a deterministic state-structured ODE system over chromosome-number classes. The formulation separates viable cells from non-viable biomass, couples local oxy-

Joint-fit numerical-search performance



B **Joint-search diagnostics by warm-start pair**

Flags, codes, numerical-start spread, and bounds describe search behavior.
They are not posterior uncertainty or biological replication.

	Starts	DEoptim flag	Iterations	Selected local step	Near-optimal starts	Selected at bounds
C01Sc01	500	500/500	26--82/500	not accepted; code 52	500 <=1% 500 <=5%	2/20
C01Sc02	500	500/500	26--90/500	not accepted; code 52	500 <=1% 500 <=5%	4/20
C02Sc01	500	500/500	26--46/500	not accepted; code 52	500 <=1% 500 <=5%	2/20
C02Sc02	500	500/500	26--76/500	not accepted; code 52	500 <=1% 500 <=5%	2/20
C03Sc01	500	500/500	26--116/500	not accepted; code 52	500 <=1% 500 <=5%	3/20
C03Sc02	500	500/500	26--73/500	not accepted; code 52	500 <=1% 500 <=5%	2/20

Supplementary Figure 4: **Joint-fit numerical-search performance across the six warm-start pairs. (A)** Objective-difference landscapes for the six 500-start joint warm-start pairs. Within each pair, starts are ranked by retained objective, the lowest retained winner is highlighted, and exact zero is displayed at 10^{-4} on the $\log_{10}$ axis. **(B)** Pair-resolved run diagnostics, including numbers of starts, recorded differential-evolution convergence flags, completed-iteration ranges relative to the target, selected local-step status and code, near-optimal start counts, and exact active-bound counts for the selected winners. The corresponding values are reported in Supplementary Table 7. These diagnostics characterize the sampled numerical searches; they are not biological replication, posterior distributions, confidence intervals, bootstrap uncertainty, profile likelihoods, held-out validation, or proof of a global optimum.

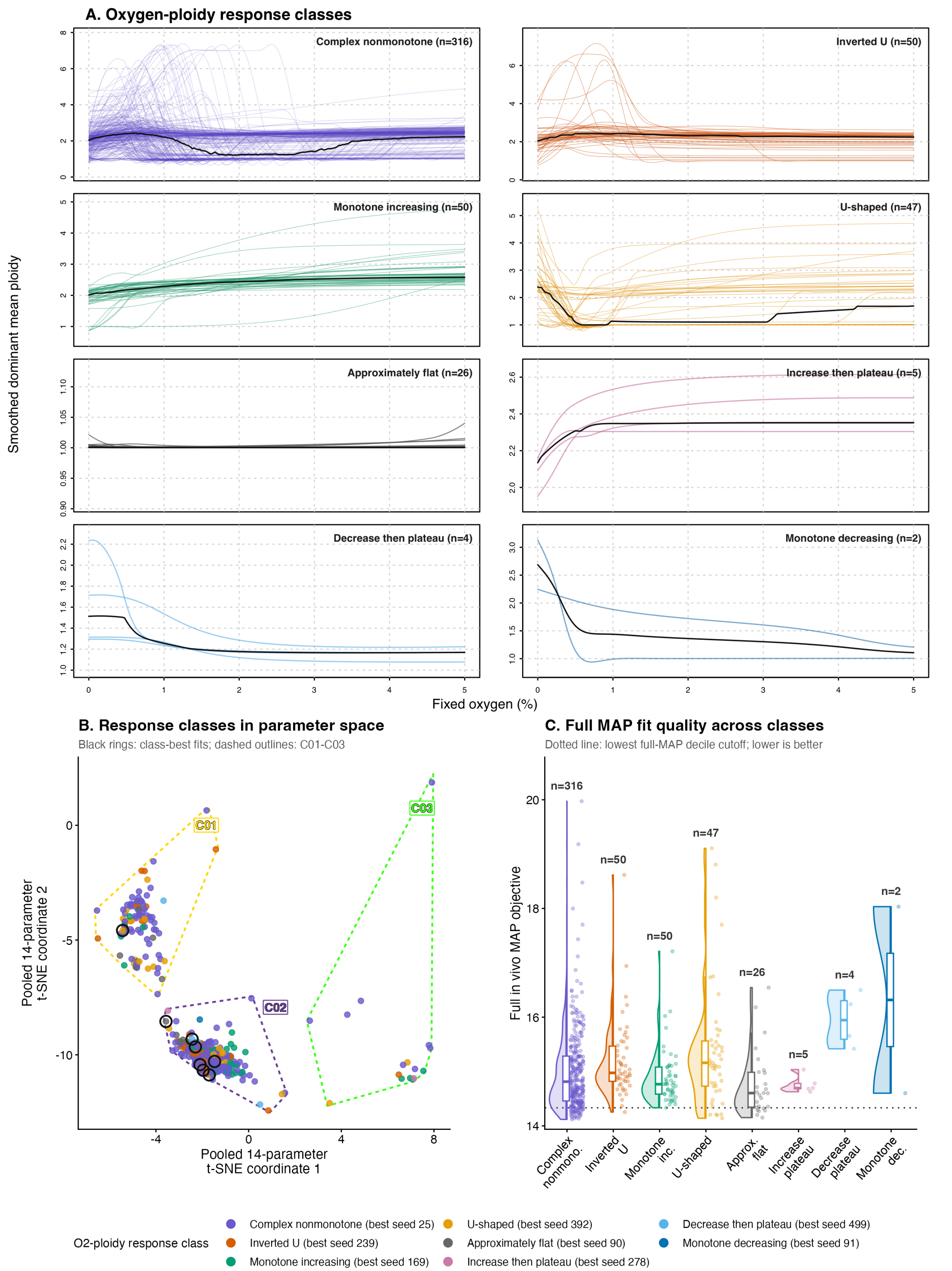
Supplementary Table 9: Fixed-O_2 response-shape and spectral-gap classifications for the 500 separate *in vivo* fitted solutions. Reliability was classified from the fraction of the 201 oxygen-grid evaluations below the prespecified spectral-gap thresholds. These categories describe asymptotic numerical diagnostics and are not additional fitted-data hypothesis tests.

Evaluation quantity	Value
Separate fitted solutions and oxygen grid	500 solutions; 201 values from 0% to 5% in 0.025-percentage-point increments
Total seed–oxygen evaluations	100,500
Unreliable criterion	At least 25% of grid values with spectral gap below 0.005
Caution criterion	Any grid value below 0.005, or at least 10% below 0.01

Response-shape class	Solutions	Fraction (%)
Complex nonmonotone	316	63.2
Inverted U-shaped	50	10.0
Monotone increasing	50	10.0
U-shaped	47	9.4
Approximately flat	26	5.2
Increase then plateau	5	1.0
Decrease then plateau	4	0.8
Monotone decreasing	2	0.4

Spectral-gap class	Solutions	Fraction (%)
Reliable	141	28.2
Caution	168	33.6
Unreliable	191	38.2

Warm-start representative	Separate *in vivo* seed	Spectral-gap class
C01Sc01	366	unreliable
C01Sc02	290	unreliable
C02Sc01	25	caution
C02Sc02	322	caution
C03Sc01	138	unreliable
C03Sc02	311	caution



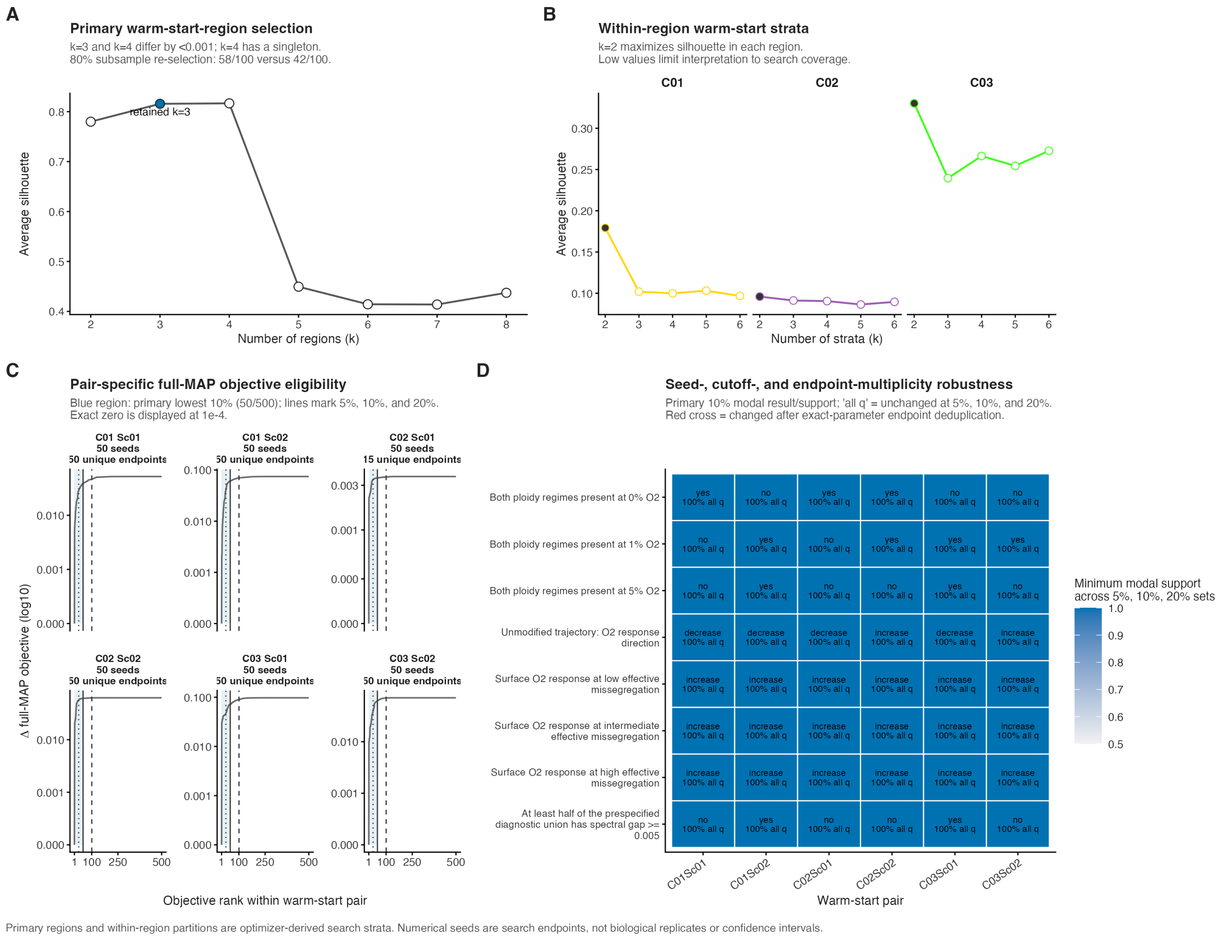
Supplementary Figure 5: **Separate-fit fixed-O_2 response classes and their fitted-landscape context. (A) Oxygen–ploidy response classes.** Analytical dominant-eigenvector ploidy curves from 500 separate *in vivo* fitted solutions were grouped by regression-smoothed response shape. The eight response classes are arranged in four rows and two columns, with a common oxygen axis and shared outer coordinate labels; the panel-specific ploidy ranges preserve visibility of both broad and nearly flat responses. Class names and sizes are printed within the upper-right corner of each panel. Individual solutions are colored, and the black curve is the pointwise class median. These pooled class counts include reliable, caution, and unreliable spectral gap strata. **(B) Response classes in**

Supplementary Table 10: Joint-ensemble fixed-O_2 response summaries for the primary lowest-objective decile. Response entries are pair medians of dominant mean ploidy at 5% minus 0% O_2 across 50 retained optimizer endpoints. Fixed-input levels are the prespecified low, intermediate, and high effective per-chromosome missegregation diagnostics. The unmodified trajectory restores each endpoint's fitted oxygen-dependent missegregation function.

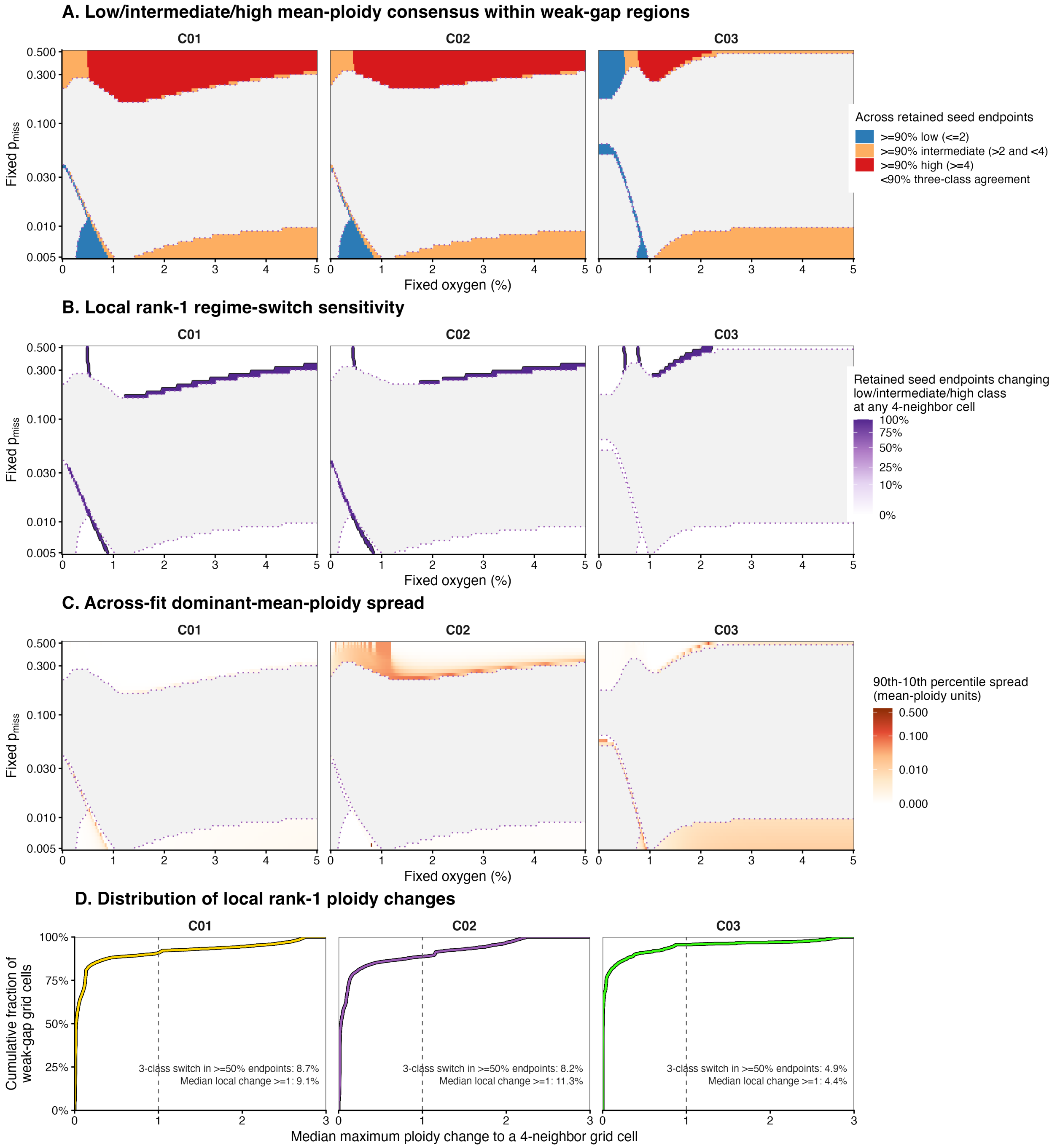
Pair	Unmodified trajectory	Fixed low	Fixed intermediate	Fixed high	Trajectory direction
C01Sc01	−1.112	1.624	0.814	4.888	decrease
C01Sc02	−0.289	0.466	0.434	0.953	decrease
C02Sc01	−1.172	1.652	0.809	4.874	decrease
C02Sc02	+0.143	1.619	0.574	1.535	increase
C03Sc01	−0.188	0.514	0.460	1.021	decrease
C03Sc02	+1.175	1.678	1.693	1.552	increase

Displayed pair	Median population-average $p_{\mathrm{miss,eff}}$ at 0% O_2	Median population-average $p_{\mathrm{miss,eff}}$ at 5% O_2
C01Sc01	0.115	0.00110
C02Sc01	0.178	0.00236
C03Sc02	0.212	0.0475

Design or robustness quantity	Value	Scope
Complete fixed-input surface per endpoint	201 × 60	Oxygen by effective-missegregation grid
Primary complete-surface evaluations	3,618,000	300 endpoints and 12,060 grid conditions per endpoint
Dense fixed-missegregation line grid	496 values	0.005–0.500 in increments of 0.001
Distinct dense-grid operator evaluations	11,465,040	Exact endpoints evaluated once before restoring seed multiplicity
Threshold-sensitivity diagnostic union	774 conditions	Reference oxygen levels and fixed-missegregation slices for additional 20% endpoints
Pair-by-diagnostic modal results unchanged across the 5%, 10%, and 20% sets	48 of 48	Six pairs and eight qualitative diagnostics
Seed-weighted versus unique-parameter comparisons with the same modal result	144 of 144	Six pairs, eight diagnostics, and three cutoffs
Within-set modal support across all qualitative audits	100%	Minimum over all pair, diagnostic, and cutoff combinations
Unique response-parameter endpoints in a 50-seed decile ensemble	15–50	Pair-specific range; C02Sc01 had 15
Pointwise ploidy-regime agreement after exact endpoint deduplication	100%	Minimum across the primary complete surfaces



Supplementary Figure 6: **Cluster-selection, objective-eligibility, and multi-seed robustness diagnostics for Figure 6.** **(A)** Average silhouette over candidate primary-region counts in the frozen pooled 14-parameter t-SNE coordinates. The retained $k = 3$ solution is within 0.001 of the numerical $k = 4$ maximum, whereas $k = 4$ creates a singleton region. This profile and the associated resampling audit condition on the saved embedding rather than re-estimating t-SNE. **(B)** Within-region average-silhouette profiles for C01, C02, and C03. The selected $k = 2$ solution is highlighted in each region; the low-to-moderate values delimit these partitions as optimizer-search strata rather than biological subtypes. **(C)** Pair-specific full joint MAP objective differences ordered across all 500 numerical endpoints. The shaded region is the primary lowest-objective decile; vertical lines mark the nested 5%, 10%, and 20% eligibility thresholds. Facet labels report both the 50 retained optimizer seeds and the number of unique exact 14-parameter endpoints that they represent. **(D)** Endpoint- and cutoff-level agreement for prespecified qualitative oxygen–CIN–ploidy diagnostics. Text reports the modal result and its exact support within the primary 50-endpoint ensemble; "all q" indicates that the modal result is unchanged in the nested 5%, 10%, and 20% objective sets, whereas "varies" indicates a threshold-dependent modal result. Fill shows the minimum modal support across those three sets. A red cross marks any primary modal result that changes when repeated exact parameter endpoints are collapsed to one equal-weight analysis unit. The corresponding cluster, objective, response, and robustness values are reported in Supplementary Tables 7 and 10. This categorical audit does not assert pointwise identity of the response surfaces. Cluster labels, objective quantiles, and support fractions describe numerical search behavior; they are not biological groups, likelihood-ratio confidence sets, posterior probabilities, or biological replication.



Supplementary Figure 7: **Weak spectral separation rarely changes the ensemble-supported low/intermediate/high dominant-ploidy regime.** All panels use the same 201 oxygen concentrations, 60 fixed effective per-chromosome missegregation probabilities, and pair-specific lowest-objective 50-endpoint ensembles as Figure 6A. The dotted purple boundary encloses grid cells where at least half of retained endpoints have dominant spectral gap below 0.005; areas outside that weak-gap region are gray. **(A)** Across-endpoint regime consensus within the weak-gap region. Stable low, intermediate, and high require at least 90% of endpoints to place dominant mean ploidy at or below 2, between 2 and 4, or at or above 4, respectively; gray within the colored region would denote less than 90% three-class agreement. **(B)** Proportion of endpoints for which the rank-1 mean-ploidy classification changes among low, intermediate, and high at any of the up to four immediately adjacent O_2–missegregation grid cells, shown with square-root color spacing and percentage labels in the original scale. The black contour marks a proportion of 0.5. **(C)** Across-endpoint 90th–10th percentile spread of rank-1 dominant mean ploidy, shown in original ploidy units with pseudo-logarithmic color spacing. **(D)** Empirical cumulative distribution of the endpoint-median maximum absolute rank-1 ploidy change to an immediately adjacent grid cell, Supplementary Table 11: Dominant-ploidy robustness within weak-spectral-gap regions of the three displayed joint ensembles. Stable classes require at least 90% endpoint agreement. Spread is the across-endpoint 90th minus 10th percentile range of dominant mean ploidy. Local-switch and local-jump quantities compare each grid cell with its immediately adjacent oxygen and fixed-missegregation cells. These grid summaries are numerical diagnostics, not biological transition probabilities.

Displayed pair	Weak-gap cells	Stable low (%)	Stable intermediate (%)	Stable high (%)	Mixed (%)	Majority local class switch (%)
C01	3,287	5.9	36.3	57.8	0	8.67
C02	2,838	7.0	37.2	55.8	0	8.17
C03	2,118	17.4	69.0	13.6	0	4.86

Displayed pair	Median spread	90th-percentile spread	Cells with median local jump at least 1 (%)
C01	0.000035	0.000733	9.13
C02	0.000317	0.0237	11.31
C03	0.00672	0.0107	4.39

gen availability to resource-dependent growth and stress-associated death, and uses chromosome missegregation and whole-genome doubling to move viable mass across chromosome-number states.

State space and compartments. We consider tumor cells stratified by modeled total autosomal chromosome count; sex chromosomes are not included in this state variable. Let

$$\mathcal{N} = \{N_{\min}, \dots, N_{\max}\}. \tag{1}$$

This set denotes the allowed autosomal chromosome-count grid. For the simulations described here, $N_{\min} = 22$ and $N_{\max} = 154$. Let $N_{\text{dip}} = 44$ denote the diploid autosomal chromosome-count reference state, and let $N_{\text{unit}} = 22$ denote the one-ploidy autosomal chromosome-count unit, so that $2N_{\text{unit}} = N_{\text{dip}}$.

The viable abundance vector is

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N\in\mathcal{N}}, \tag{2}$$

where $x_N^L(t)$ is the number of viable cells with modeled autosomal chromosome count N at time t. The ODE state is indexed only by autosomal chromosome count and by live/dead compartment. Unless explicitly marked as a dead compartment, chromosome-number distributions in the model are defined over the viable population. The total viable population size is

$$N_{\text{tot}}(t) = \sum_{N\in\mathcal{N}} x_N^L(t). \tag{3}$$

To represent non-viable biomass contributing to macroscopic tumor volume, we track two dead pools:

$$\mathbf{x}^{D,\text{hyp}}(t) = \{x_N^{D,\text{hyp}}(t)\}_{N\in\mathcal{N}}, \qquad \mathbf{x}^{D,\text{CIN}}(t) = \{x_N^{D,\text{CIN}}(t)\}_{N\in\mathcal{N}}. \tag{4}$$

Here $\mathbf{x}^{D,\text{hyp}}$ receives cells assigned to the modeled pathway for death associated with resource stress, and $\mathbf{x}^{D,\text{CIN}}$ receives non-viable daughter mass generated by chromosome missegregation or WGD; the superscript CIN denotes missegregation-origin loss or out of grid WGD. The superscript hyp is an internal label for a model assigned compartment. It does not indicate that the biological cause of death was directly observed or uniquely identified. Their sum is

$$\mathbf{x}^D(t) = \mathbf{x}^{D,\text{hyp}}(t) + \mathbf{x}^{D,\text{CIN}}(t). \tag{5}$$

The model's bulk tumor burden is therefore

$$N_{\text{burden}}(t) = \sum_{N \in \mathcal{N}} \left(x_N^L(t) + x_N^{D,\text{hyp}}(t) + x_N^{D,\text{CIN}}(t) \right), \tag{6}$$

which includes viable cells and dead biomass. This distinction is used throughout: tumor-volume observations are compared with total burden, whereas single-cell chromosome-number observations sample the viable distribution.

Environmental stress functions. Let $O_2(t)$ denote the mean-field approximation of the effective oxygen level experienced by the tumor population. Oxygen/resource stress is represented by the bounded Hill function

$$h(O_2) = \frac{O_{2,c}^{n_O}}{O_{2,c}^{n_O} + O_2^{n_O}}, \tag{7}$$

where $O_{2,c}$ is the critical oxygen level at which the stress term is half-maximal, and n_O controls the steepness of the transition from low stress at high oxygen to strong stress under oxygen limitation. A Hill-type stress function is biologically justified because the literature indicates that oxygen-dependent cell fate is graded but threshold-like rather than linear: "The severity of hypoxia determines whether cells become apoptotic or adapt to hypoxia and survive",[26] and recent spheroid work identifies a "critical survival pO_2";[27] accordingly, Eq. (7) provides a smooth, bounded approximation to an oxygen-linked resource-stress threshold.

Proliferation and stress-associated death rates. Given chromosome count N and local oxygen level O_2, the effective division rate is

$$\lambda_{\text{eff}}(N, O_2) = \lambda_{\max} \frac{1}{1 + \alpha_{O_2}\, h(O_2) \left(\frac{N}{N_{\text{dip}}}\right)^{\gamma_{\text{growth}}}}. \tag{8}$$

Here α_{O_2} controls the strength of resource-stress-dependent growth damping, and γ_{growth} controls how strongly this damping increases with chromosome content. When oxygen-linked stress is small, $h(O_2) \approx 0$ and the reciprocal factor approaches one; as oxygen decreases, the reciprocal factor suppresses proliferation more strongly, especially for high-chromosome-count states.

Stress-associated death is applied as an explicit live-to-dead flow. The per-capita death hazard is

$$\mu_{\text{eff}}(N, O_2) = \mu_{\text{hp}}\, h(O_2) \left(\frac{N}{N_{\text{dip}}}\right)^{\gamma_\mu}. \tag{9}$$

Thus μ_{hp} sets the stress-associated death scale, $h(O_2)$ makes death depend on the same oxygen-linked stress function (Eq. (7)), and γ_μ controls the chromosome-number penalty on stress-associated death.

Cell division, WGD, and chromosomal transitions. With the stress-modulated rates defined, divisions are represented by a live-state generator matrix

$$G_{\mathrm{div}}(O_2) \in \mathbb{R}^{R\times R}, \qquad R = N_{\max} - N_{\min} + 1. \tag{10}$$

Let

$$L(O_2) = \mathrm{diag}\big(\lambda_{\mathrm{eff}}(N, O_2)\big) \tag{11}$$

be the diagonal matrix of state-specific division rates. A division follows the ordinary chromosome-missegregation branch with probability $1 - p_{\mathrm{wgd}}$ and follows a whole-genome-doubling branch with probability p_{wgd}. The corresponding division generator is

$$G_{\mathrm{div}}(O_2) = (1 - p_{\mathrm{wgd}})\, B(O_2)L(O_2) + p_{\mathrm{wgd}}\, W\, L(O_2) - L(O_2), \tag{12}$$

where $B(O_2)$ is the viable offspring-allocation matrix generated by chromosome missegregation and post-missegregation viability filtering, W maps a mother state N to the within-grid doubled state $2N$, and the final $-L(O_2)$ term removes the dividing mother cell. The ordinary branch B accounts for the two daughter branches of a division and therefore has total mass up to two before viability losses. By contrast, the WGD branch is implemented as an effective tracked-lineage conversion: W has unit mass in the doubled state, so a WGD event replaces the mother lineage by one viable doubled lineage when $2N$ lies in the modeled grid. Out-of-grid WGD mass is not retained in the viable state space and is routed to the CIN-associated dead compartment.

Chromosome missegregation and daughter viability. For a mother cell with chromosome count N, the per-chromosome missegregation probability increases with the stress-associated death hazard defined in Eq. (9):

$$p_{\mathrm{mis}}(N, O_2) = p_{\mathrm{mis},base} + p_{\mathrm{misseg}} \frac{\mu_{\mathrm{eff}}(N, O_2)}{\mu_{\mathrm{eff}}(N, O_2) + k_{o,\mathrm{mis}}}, \tag{13}$$

with values clipped to $[0, 1]$. Here $p_{\mathrm{mis},base}$ is the baseline per-chromosome missegregation probability under low stress, p_{misseg} is the maximal stress-induced increment above baseline, and $k_{o,\mathrm{mis}}$ is the death-hazard scale at which the stress-induced increase in missegregation is half-maximal.

Let m denote the number of missegregated chromosome copies in a division of a mother cell with chromosome count N. We model

$$m \sim \mathrm{Binomial}\{N, p_{\mathrm{mis}}(N, O_2)\}. \tag{14}$$

For each value of m, the two daughter branches are assigned coarse chromosome-count shifts $+m$ and $-m$. Before applying the daughter-viability modifier, the shift kernel is

$$d_N(\Delta; O_2) = \sum_{m=0}^{N} \binom{N}{m} \{p_{\mathrm{mis}}(N, O_2)\}^m \{1 - p_{\mathrm{mis}}(N, O_2)\}^{N-m} \left[\mathbf{1}\{\Delta = m\} + \mathbf{1}\{\Delta = -m\}\right]. \tag{15}$$

This kernel sums over two daughter branches, so its total mass is two before viability filtering.

For a daughter with chromosome-count shift Δ, post-missegregation survival is approximated by the symmetric ploidy-dependent modifier

$$S_N(\Delta) = s_N^{|\Delta|}, \qquad s_N = s_{\max} \exp\left[-\beta_{\text{buf}} \left(\frac{N_{\text{dip}}}{N}\right)^{n_{\text{exp}}}\right]. \tag{16}$$

The parameter $s_{\max}$ is the maximum per-copy survival factor, β_{buf} controls the severity of post-missegregation viability loss at lower ploidy, and n_{exp} controls the steepness of the ploidy dependence. This form lets higher-chromosome-count states tolerate a given chromosome-count shift more readily than lower-chromosome-count states. This survival modifier enters the retained-grid offspring allocation and dead-pool accounting described in the boundary-handling subsection below.

Boundary handling on the chromosome-count domain. The chromosome-count state space is finite. We define the viable chromosome-count domain as

$$\mathcal{N} = \{N_{\min}, N_{\min} + 1, \ldots, N_{\max}\},$$

where $N_{\min} = 22$ and $N_{\max} = 154$ in the simulations described here. The viable state vector

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}$$

therefore tracks only cells whose modeled autosomal chromosome count lies within this retained grid.

This finite state space requires an explicit rule for division events that would produce daughter or doubled-lineage states outside $\mathcal{N}$. We use absorbing boundary handling on the viable chromosome-count grid. That is, daughter cells or WGD-derived lineages whose chromosome count would fall below $N_{\min}$ or exceed $N_{\max}$ are removed from the viable offspring allocation and routed to the CIN-associated dead compartment.

For ordinary missegregating divisions, the viable offspring-allocation matrix is defined only over retained daughter states $N' \in \mathcal{N}$:

$$B_{N',N}(O_2) = d_N(N' - N; O_2) S_N(N' - N), \qquad N, N' \in \mathcal{N}.$$

Here $d_N(\Delta; O_2)$ specifies the probability of generating a daughter chromosome-count shift Δ, and $S_N(\Delta)$ specifies the probability that a daughter with that shift remains viable. Any probability mass corresponding to $N + \Delta \notin \mathcal{N}$ is excluded from $B_{\cdot,N}$. Similarly, for whole-genome-doubling events, the doubling operator W retains only within-grid transitions $N \mapsto 2N \in \mathcal{N}$; out-of-grid WGD offspring mass is excluded from $W_{\cdot,N}$ and routed to the CIN-associated dead compartment.

Because the viable offspring allocation can be subunitary, either because buffering reduces daughter survival or because daughter or doubled-lineage states fall outside the retained chromosome-count grid, we define the CIN-associated event-level nonviable inflow as

$$\phi_{\text{CIN},N}(\mathbf{x}^L, O_2) = \lambda_{\text{eff}}(N, O_2) x_N^L \left\{ (1 - p_{\text{wgd}}) \left[2 - \sum_{N' \in \mathcal{N}} B_{N',N}(O_2)\right] + p_{\text{wgd}} \left[1 - \sum_{N' \in \mathcal{N}} W_{N',N}\right] \right\}. \tag{17}$$

The first bracket is the ordinary-division deficit between two potential daughters and the viable daughter mass retained within $\mathcal{N}$. The second bracket accounts for WGD events for which the doubled state $2N$ lies outside the modeled grid. This term is distinct from the modeled continuous death hazard associated with resource stress, $\mu_{\text{eff}}(N, O_2)$; it is added to the CIN-associated dead compartment $x^{D,\text{CIN}}$.

Thus, the chromosome-count boundaries are absorbing for viable offspring mass, with out-of-domain and non-buffered daughter mass transferred to the dead compartment.

Oxygen demand and supply dynamics. To couple tumor size to oxygen demand, we define the chromosome-number-weighted oxygen-demand proxy

$$N_{\text{eff}}(t) = \sum_N x_N^L(t) \left(\frac{N}{N_{\text{dip}}}\right)^{\eta}, \tag{18}$$

where η controls how oxygen demand scales with chromosome content. The reference scale N_{ref} is a fixed cell-count scale used to nondimensionalize demand, so that $N_{\text{eff}}/N_{\text{ref}}$ is a relative effective tumor size.

The oxygen supply target is the phenomenological supply–demand relation

$$O_2^*(N_{\text{eff}}) = \max\left(O_{2,\min},\, S_0 - \kappa_O \log\left(1 + \frac{N_{\text{eff}}}{N_{\text{ref}}}\right)\right), \tag{19}$$

where S_0 is the low-burden oxygen supply level, $O_{2,\min}$ is the oxygen floor, and κ_O controls the oxygen-drop amplitude per unit log-burden increase. The logarithmic term is dimensionless because of the normalization by N_{ref}. Oxygen values are interpreted as percentages and are restricted to the physical interval $[0, 100]$.

Effective oxygen is updated by first-order relaxation toward this target:

$$\alpha_\tau = 1 - \exp\left(-\frac{\Delta t}{\tau_{O_2}}\right), \qquad O_{2,k+1} = O_{2,k} + \alpha_\tau (O_2^*(N_{\text{eff},k}) - O_{2,k}). \tag{20}$$

Thus oxygen changes gradually toward the demand-adjusted target rather than instantaneously matching it at each simulation step.

Discrete-time simulation scheme. The simulations use a time step of $\Delta t = 0.05$ days. At each step $k \to k+1$, the oxygen state is first relaxed toward the supply–demand target in Eq. (20), using the current viable population to calculate $N_{\text{eff},k}$. The resulting relaxed value, $O_{2,k+1}$, is then used to evaluate division-linked chromosome transitions and explicit stress-associated death for that population step:

$$\mathbf{x}_{k+1}^L = \mathbf{x}_k^L + \Delta t\, G_{\text{div}}(O_{2,k+1})\mathbf{x}_k^L - \Delta t\, \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L, \tag{21}$$

where $\odot$ denotes element-wise multiplication over chromosome-count states.

Mass that becomes nonviable through the modeled stress pathway or failed divisions is transferred into the dead compartments, which contribute to tumor burden. The dead pool assigned by the model to resource stress is updated as

$$\mathbf{x}_{k+1}^{D,\text{hyp}} = \mathbf{x}_k^{D,\text{hyp}} + \Delta t\, \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L - \Delta t\, k_{\text{clear}} \mathbf{x}_k^{D,\text{hyp}}, \tag{22}$$

and the CIN-associated dead pool is updated as

$$\mathbf{x}_{k+1}^{D,\mathrm{CIN}} = \mathbf{x}_k^{D,\mathrm{CIN}} + \Delta t\, \boldsymbol{\phi}_{\mathrm{CIN}}(\mathbf{x}_k^L, O_{2,k+1}) - \Delta t\, k_{\mathrm{clear}}\mathbf{x}_k^{D,\mathrm{CIN}}. \tag{23}$$

Here k_{clear} is the first-order clearance rate for dead biomass, and $\boldsymbol{\phi}_{\mathrm{CIN}}$ is defined component-wise in Eq. (17). Together, the oxygen relaxation and Eqs. (21)–(23) define the coupled viable/dead population dynamics used in the simulations.

1.3.1 Model calibration and observation models

Model calibration uses a common transformed-parameter optimizer for *in vivo*, *in vitro*, and joint fits. Natural-scale parameter specifications provide initialization values and admissible bounds; each component maps its fitted parameters to the optimizer scale before evaluating its objective. The mechanistic dynamics described above are shared, while the observation models differ between the *in vivo* and *in vitro* data.

***In vivo* observation model and likelihood.** The *in vivo* component uses three data modalities: longitudinal tumor burden, terminal necrosis where matched histology is available, and terminal single-cell chromosome-number profiles.

Tumor burden. For each mouse $i \in \mathcal{I}$, where $\mathcal{I}$ denotes the eight retained tumors used for calibration (section 1.1), tumor burden is measured at observation times $\{t_{ij}^{(V)}\}_{j=1}^{m_i}$, yielding observed tumor volumes

$$V_{ij}^{\mathrm{obs}} \equiv V_i^{\mathrm{obs}}(t_{ij}^{(V)}). \tag{24}$$

Let $x_{i,N}^L(t)$, $x_{i,N}^{D,\mathrm{hyp}}(t)$, and $x_{i,N}^{D,\mathrm{CIN}}(t)$ denote viable abundance, dead abundance assigned by the model to loss associated with resource stress, and CIN-associated dead abundance, respectively, for tumor i in chromosome-count state N at time t. The model predicts tumor volume by scaling total burden:

$$\widehat{V}_{ij} = \sum_N \left(x_{i,N}^L(t_{ij}^{(V)}) + x_{i,N}^{D,\mathrm{hyp}}(t_{ij}^{(V)}) + x_{i,N}^{D,\mathrm{CIN}}(t_{ij}^{(V)}) \right) v_{\mathrm{cell}}, \tag{25}$$

where each cell is assigned the effective volume

$$v_{\mathrm{cell}} = \frac{1}{\rho_{2N}}. \tag{26}$$

This observation model does not impose a ploidy-dependent cell-volume correction: low- and high-ploidy cells contribute the same effective per-cell volume to the bulk burden. Tumor-burden observations are modeled on the logarithmic scale,

$$\log V_{ij}^{\mathrm{obs}} \sim \mathcal{N}\left(\log \widehat{V}_{ij}, \sigma_{\mathrm{burden}}^2 \right), \tag{27}$$

where σ_{burden} is the burden measurement-noise scale. For longitudinal tumor burden, the fitted loss is a Gaussian negative log-likelihood on the logarithmic scale, averaged first within tumor and then across tumors. Let $\mathcal{J}_i$ denote the retained observation times for tumor i after excluding day

0. Then

$$\mathcal{L}_B(\theta) = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}}\frac{1}{|\mathcal{J}_i|}\sum_{j\in\mathcal{J}_i}\left[\log\sigma_{\text{burden}} + \frac{1}{2}\log(2\pi) + \frac{\left(\log V_{ij}^{\text{obs}} - \log \widehat{V}_{ij}\right)^2}{2\sigma_{\text{burden}}^2}\right]. \tag{28}$$

In implementation, each volume entering a logarithm is replaced by $\max(V, \varepsilon_{\log B})$ with $\varepsilon_{\log B} = 10^{-12}$ mm^3; this numerical floor is suppressed in Eqs. (27)–(28) for readability. Because σ_{burden} is estimated jointly, the $\log\sigma_{\text{burden}}$ term is retained in the likelihood contribution. Omitting it would allow the optimizer to reduce the burden loss by inflating the assumed measurement noise.

Terminal necrosis. For tumors with matched terminal histology, necrosis is compared with the simulated terminal fraction of total dead biomass. This is an observation-model term, not an additional live-to-dead flux. The stress-associated death hazard $\mu_{\text{eff}}(N, O_2)$ contributes to $\mathbf{x}^{D,\text{hyp}}$, while mitotic nonviability and boundary-dropped offspring contribute to $\mathbf{x}^{D,\text{CIN}}$. The terminal necrosis comparison uses the sum of these two dead pools.

For tumor i, define the terminal total-dead volume and terminal total tumor volume as

$$\widehat{V}_i^{D,\text{tot}}(\theta) = v_{\text{cell}}\sum_N\left[x_{i,N}^{D,\text{hyp}}(T_i^{(H)};\theta) + x_{i,N}^{D,\text{CIN}}(T_i^{(H)};\theta)\right], \qquad \widehat{V}_i^{\text{tot}}(\theta) = v_{\text{cell}}\sum_N\Big[x_{i,N}^{L}(T_i^{(H)};\theta) + x_{i,N}^{D,\text{hyp}}(T_i^{(H)};\theta) + \tag{29}$$

The predicted terminal necrotic fraction is

$$\widehat{f}_i^{\text{nec}}(\theta) = \frac{\widehat{V}_i^{D,\text{tot}}(\theta)}{\widehat{V}_i^{\text{tot}}(\theta)}. \tag{30}$$

Because the same effective cell-volume factor appears in both numerator and denominator, it cancels in this ratio. Thus the fitted necrosis term constrains the terminal total-dead fraction of the simulated tumor burden, after both dead pools have accumulated their respective inflows and have been cleared by the shared first-order clearance process.

Let $f_i^{\text{nec,obs}}$ be the observed terminal necrotic fraction for tumors in the matched necrosis set $\mathcal{I}_{\text{nec}}$. Fractions are compared on a clipped logit scale. Define

$$\operatorname{logit}_\epsilon(f) = \log\left(\frac{\widetilde{f}}{1-\widetilde{f}}\right), \qquad \widetilde{f} = \min\{1-\epsilon_{\text{nec}}, \max(\epsilon_{\text{nec}}, f)\}. \tag{31}$$

The terminal necrosis loss is the mean standardized squared discrepancy on this scale,

$$\mathcal{L}_{\text{nec}}(\theta) = \frac{1}{|\mathcal{I}_{\text{nec}}|}\sum_{i\in\mathcal{I}_{\text{nec}}}\left[\frac{\operatorname{logit}_\epsilon(\widehat{f}_i^{\text{nec}}(\theta)) - \operatorname{logit}_\epsilon(f_i^{\text{nec,obs}})}{\sigma_{\text{nec}}}\right]^2. \tag{32}$$

Here σ_{nec} sets the tolerated discrepancy in terminal necrosis on the logit-fraction scale, and ϵ_{nec} prevents fractions at 0 or 1 from producing infinite logits. In the run configuration these quantities are named `sigma_necrosis_logit` and `necrosis_fraction_eps`, respectively.

Ploidy. Each mouse has an initial ploidy group $g_i \in \{2N, 4N\}$ and harvest time $T_i^{(H)}$. Retained burden observations are nonzero-time measurements up to harvest, so the number and timing of observations may differ across mice.

For each tumor $i \in \mathcal{I}$, endpoint single-cell copy-number profiling provides modeled total autosomal chromosome-number estimates. If n_i cells are profiled from tumor i, the observed chromosome-number values are denoted

$$z_{i\ell}, \qquad \ell = 1, \ldots, n_i. \tag{33}$$

These single-cell observations sample the viable population at harvest. The model-predicted viable chromosome-number distribution at harvest is

$$\widehat{\pi}_{i,N}(\theta) = \frac{x^L_{i,N}(T_i^{(H)}; \theta)}{\sum_{N'} x^L_{i,N'}(T_i^{(H)}; \theta)}. \tag{34}$$

This distribution represents the predicted probability that a viable cell sampled from tumor i at harvest has true chromosome count N.

Observed single-cell chromosome-number estimates are then modeled conditional on this predicted viable distribution. Specifically, each observed value $z_{i\ell}$ is treated as a noisy measurement of an unobserved true chromosome-count state N, with mixture weights given by $\widehat{\pi}_{i,N}(\theta)$:

$$p(z_{i\ell} \mid \theta) = \sum_N \widehat{\pi}_{i,N}(\theta)\, \mathcal{N}\big(z_{i\ell}; N, \sigma^2_{\text{ploidy}}\big)\,. \tag{35}$$

where σ_{ploidy} represents measurement variability in single-cell chromosome-number estimates. For terminal single-cell chromosome-number data, the fitted loss first computes a tumor-level average negative log-likelihood. Define

$$\ell_i(\theta) = -\frac{1}{n_i} \sum_{\ell=1}^{n_i} \log p(z_{i\ell} \mid \theta). \tag{36}$$

Let $\mathcal{I}_{2N}$ and $\mathcal{I}_{4N}$ denote the retained tumors initiated from the near-diploid and near-tetraploid lineages, respectively. With both cohorts present, the endpoint contribution gives equal weight to the two initial-ploidy cohorts and equal weight to tumors within each cohort:

$$\mathcal{L}_C(\theta) = \frac{1}{2}\left[\frac{1}{|\mathcal{I}_{2N}|} \sum_{i \in \mathcal{I}_{2N}} \ell_i(\theta) + \frac{1}{|\mathcal{I}_{4N}|} \sum_{i \in \mathcal{I}_{4N}} \ell_i(\theta)\right]. \tag{37}$$

Maximum-a-posteriori objective. The *in vivo* component is formulated as a maximum-a-posteriori objective combining burden likelihood, chromosome-number likelihood, terminal necrosis loss, and soft-prior penalties:

$$\mathcal{L}_{\text{vivo}}(\theta) = \mathcal{L}_B(\theta) + \mathcal{L}_C(\theta) + \lambda_{\text{nec}}\, \mathcal{L}_{\text{nec}}(\theta) + \lambda_{\text{prior}}\, \mathcal{L}_{\text{prior}}(\theta), \tag{38}$$

where θ denotes the fitted parameter vector for this component, $\mathcal{L}_B$ is the burden contribution, $\mathcal{L}_C$ is the terminal chromosome-number contribution, $\mathcal{L}_{\text{nec}}$ is the terminal total-dead-fraction loss, λ_{nec} is its relative objective weight, and $\mathcal{L}_{\text{prior}}$ is the soft-prior penalty.

For each prior-constrained parameter, let $z_i = T_i(\theta_i)$ denote the calibration-scale value used by the optimizer. The soft-prior penalty is

$$\mathcal{L}_{\text{prior}}(\theta) = \frac{1}{2} \sum_{i \in \mathcal{P}} \left(\frac{z_i - \mu_i}{\sigma_i} \right)^2 , \tag{39}$$

where $\mathcal{P}$ is the subset of parameters with configured soft priors. The transform T_i is the parameter-specific optimizer transform (including $\log_{10}$ transforms for several positive parameters and the identity transform where configured), and μ_i and σ_i are the prior center and scale on that same calibration scale.

***In vitro* observation model and likelihood.** The *in vitro* component uses the same chromosome-number transition model but applies it to passage-structured cell-line data. Each passage segment s has an initial cohort label, a fixed oxygen level during the segment, and a duration Δt_s; available growth observations are represented as segment-level growth-rate data rather than being restricted to a single initial–final cell-count pair. The model propagates the corresponding viable chromosome-number distribution through the segment and compares the simulated passage outcome with three measured data modalities: passage growth-rate observations, passage-level chromosome-number observations, and flow-density measurements. For each passage, the propagated chromosome-number distribution used for reseeding and endpoint comparison was taken from the simulated time point whose live-cell count was closest to the observed final or next-passage initial cell count.

For growth-rate data, let g_s^{obs} be the observed passage growth rate and $\widehat{g}_s(\theta)$ the model-predicted growth rate for segment s. The averaged growth log-likelihood is

$$\bar{\ell}_{\text{growth}}(\theta) = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \log \mathcal{N}\Big(g_s^{\text{obs}}; \widehat{g}_s(\theta), \sigma_{\text{growth}}^2 \Big) , \tag{40}$$

where $\mathcal{S}_g$ is the set of segments with observed growth-rate data and σ_{growth} is the growth measurement-noise scale.

For chromosome-number observations in segment s, let $\pi_{s,N}$ denote the predicted viable chromosome-number distribution at the comparison time and let $z_{s\ell}$ be the observed integer chromosome-number value for cell ℓ. The implementation first smooths the predicted distribution on the finite modeled chromosome grid $\mathcal{N}$. For an observed-grid state m and true modeled state N, define the column-normalized Gaussian kernel

$$K_{\sigma_{\text{kary}}}(m \mid N) = \frac{\phi\Big(m; N, \sigma_{\text{kary}}^2\Big)}{\sum_{r \in \mathcal{N}} \phi\big(r; N, \sigma_{\text{kary}}^2\big)}, \qquad m, N \in \mathcal{N}, \tag{41}$$

where $\phi(\cdot; N, \sigma_{\text{kary}}^2)$ is a Gaussian density evaluated only on the discrete chromosome grid. The smoothed discrete probability mass is

$$\widetilde{\pi}_{s,m} = \sum_{N \in \mathcal{N}} K_{\sigma_{\text{kary}}}(m \mid N) \pi_{s,N}. \tag{42}$$

The averaged chromosome-number log-likelihood is then

$$\bar{\ell}_{\text{ploidy}}(\theta) = \frac{1}{|\mathcal{S}_z|} \sum_{s \in \mathcal{S}_z} \frac{1}{n_s} \sum_{\ell=1}^{n_s} \log\left[\max\left\{\widetilde{\pi}_{s,z_{s\ell}}, 10^{-12}\right\}\right], \tag{43}$$

where $\mathcal{S}_z$ is the set of passage comparisons with chromosome-number observations, n_s is the number of observations in segment s, and σ_{kary} is the chromosome-number smoothing scale for the *in vitro* karyotype term. Observations that do not match a modeled grid state receive the same 10^{-12} probability floor. Thus, this term is a discrete grid likelihood after finite-grid kernel smoothing, not a continuous Gaussian-mixture density.

For flow-density data, let ω_{sp}^{obs} be the normalized observed density mass at flow grid point p for segment s, and let $\widehat{f}_s(p;\theta)$ be the model-predicted density after Gaussian smoothing of the simulated viable chromosome-state distribution onto the observed flow grid and renormalization over that grid. The Gaussian width was set separately from the fitted karyotype smoothing parameter to the larger of twice the median flow-grid spacing and 1/22 ploidy units; for the seed-10 grid this was 0.0804020100502587. The averaged flow log-likelihood is

$$\bar{\ell}_{\text{flow}}(\theta) = \frac{1}{|\mathcal{S}_f|} \sum_{s \in \mathcal{S}_f} \sum_{p \in \mathcal{G}_s} \omega_{sp}^{\text{obs}} \log\left[\max\left\{\widehat{f}_s(p;\theta), 10^{-12}\right\}\right], \tag{44}$$

where $\mathcal{S}_f$ is the set of flow-profile comparisons and $\mathcal{G}_s$ is the observed flow grid for segment s. The 10^{-12} floor is a numerical safeguard applied after smoothing and normalization.

The fitted *in vitro* objective is the negative weighted mean log-likelihood

$$\mathcal{L}_{\text{vitro}}(\theta) = -\left[w_{\text{growth}}\,\bar{\ell}_{\text{growth}}(\theta) + w_{\text{ploidy}}\,\bar{\ell}_{\text{ploidy}}(\theta) + w_{\text{flow}}\,\bar{\ell}_{\text{flow}}(\theta)\right]. \tag{45}$$

The three terms compare the simulated passage lineages with growth-rate, chromosome-number, and flow-density data, respectively. The reported fits used equal component weights, $w_{\text{growth}} = w_{\text{ploidy}} = w_{\text{flow}} = 1$ (Supplementary Table 13).

Joint fitting of *in vivo* and *in vitro* data. The joint fit uses a single optimizer vector while evaluating separate transformed parameter vectors for the *in vivo* and *in vitro* likelihood components. Soft-coupled parameters are represented by a center–delta parameterization: for each soft-coupled parameter, the optimizer carries one transformed center variable, c_Q, and one additional offset variable, δ_Q. The context-specific transformed values are reconstructed from these two optimizer variables during objective evaluation. Parameters that appear only in one observation model are retained as context-specific observation-model parameters. In the analysis reported here, all 14 overlapping active biological parameters—$O_{2,c}$, μ_{hp}, p_{misseg}, $k_{o,\text{mis}}$, $s_{\max}$, β_{buf}, $n_{\exp}$, n_O, α_{O_2}, γ_{growth}, $\lambda_{\max}$, $p_{\text{mis,base}}$, p_{wgd}, and γ_μ—were soft-coupled across contexts; no overlapping active biological parameter was hard-shared (Supplementary Table 12). Thus, the strength and chromosome-number dependence of resource-stress growth suppression were allowed to differ between the *in vivo* and *in vitro* components, while regularization discouraged unsupported divergence.

The minimized joint objective is

$$\mathcal{L}_{\text{joint}}(\theta) = w_{\text{vivo}}\mathcal{L}_{\text{vivo}}(\theta_{\text{vivo}}) + w_{\text{vitro}}\mathcal{L}_{\text{vitro}}(\theta_{\text{vitro}}) + \mathcal{L}_{\text{soft}}(\theta), \tag{46}$$

where θ_{vivo} and θ_{vitro} are the context-specific parameter vectors reconstructed from the joint optimizer vector and passed to the two likelihood components. The reported analysis used $w_{\text{vivo}} = w_{\text{vitro}} = 1$ (Supplementary Table 13). For each soft-coupled natural-scale parameter Q, let

$$z_Q = T_Q(Q)$$

denote its transformed optimizer-scale value, where T_Q is the transformation used for that parameter during optimization. The separate *in vivo* and *in vitro* fits define context-specific transformed bounds $[l_Q^{\text{vivo}}, u_Q^{\text{vivo}}]$ and $[l_Q^{\text{vitro}}, u_Q^{\text{vitro}}]$. In the joint fit, these component-specific bounds are used only to define a single active joint admissible interval:

$$l_Q^{\text{joint}} = \min\bigl(l_Q^{\text{vivo}}, l_Q^{\text{vitro}}\bigr), \qquad u_Q^{\text{joint}} = \max\bigl(u_Q^{\text{vivo}}, u_Q^{\text{vitro}}\bigr). \tag{47}$$

Thus, the original component-specific bounds record the provenance of the separate fits, whereas the joint optimizer uses the union interval $[l_Q^{\text{joint}}, u_Q^{\text{joint}}]$ as the active transformed-scale admissible range. During objective evaluation, the transformed values passed to the two components are reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \qquad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \tag{48}$$

The reconstructed values must satisfy

$$z_Q^{\text{vivo}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}], \qquad z_Q^{\text{vitro}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}]. \tag{49}$$

Parameter vectors that do not yield admissible reconstructed transformed values are outside the joint feasible domain and are not evaluated as valid likelihood points. The soft-coupling penalty is applied to the offset variable,

$$\mathcal{L}_{\text{soft}}(\theta) = \sum_{Q \in \mathcal{S}} \frac{c^2}{2} \left[1 - \exp\left\{-\left(\frac{\delta_Q}{c\sigma_Q}\right)^2\right\}\right], \tag{50}$$

where $\mathcal{S}$ is the set of soft-coupled parameters, σ_Q is the transformed-scale soft-coupling scale, and c controls the transition from near-quadratic behavior to saturation. For the reported fits, $\sigma_Q = 0.65$ for all 14 parameters and $c = 0.4$ (Supplementary Table 13). For $|\delta_Q| \ll c\sigma_Q$, each term approaches $\frac{1}{2}(\delta_Q/\sigma_Q)^2$; for large separations it saturates at $c^2/2$, limiting the influence of a single strongly context-specific parameter. Since

$$z_Q^{\text{vivo}} - z_Q^{\text{vitro}} = \delta_Q,$$

this penalty regularizes the transformed difference between the *in vivo* and *in vitro* parameter values. For log-transformed parameters, this corresponds to regularization of fold differences rather than absolute natural-scale differences.

Differential-evolution initial-population and optimizer-endpoint analysis. For Figure 5D, "initial distribution" denotes the population passed to differential evolution at the start of each joint optimization. It is not a distribution induced from the configured fitting intervals and is not obtained by weighting candidate values with the Welsch objective penalty. One endpoint ensemble was selected within each of C01, C02, and C03. Each family contained two candidate joint pairs, and the retained pair was the one whose original standalone separate-*in vivo* member had the lower separate-*in vivo* objective. This selected separate-*in vivo* seeds 366, 25, and 311, respectively; each remained paired with the common separate-*in vitro* seed10 anchor (Supplementary Tables 7 and 8).

The initial populations were deterministically reconstructed with the fitting code and saved run inputs. For each selected pair, joint seeds 1–500 were reset individually and the same `joint_deoptim_initial_popu` routine used by the fit was called. The optimizer contained 40 coordinates, so the enforced population size was

$$N_P = \max(80, 10 \times 40) = 400.$$

Warm-start initialization was enabled with $\sigma_N = 0.1216$. For an ordinary transformed coordinate with warm-start value z_0, the routine sampled a normal distribution centered at z_0, with standard deviation $\sigma_N |z_0|$, truncated to the active optimizer bounds. When $|z_0| \leq 10^{-12}$, the active bound width replaced $|z_0|$ as the scale reference. For each soft-coupled parameter, the center was sampled first; the feasible bounds for the transformed context difference δ_Q were then recomputed from that sampled center so that both reconstructed context values remained inside the joint union interval. The delta coordinate was sampled from a truncated normal centered on the warm-start delta after clipping that mean to its center-dependent feasible interval, using the same scale rule. The first member of every 400-member population was replaced by the exact warm-start vector. Thus, each family–parameter initial distribution contained 200,000 values (500 seeds times 400 population members; Supplementary Table 8).

For every soft-coupled coordinate, the transformed context values were reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \qquad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \tag{51}$$

For $\log_{10}$-transformed parameters, natural values were obtained as

$$Q_{\text{vivo}} = 10^{z_Q^{\text{vivo}}}, \qquad Q_{\text{vitro}} = 10^{z_Q^{\text{vitro}}}; \tag{52}$$

identity-scale coordinates already equaled the natural context values. Figure 5D displays these two natural-scale marginals separately rather than plotting their ratio. For every selected pair and coupled parameter, all 500 endpoints were required to have finite positive context values, to satisfy the joint feasible-domain check, and to require no post-optimization projection (Supplementary Table 8).

For each context-specific DE initial population and its corresponding endpoint ensemble, the natural-scale median and type-8 5th and 95th percentiles were calculated. The percentiles were retained for the numerical concentration audit and its source tables, but Figure 5D displays only optimizer-endpoint medians, as asterisks on the shared coordinate axis; initial-population median points and separate percentile spans are not drawn. Kernel densities were evaluated on a 401-point parameter-specific grid shared by the two contexts, both numerical roles, and all three families. Density estimation used the configured optimizer scale—$\log_{10}$ for log-transformed parameters and identity otherwise—while logarithmic axes were labeled with natural parameter values. Each density was divided by its own maximum; therefore only horizontal location, width, and shape, not vertical peak height, were compared (Supplementary Table 8).

The row order in Figure 5D used the single best retained joint winner from each C family.

For parameter Q and family $f \in \{\mathrm{C01}, \mathrm{C02}, \mathrm{C03}\}$, the natural-scale ratio was $R_{Qf} = Q_f^{\mathrm{vivo}}/Q_f^{\mathrm{vitro}}$. Parameters were ordered by increasing $\left|\frac{1}{3}\sum_f R_{Qf} - 1\right|$, with parameter name as the deterministic tie breaker. This ordering is a display rule for proximity to cross-context parity, not an inferential test.

Boundary proximity in Figure 5D was defined from the complete joint-union fitting interval, not from a distribution quantile. For lower and upper bounds L_Q and U_Q on the optimizer coordinate, the two red background regions were

$$[L_Q,\, L_Q + 0.05(U_Q - L_Q)] \quad \text{and} \quad [L_Q + 0.95(U_Q - L_Q),\, U_Q]\,. \tag{53}$$

Thus each region occupies exactly 5% of the plotted horizontal interval. For $\log_{10}$-optimized parameters these limits were calculated in $\log_{10}$ optimizer space and the tick labels were then converted back to natural units; identity-scale parameters required no conversion. The three retained families used the same joint-union bounds for each parameter.

As an auxiliary directional audit, the paired values were also transformed to $\log_2(Q_{\mathrm{vivo}}/Q_{\mathrm{vitro}})$. A parameter-family contrast was called higher *in vivo* when its endpoint 5th percentile exceeded zero, lower *in vivo* when its 95th percentile was below zero, and overlapping equality otherwise. Cross-family direction agreement required the same non-overlap classification in all three families. Paired-ratio fitted-solution concentration relative to initialization was summarized as

$$R_{W,Qf} = \frac{q_{0.95,Qf}^{\mathrm{endpoint}} - q_{0.05,Qf}^{\mathrm{endpoint}}}{q_{0.95,Qf}^{\mathrm{DE\ init}} - q_{0.05,Qf}^{\mathrm{DE\ init}}}\,. \tag{54}$$

We additionally reported, separately for *in vivo* and *in vitro*, the fraction of the 500 endpoints that lay at the corresponding active parameter bound (Supplementary Table 8).

The fixed red edge bands show whether a fitted marginal approaches the extremes of the declared joint fitting domain, while the source-table 5th–95th percentile summaries and $R_{W,Qf}$ provide the separate numerical-concentration audit. Both are conditional on the tested warm starts, bounds, objective, and search settings. Neither the initial populations nor the endpoints are posterior or confidence distributions; they do not add biological replication or establish structural identifiability. In particular, a zero endpoint span may result from many numerical starts converging to the same endpoint, whereas high active-bound occupancy indicates optimizer pressure against the declared fitting domain rather than a resolved biological optimum.

Warm-start initialization for the joint fit. The main joint analysis used three landscape-informed separate *in vivo* fits as warm-start representatives. The primary regions were obtained by k-means clustering of the 500 final *in vivo* positions in the pooled 14-parameter t-SNE embedding described below, not from the *in vivo*-only 18-parameter landscape in Figure 4B. Candidate partitions with $k = 2, \ldots, 8$ were fitted with 100 starts and evaluated by average silhouette.[28] Although $k = 4$ gave the numerical maximum (0.816651), it improved on $k = 3$ by only 0.000957 and created a singleton region. We therefore retained the smaller non-singleton $k = 3$ partition (average silhouette 0.815694; region sizes 99, 385, and 16) as a parsimonious warm-start coverage design. This choice was audited with 100 one-start reruns on the frozen coordinates, 100 80%-endpoint subsamples drawn without replacement with re-evaluation of $k = 2, \ldots, 8$, and comparison with a $k = 3$ partition fitted directly in standardized 14-parameter endpoint space (Supplementary Figure 6;

Supplementary Table 7). The archived reproducibility bundle retains all 228,000 frozen t-SNE coordinates but not the corresponding 228,000 standardized 14-parameter input vectors; t-SNE seed and perplexity sensitivity therefore could not be re-estimated from the archived bundle, and the coordinate-space audits condition on the saved embedding. These checks assess reproducibility of an optimizer-search partition; they do not validate biological subtypes or make t-SNE coordinates mechanistic variables.

Within each primary region, the separate-*in vivo* endpoint with the lowest standalone objective was retained. This selected seeds 366, 25, and 311 for C01, C02, and C03, respectively. Each was paired with the same lowest-objective separate *in vitro* fit (seed 10) on the transformed optimizer scale. Thus, the main design explored three *in vivo* primary regions against one fixed *in vitro* anchor rather than a balanced cross-product of both landscapes (Supplementary Table 7).

For each soft-coupled parameter Q, let $z_Q^{\mathrm{vivo},0}$ and $z_Q^{\mathrm{vitro},0}$ denote the transformed values from one such pair. The initial center and offset are set to

$$c_Q^0 = \frac{z_Q^{\mathrm{vivo},0} + z_Q^{\mathrm{vitro},0}}{2}, \qquad \delta_Q^0 = z_Q^{\mathrm{vivo},0} - z_Q^{\mathrm{vitro},0}. \tag{55}$$

This initialization exactly reconstructs the two separate-fit transformed values:

$$z_Q^{\mathrm{vivo},0} = c_Q^0 + \frac{\delta_Q^0}{2}, \qquad z_Q^{\mathrm{vitro},0} = c_Q^0 - \frac{\delta_Q^0}{2}.$$

Parameters that appear only in one component, such as context-specific observation-model parameters, are initialized from the corresponding separate-fit solution.

The warm start must be representable under the active joint union bounds in Eq. (47); otherwise, initialization fails explicitly. This ensures that the joint fit begins from the selected separate-fit solutions only when those solutions are valid under the declared joint feasible domain.

Optimization and data inclusion. Optimization attempted a parallel differential-evolution global stage followed by serial L-BFGS-B refinement; a local result was retained only when it passed the implementation checks and improved the objective. Separate *in vivo* and *in vitro* searches each used 500 numerical seeds. In the main joint analysis, each of the three retained warm-start pairs used 500 numerical seeds, for 1,500 optimizer runs. All 1,500 selected runs recorded an attempted local stage with convergence code 52 and `optimizer_local_accepted=FALSE`; the reported joint solutions are therefore the retained differential-evolution endpoints. Every selected joint differential-evolution run ended by the configured relative-tolerance/step-tolerance early-stopping rule within the 500-iteration target. The separate *in vitro* convergence summary labeled 0/500 differential-evolution runs as converged; the best retained seed-10 local step was accepted but returned convergence code 1. In the separate *in vivo* search, the convergence flag was recorded for all 500 starts, 29 objectives were within 1% and 241 were within 5% of seed 25, and the selected seed-25 local step was accepted with code 52 while two of 20 active parameters were exactly at bounds. The corresponding diagnostics are summarized in Figure 3G, Supplementary Figures 4 and 2, and Supplementary Tables 6, 4, and 7. In all cases, "best" denotes the lowest retained objective among the supplied searches, not a proven global optimum. Seed-to-seed spread was used to assess numerical search robustness and was not interpreted as biological replication or inferential uncertainty.

For the Figure 6A multi-seed analysis, eligibility was defined independently within each warm-start pair because the six pairs had different objective baselines. A retained endpoint first had to have a finite recorded full joint MAP objective, a valid unique seed identifier, exactly one finite value for each of the 14 context-specific *in vivo* parameters, feasibility both before projection and at the reported solution, no applied parameter projection, and an available local model configuration. Eligible endpoints were then ordered by the complete retained joint MAP objective, including the fitted-data terms and soft-coupling penalty. The primary ensemble was the lowest objective decile within each pair (50 of 500 endpoints); nested lowest-5% (25 endpoints) and lowest-20% (100 endpoints) sets were used as threshold-sensitivity analyses. Within an eligibility set, endpoints received equal weight; objective values determined inclusion but were not converted into posterior-like weights. The decile rule is an operational numerical-fit filter that retains 50 endpoints per pair for distributional summaries while excluding the upper 90% of each pair-specific objective distribution; it is not a likelihood-ratio cutoff, posterior credible set, or confidence region. Every retained operator evaluation additionally had to return finite dominant-mode summaries with status `ok`, and the realized forced missegregation probability had to agree with the requested value within 10^{-8}. To prevent repeated convergence to an identical numerical solution from being mistaken for independent parameter evidence, a second summary formed an exact signature from the ordered natural-scale values of all 14 *in vivo* response parameters within each warm-start pair, retained the lowest-objective representative of each signature, and weighted each unique endpoint once. Seed-weighted and unique-endpoint-weighted modal claims were compared at all three objective thresholds; the complete 5% and 10% surfaces and all three unmodified trajectories were additionally compared pointwise. Exact inclusion, exclusion, endpoint multiplicity, threshold, and deduplication diagnostics are reported in Supplementary Figure 6 and Supplementary Tables 7 and 10.

Optimizer-ensemble sensitivity analyses. A broader archived sensitivity analysis, separate from the three-pair Figure 5 display, retained both completed candidate-pair ensembles within each primary family. This audit used all 3,000 retained joint endpoints; all 42,000 parameter-specific ratio records were feasible at the reported solution, had positive finite context ratios, and required no projection (Supplementary Table 7). For coupled parameter Q, warm-start family $f \in \{\mathrm{C01}, \mathrm{C02}, \mathrm{C03}\}$, within-family candidate pair $r \in \{1, 2\}$, and numerical start $s \in \{1, \ldots, 500\}$, the contrast was

$$D_{Qfrs} = \log_2\left(\frac{Q^{\mathrm{vivo}}_{frs}}{Q^{\mathrm{vitro}}_{frs}}\right). \tag{56}$$

Within each of the six warm-start pairs, the 500 endpoints contributed equal weight to the pair median shown in Supplementary Figure 3A. Between-pair consensus in Supplementary Figure 3B used the pair as the analysis unit and classified natural-scale ratios as lower *in vivo* for $0 < Q_{\mathrm{vivo}}/Q_{\mathrm{vitro}} \leq 0.8$, approximately equal for $0.8 < Q_{\mathrm{vivo}}/Q_{\mathrm{vitro}} < 1.2$, and higher *in vivo* for ratios at least 1.2. Direction sensitivity was additionally evaluated after restricting each pair to its 50 lowest-objective endpoints and after retaining the lowest-objective representative of each exact 28-value context-specific parameter signature within a pair, leaving 947 unique endpoints (Supplementary Table 7). These analyses characterize optimization and warm-start sensitivity; they are not posterior uncertainty estimates.

Retained *in vivo* observation times satisfy $0 < t_{i,1} < \cdots < t_{i,n_i} \leq T_i^{(H)}$, so day-0 burden points are excluded and longitudinal follow-up is administratively truncated at harvest. No censoring likelihood was used. Only tumors with both burden and terminal chromosome-number data are retained. Terminal endpoint discrepancies are evaluated directly from the single-cell chromosome-number observations under the Gaussian-mixture likelihood in Eq. (35), and tumor-level endpoint losses are aggregated equally within each initial-ploidy cohort before applying cohort balancing as in Eq. (37).

Representative in vivo trajectory and terminal-distribution visualization. The lowest-objective retained separate *in vivo* fit (seed 25) was propagated at one-day reporting intervals for the eight paired tumors. The four 2N-initiated and four 4N-initiated trajectories were summarized separately (Supplementary Tables 1 and 2). At each day, the viable model state, the dead model state assigned to loss associated with resource stress, and the CIN-associated dead model state were converted to tumor-volume components and averaged over the tumor trajectories available on that day before being plotted as stacked areas. The fitted effective oxygen state was plotted directly in percent. Viable-cell mean ploidy was calculated within each tumor as

$$\overline{P}_i(t) = \frac{\sum_{N\in\mathcal{N}} N\, x^L_{i,N}(t)}{N_{\text{unit}} \sum_{N\in\mathcal{N}} x^L_{i,N}(t)}, \qquad N_{\text{unit}} = 22. \tag{57}$$

The tumor-specific effective-O_2 and $\overline{P}_i(t)$ trajectories were then averaged over the same available trajectories at each day. Both cohort means were displayed without standardization on the numerical 0–5 left axis, while burden used the right axis. Sharing a coordinate axis is a visual alignment only and does not imply common physical units. Both longitudinal panels use a 0–100-day x-axis; model curves stop at the last available modeled time rather than being extrapolated to day 100.

Observed tumor burden was summarized only at recorded measurement days. For cohort c, day t, and n_{ct} observed tumors, the black curve is the arithmetic mean $\overline{B}_c^{\text{obs}}(t)$, and the gray band is $\overline{B}_c^{\text{obs}}(t) \pm s_c(t)$, where $s_c(t)$ is the between-tumor sample standard deviation, truncated at zero for the lower display boundary. All available observations at each recorded day contributed to these descriptive summaries. The band is not measurement error, a confidence interval, or model uncertainty. The 120 selected burden measurements were cross-checked by tumor and day between the independently prepared Figure 1 and Figure 4 analysis tables, with exact agreement (Supplementary Table 1).

For the terminal comparison, the observed single-cell chromosome-number values and predicted terminal chromosome-state weights underlying the seed-25 report's Figure 2.2 were converted to ploidy by division by $N_{\text{unit}} = 22$. Within each cohort and source, each tumor was normalized to unit total weight and then assigned one-quarter of the cohort distribution. This equal-tumor weighting prevents tumors with more profiled cells from dominating the observed cohort distribution. Observed and predicted distributions were displayed on the common ploidy range 1–7 with weighted box summaries (Supplementary Tables 1 and 2).

Fixed-oxygen attractor analysis. For each of the 500 separate *in vivo* parameter sets, the chromosome-number model was evaluated at fixed oxygen without parameter refitting. At each oxygen level, the chromosome-transition generator was combined with the state-specific effective death rate to form

$$M(O_2) = G_{\text{div}}(O_2) - \text{diag}\{\mu_{\text{eff}}(N, O_2)\}\,. \tag{58}$$

The eigenvalue of M with the largest real part, λ_1, was calculated numerically, and the real part of its associated eigenvector was retained. The eigenvector sign was oriented to have a non-negative sum, negative numerical components were truncated to zero, and the vector was normalized to unit mass. With normalized dominant eigenvector v_N, dominant mean ploidy was

$$\overline{P}_{\text{dom}}(O_2) = \frac{1}{N_{\text{unit}}} \sum_{N \in \mathcal{N}} N v_N. \tag{59}$$

The spectral gap was defined as the difference between the largest and second-largest eigenvalue real parts, $\text{Re}(\lambda_1) - \text{Re}(\lambda_2)$. The operator was evaluated at 201 fixed oxygen concentrations from 0% through 5% O_2 in increments of 0.025 percentage points. The continuous dominant mean ploidy from Eq. (59), without thresholding at ploidy 2, was retained for the Figure 4B association analysis (Supplementary Tables 3 and 4). These are analytical evaluations of the fitted linear operator, not stochastic simulations or oxygen-intervention experiments.

Post-fit continuous single-parameter ploidy-association analysis. At each of the 201 oxygen concentrations, the analysis units were the 500 parameter sets obtained from distinct numerical optimizer seeds, not biological replicates or posterior draws. Each of 18 prespecified fitted *in vivo* parameters was analyzed separately; the observation/volume parameters ρ_{2N} and σ_{burden} were excluded (Supplementary Tables 3 and 4). Let $R_{\theta,ij}$ be the average rank of fitted parameter θ across the 500 endpoints and let $R_{P,ij}$ be the average rank of continuous dominant mean ploidy at oxygen value $O_{2,j}$. Spearman association was calculated as

$$\rho_\theta(O_{2,j}) = \frac{\text{cov}(R_{\theta,ij}, R_{P,ij})}{\text{sd}(R_{\theta,ij})\,\text{sd}(R_{P,ij})}. \tag{60}$$

Ties received average ranks. Natural-scale parameter values were used for this calculation; applying the configured $\log_{10}$ transformation to strictly positive parameters gives the same ranks and therefore the same ρ.

For each parameter, the displayed peak association magnitude was

$$M_\theta = \max_{j=1,\ldots,201} \left|\rho_\theta(O_{2,j})\right|. \tag{61}$$

Let j_θ^* be the lowest oxygen-grid index attaining M_θ. Figure 4B rows were first divided by the sign of $\rho_\theta(O_{2,j_\theta^*})$, with the positive group displayed before the negative group, and were then ordered by decreasing M_θ within each group; configured parameter order was used only to break an exact tie. The adjacent dot position reports M_θ, while its fill reports the signed value $\rho_\theta(O_{2,j_\theta^*})$. Selecting a maximum over 201 oxygen values emphasizes the strongest oxygen-specific association and does not measure persistence across the grid. The full signed heatmap is therefore retained to show whether that peak direction persists or reverses with oxygen. Neither M_θ nor the signed peak correlation is a causal effect, hypothesis-test statistic, multivariable importance estimate, or out-of-sample prediction measure.

Separate *in vivo* parameter-landscape embedding and exploratory clustering. For Figure 4C, the differential-evolution initial population was deterministically reconstructed for each of the 500 separate *in vivo* fits from its optimizer seed, ordered 20-variable parameter table, transformed bounds, and effective initialization rule. Although the saved configuration recorded `de_init_mode=hybrid`, the single-stage backend explicitly used uniform initialization; all 500 fit logs recorded `init_mode=uniform`, zero local candidates, and 255 uniform candidates after the parameter-table initializer. The full 20-variable population was reconstructed before feature selection to preserve the optimizer random-number sequence. Candidate 1, which reproduced the parameter-table initializer, was then removed from every 256-member population. The observation/calibration variables ρ_{2N} and σ_{burden} were excluded, leaving 18 mechanistic parameters and 127,500 retained initial-population rows. These were combined with the 500 best endpoints, giving 128,000 *in vivo*-only parameter vectors. Each parameter was transformed on its configured natural or $\log_{10}$ optimizer scale and z-scored across the combined data. The vectors were embedded in two dimensions by t-SNE using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations (Supplementary Table 4).

The 500 best endpoints were retained and k-means clustering was applied to their two-dimensional t-SNE coordinates for $k = 2, \ldots, 8$. The value maximizing the average silhouette coefficient was selected; this procedure selected $k = 3$ with average silhouette 0.655 and cluster sizes of 43, 412, and 45 solutions (Supplementary Table 4). Figure 4B displays each original parameter value on a shared $\log_{10}$ coordinate, independently of whether its optimizer transformation was logarithmic or identity. Strictly positive values were mapped to $\log_{10}(x)$. Because one configured fitting interval had a zero lower bound, a dedicated display coordinate one decade below the smallest positive bound, initial value, or fitted endpoint was labeled "0"; in this analysis the display floor was 10^{-7}. The upper pink half violin is a kernel-density estimate of the 500 best retained endpoints on the displayed $\log_{10}$ coordinate, and its green point marks seed 25, verified as rank 1 under the canonical 500-seed fit objective. The lower gray glyph spans the configured lower and upper fitting bounds and its black point marks the configured initial value. These configured ranges and optimizer endpoints are not measurement bounds, posterior draws, confidence intervals, or biological replicates.

For Supplementary Figure 1, each natural-scale endpoint was instead transformed on its configured optimizer scale and mapped to the unit interval defined by its configured transformed lower and upper bounds. The plotted prior-referenced position was $z_{\text{prior}} = (u - 0.5)/\sqrt{1/12}$, where u is the bound-scaled value; zero therefore denotes the midpoint of the configured interval and one unit is the standard deviation of a uniform distribution over that interval. The endpoint values were stratified by the three exploratory t-SNE groups and shown in the same positive-peak then negative-peak order as Figure 4B, with decreasing M_θ within each group. These summaries remain distributions of optimizer-derived fitted endpoints, not a Bayesian posterior; their percentiles are not credible intervals, measurement bounds, or biological confidence intervals. Solid t-SNE boundaries are convex hulls of all assigned endpoints and therefore contain every plotted member, but they remain descriptive membership outlines. Because the clusters and displayed parameters were derived from the same fitted ensemble, cluster-associated parameter differences are descriptive fitted-landscape structure rather than independent validation, biological subtype evidence, or causal parameter effects (Supplementary Table 4).

For the exploratory Figure 4D stratification, the three cluster distributions were compared separately for each of the 18 parameters with a Kruskal–Wallis rank-sum test on z_{prior} (Supplementary Table 4). This transformation is monotone within each parameter, so it preserves the ranks and test statistic obtained from the corresponding configured transformed parameter values. The 18 omnibus P values were adjusted together by the Benjamini–Hochberg procedure. For parameter θ, the displayed effect size was

$$\epsilon_\theta^2 = \max\left\{0, \frac{H_\theta - k + 1}{n - k}\right\}, \tag{62}$$

where H_θ is the Kruskal–Wallis statistic, $k = 3$ groups, and $n = 500$ fitted endpoints. Parameters with adjusted $q < 0.05$ were ranked by decreasing ϵ_θ^2; adjusted q and then the Figure 4B display order were prespecified tie breakers, and the first six were shown. This test uses numerical optimizer endpoints rather than biological replicates. Moreover, because the tested parameters were also inputs to the t-SNE and cluster assignment, the adjusted values and effect sizes describe internal cluster separation and cannot serve as independent validation or biological-subtype evidence.

Pooled parameter-landscape embedding and joint-fit warm-start selection. A separate pooled landscape was constructed from optimizer population samples labeled as initial points and the final fitted solutions from the separate *in vivo* and *in vitro* calibrations. This landscape was used to design the joint-fitting warm starts shown in Figure 5C; it is a joint-fitting initialization result rather than an embedding of optimized joint-fit endpoints, and it is distinct from the 18-parameter *in vivo*-only landscape above. Fourteen parameters shared by the two contexts were used: $\lambda_{\max}$, $p_{\text{mis,base}}$, p_{misseg}, $k_{o,\text{mis}}$, $s_{\max}$, β_{buf}, n_{exp}, p_{wgd}, α_{O_2}, γ_{growth}, μ_{hp}, γ_μ, $O_{2,c}$, and n_O. Parameters were transformed on their configured natural or $\log_{10}$ scales and standardized jointly across the pooled data. By configuration, the first stored population row for each seed was removed in both contexts. In the *in vivo* export that row represents an optimizer initializer, whereas the *in vitro* export does not distinguish it from an ordinary sampled population row; this asymmetric provenance is a limitation of the warm-start landscape, not of the component likelihoods. The resulting 228,000 parameter vectors—127,500 *in vivo* population points, 99,500 *in vitro* population points, and 500 final solutions from each context—were embedded in two dimensions by t-SNE[29] using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations. The retained t-SNE axes are unitless visualization coordinates that preserve local neighborhood structure only approximately; neither axis is a fitted parameter, a latent biological variable, or a mechanistic "distance." The primary-region selection procedure, stability audits, and three retained warm-start representatives are specified above, in Supplementary Figure 6, and in Supplementary Table 7.

Weak-gap dominant-ploidy regime robustness. For Supplementary Figure 7, we reused the same 201 oxygen values, 60 log-spaced fixed effective per-chromosome missegregation probabilities, three displayed warm-start pairs, and pair-specific lowest-objective 50-endpoint ensembles used in Figure 6A. Exact duplicate 14-parameter endpoints were eigendecomposed once, giving 50, 15, and 50 unique endpoint calculations for C01, C02, and C03, and the original optimizer-seed multiplicities were restored before pointwise aggregation (Supplementary Tables 10 and 11). The rank-1 eigenvector was oriented to have positive total mass and normalized with the same nonnegative rule used for Figure 6A. Its chromosome-number-weighted mean divided by N_{unit} defined dominant mean ploidy $P_{es}(O_2, p)$ for endpoint e, pair s, oxygen concentration O_2, and forced effective missegregation probability p. Recalculated pointwise rank-1 medians and 10th and 90th percentiles agreed with the corresponding Figure 6A source surface to numerical tolerance. The legacy fraction with $P_{es} \geq 2$ was retained only as an additional source-surface crosscheck and was not used to define the new high class.

A grid cell was classified as weak-gap when at least half of its 50 endpoints had $\mathrm{Re}(\lambda_1)-\mathrm{Re}(\lambda_2) < 0.005$, matching the Figure 6A overlay. For each endpoint, dominant mean ploidy was classified as low when $P_{es} \leq 2$, intermediate when $2 < P_{es} < 4$, and high when $P_{es} \geq 4$. At each grid cell, q_{low}, $q_{\mathrm{intermediate}}$, and q_{high} were the corresponding fractions among the 50 accepted optimizer-seed endpoints, and $\max(q_{\mathrm{low}}, q_{\mathrm{intermediate}}, q_{\mathrm{high}})$ was regime consensus. A cell was assigned a stable class when the corresponding endpoint fraction was at least 0.9 and was called mixed otherwise. Exact across-fit dispersion was reported as the 90th minus 10th percentile of P_{es} (Supplementary Table 11).

Local grid sensitivity was calculated independently for every endpoint before aggregation. The neighborhood of a grid cell contained its available immediately adjacent oxygen values and immediately adjacent fixed-missegregation values, for at most four neighbors. The endpoint-level regime-switch indicator equaled one when any neighbor changed among the low, intermediate, and high classes defined above. The corresponding local-change amplitude was the largest absolute difference in P_{es} among those neighbors. Supplementary Figure 7B reports the endpoint fraction with a local three-class switch, panel C reports the across-endpoint percentile spread at the focal cell, and panel D reports the empirical cumulative distribution of the endpoint-median local-change amplitude across weak-gap cells. Boundary cells used only their available neighbors. These grid-local quantities measure sensitivity to one numerical evaluation step; they are not derivatives, finite-time transition rates, probabilities that cells switch state, or biological confidence levels. Higher eigenmodes were retained in the reproducibility audit but were not used as alternative population distributions because non-dominant eigenvectors may be signed or complex.

Fixed-oxygen response-curve classification and reliability. For each final *in vivo* parameter set, dominant mean ploidy and spectral gap were evaluated from 0% to 5% O_2 in increments of 0.025 percentage points, giving 201 oxygen values per solution and 100,500 seed–oxygen evaluations. Each response curve was smoothed with a degree-2 Gaussian-family LOESS fit with span 0.20 under the versioned rule `loess_persistent_v1`. A smoothing spline with `spar`=0.65 was used if LOESS failed or produced values outside the implemented plausibility envelope; if smoothing still failed, the raw finite values were retained. Finite differences of the smoothed curve were classified as positive, negative, or effectively zero using a tolerance equal to the larger of 10^{-6} and 10^{-4} times the fitted ploidy range. Persistent segments were required to contain at least three grid intervals, span at least 2% of the oxygen range, and have amplitude at least the larger of 0.01 ploidy units and 3% of the fitted range. A terminal plateau required at least 10% of the oxygen range after the last persistent segment and a trailing amplitude no greater than the segment-amplitude threshold. Curves with a fitted range no greater than 0.05 were classified as approximately flat. Remaining curves were classified from the ordered persistent slope signs as monotone increasing, monotone decreasing, a single increase or decrease followed by a plateau, U-shaped, inverted U-shaped, or complex nonmonotone. For a curve with multiple persistent signs to be reduced to a monotone class, the total reverse-direction amplitude could not exceed 5% of the larger of the forward amplitude and fitted range (Supplementary Table 9).

For seed-level reliability summaries, a response curve was flagged as unreliable when at least 25% of its oxygen-grid values had a spectral gap below 0.005. Curves were flagged for caution when any grid value had a gap below 0.005 or when at least 10% of grid values had a gap below 0.01; all remaining curves were labeled reliable (Supplementary Table 9). These diagnostics indicate weak separation between the first two eigendirections and are not additional fitted-data hypothesis tests.

For the pair-specific response surfaces in Figure 6A, every endpoint in the pair-specific lowest-objective decile supplied one *in vivo* parameter vector, giving 50 vectors per pair and 300 vectors across the six pairs. Each vector was evaluated on the same 201-point oxygen grid and on 60 log-spaced effective per-chromosome missegregation probabilities from 0.005 through 0.5, giving 12,060 operator evaluations per endpoint and 3,618,000 evaluations for the primary ensemble. To isolate this probability as a standardized input axis, $p_{\mathrm{mis,base}}$ was set to the grid value and the stress-linked increment p_{misseg} was set to zero; all other parameters retained that endpoint's fitted values. This construction makes the per-chromosome missegregation probability constant across chromosome states at each grid point. The realized population-average probability was recalculated from the dominant eigenvector as $\sum_N v_N p_{\mathrm{mis}}(N, O_2)$ and was required to agree with the requested input to within 10^{-8}. The main panel displays the C01Sc01, C02Sc01, and C03Sc02 subsets in one row, with the abbreviated labels C01, C02, and C03; all six pair-level surfaces remain in the analytical data products and robustness analyses (Supplementary Table 10).

At each oxygen–missegregation grid point, the displayed surface is the median dominant mean ploidy across the 50 eligible endpoints. Purple diagonal hatching marks positions where at least half of the endpoints had spectral gap below 0.005, and a purple dotted contour delimits each such region; white crosses mark sampled positions where fewer than 80% of endpoints agreed on whether dominant mean ploidy was at least 2. The unmodified fixed-O_2 trajectory was also calculated separately for all 50 endpoints: the black curve is the pointwise median of the model-implied population-average missegregation probability and the gray band is its 10th–90th percentile interval. The surface therefore summarizes a standardized post-fit response ensemble, whereas the trajectory indicates the portion of that plane traversed by the corresponding unmodified fitted functions (Supplementary Table 10).

Figure 6B applies the same standardized fixed-input operator construction at higher missegregation resolution. The fixed-probability grid was $p_i = 0.005, 0.006, \ldots, 0.500$, giving 496 values at an interval of 0.001. For each p_i, one curve connects the 201 pointwise medians $\{\widetilde{P}(O_{2,j}, p_i) : j = 1, \ldots, 201\}$, where $\widetilde{P}$ is the median dominant mean ploidy across the 50 eligible endpoints in that warm-start pair. Values at all 496×201 conditions were calculated directly; no interpolation was used. Exact duplicate 14-parameter endpoints give identical operator outputs within a pair, so one representative of each exact signature was evaluated and its original optimizer-seed multiplicity was restored before calculating the 50-endpoint pointwise median. The three displayed pairs contained 50, 15, and 50 exact unique parameter endpoints, respectively, requiring 11,465,040 distinct operator evaluations while retaining the original 50-endpoint weighting in every pair. The fixed value p_i, rather than the unmodified trajectory value, determines the curve color. A $\log_{10}$ transformation controls color spacing, while colorbar labels report the original probability values. For each pair and oxygen value, the red summary curve was calculated as $\overline{P}(O_{2,j}) = 496^{-1} \sum_{i=1}^{496} \widetilde{P}(O_{2,j}, p_i)$. This arithmetic mean assigns equal weight to every displayed fixed probability and is a visualization summary of the forced-input grid rather than a fitted or probability-weighted model trajectory. To make representative conditional trajectories directly traceable through the dense curve family, the curves at $p_{\mathrm{miss,eff}} = 0.01$, 0.10, 0.20, and 0.30 were overlaid in black using solid, dash–dot–dot, dot–dash, and dotted lines, respectively; these overlays repeat existing evaluated curves and do not constitute additional model calculations. The spectral-gap boundary remains displayed in panel A but is not repeated in panel B. The C01, C02, and C03 subpanels within Figure 6B share the oxygen, ploidy, and color scales (Supplementary Table 10).

The nested lowest-5% ensemble used the same complete surfaces. To test sensitivity to widening the objective filter to 20% without generating unused full-resolution maps for ranks 51–100, those additional endpoints were evaluated on the complete 201-point unmodified trajectory and on a prespecified diagnostic union: all 60 missegregation values at 0%, 1%, and 5% O_2, together with missegregation slices of 0.005, 0.0481 (the grid point nearest 0.05), and 0.5 across all 201 oxygen values. This 774-point union identifies the presence of both ploidy regimes at the three reference oxygen levels, the direction of oxygen response at three missegregation levels, and the evaluated-domain spectral-gap reliability fraction. Dominant mean ploidy at least 2 defined the high-ploidy regime. An O_2-response direction was labeled increasing or decreasing when the 5%-minus-0% change exceeded $+0.05$ or was below -0.05 ploidy units, respectively, and approximately flat otherwise. Qualitative claim support was tabulated separately for the 5%, 10%, and 20% objective sets, both over optimizer seeds and after exact parameter-endpoint deduplication. These empirical support fractions quantify agreement among numerical optimizer endpoints and are not biological confidence levels or posterior probabilities (Supplementary Table 10).

Embedding of oxygen-response classes and fit objectives. For the response-class diagnostic, the 500 final *in vivo* coordinates were taken from the pooled 14-parameter joint-warm-start t-SNE embedding described immediately above; no second t-SNE embedding was estimated. Supplementary Figure 5B therefore uses two-dimensional pooled-embedding t-SNE coordinates and does not use the *in vivo*-only 18-parameter Figure 4B coordinates. The coordinate labels denote only the first and second axes returned by t-SNE: they are unitless, may change with embedding settings, and have no direct mechanistic or biological interpretation. These endpoints were joined by optimizer seed to their smoothed fixed-O_2 response classes and complete fitted MAP objective values. In the parameter-space panel, one black ring marks the lowest-objective endpoint within each of the eight response classes; the complete objective distributions remain visible in the adjacent panel, where a dotted horizontal line marks the empirical lowest-decile cutoff across all 500 endpoints. The class and objective overlays were used descriptively to assess landscape overlap, not as independent tests of class separation (Supplementary Tables 9 and 4).

Visualization and reporting. For visualization and reporting in cell-number units, observed tumor volume (Eq. (24)) is converted to an approximate cell-count range using the specified diploid packing-density bounds:

$$N_{ij}^{\mathrm{obs,low}} = V_{ij}^{\mathrm{obs}}\,\rho_{2N,\min}, \qquad N_{ij}^{\mathrm{obs,mid}} = V_{ij}^{\mathrm{obs}}\,\sqrt{\rho_{2N,\min}\rho_{2N,\max}}, \qquad N_{ij}^{\mathrm{obs,high}} = V_{ij}^{\mathrm{obs}}\,\rho_{2N,\max}. \quad (63)$$

This conversion is used for plotting/interpretation only and is not part of the fitted likelihood, which is evaluated directly in the volume domain via Eq. (27).

1.4 Parameter Tables

The active fitted parameters and their calibration bounds are summarized in Table 12; fixed numerical quantities are summarized in Table 13. Values and bounds are reported on the natural scale. The bounds encode the configuration of the reported analyses and should not be interpreted as likelihood-based confidence intervals. Some ranges were informed by published biological scales, whereas parameters without reliable direct measurements were assigned engineering bounds for numerical calibration.

For the reported joint fits, the *in vivo*/*in vitro* bounds for each overlapping active biological parameter are merged into the joint union interval in Eq. (47), and all 14 such parameters are represented by the center–delta construction in Eq. (48). Parameters used only by one context retain their component-specific bounds.

Supplementary Table 12: Active fitted parameters and natural-scale bounds used by the separate and joint analyses reported here. A dash means that the parameter is not active in that component. For the 14 overlapping biological parameters, the joint admissible interval is the union of the separate *in vivo* and *in vitro* intervals. These bounds are calibration constraints, not confidence intervals.

Parameter	*In vivo* bounds	*In vitro* bounds	Joint bounds	Role in the reported model
$\lambda_{\max}$	0.005–0.300	0.005–1.500	0.005–1.500	Maximum division rate (day^{-1}); soft-coupled.
$p_{\mathrm{mis,base}}$	10^{-6}–0.100	10^{-6}–0.005	10^{-6}–0.100	Baseline per-chromosome missegregation probability; soft-coupled.
p_{misseg}	0.005–0.500	0.005–0.400	0.005–0.500	Maximal stress-induced missegregation increment; soft-coupled.
$k_{o,\mathrm{mis}}$	0.001–0.020	10^{-5}–0.020	10^{-5}–0.020	Death-hazard half-saturation scale for stress-induced missegregation (day^{-1}); soft-coupled.
$s_{\max}$	0–1	0–1	0–1	Maximum per-copy post-missegregation survival factor; soft-coupled.
β_{buf}	0.01–10	0.01–10	0.01–10	Ploidy-dependent post-missegregation viability-loss strength; soft-coupled.
$n_{\exp}$	0.1–10	0.1–10	0.1–10	Exponent controlling ploidy dependence of post-missegregation survival; soft-coupled.
p_{wgd}	10^{-5}–10^{-3}	10^{-5}–10^{-3}	10^{-5}–10^{-3}	Constant per-division WGD probability; soft-coupled.
α_{O_2}	0.5–4.0	0.5–4.0	0.5–4.0	Strength of resource-stress-dependent growth damping; soft-coupled.
γ_{growth}	1.5–4.5	1.5–4.5	1.5–4.5	Chromosome-number dependence of growth damping; soft-coupled.
μ_{hp}	0.005–0.300	0.005–0.300	0.005–0.300	Stress-associated death scale (day^{-1}); soft-coupled.
γ_{μ}	1.2–3.5	1.2–3.5	1.2–3.5	Chromosome-number penalty on stress-associated death; soft-coupled.
$O_{2,\mathrm{c}}$	0.1–2.5	10^{-4}–2.5	10^{-4}–2.5	Critical level for the oxygen-linked resource-stress function (% O_2); soft-coupled.
n_O	0.5–5.0	10^{-4}–5.0	10^{-4}–5.0	Hill exponent for the oxygen-linked resource-stress function; soft-coupled.
S_0	2.0–5.0	–	*In vivo* only	Low-burden effective oxygen supply level (% O_2).
κ_O	0.1–1.2	–	*In vivo* only	Oxygen-drop amplitude in the supply–demand target.
η	1.0–2.0	–	*In vivo* only	Chromosome-number-weighted oxygen-demand exponent.
ρ_{2N}	5.0×10^4–5.0×10^5	–	*In vivo* only	Diploid-cell density (cells mm^{-3}).
k_{clear}	0.005–0.300	–	*In vivo* only	Dead-biomass clearance rate (day^{-1}).
σ_{burden}	0.100–0.150	–	*In vivo* only	Log-scale burden observation standard deviation.
σ_{growth}	–	0.001–1.000	*In vitro* only	Growth-rate observation standard deviation.
σ_{kary}	–	0.01–10.0	*In vitro* only	Chromosome-number observation smoothing/error scale.

Parameter	*In vivo* bounds	*In vitro* bounds	Joint bounds	Role in the reported model
$\mu_{2N,0}$	–	45.725–49.725	*In vitro* only	Initial near-diploid Gaussian chromosome-count mean.
$\sigma_{2N,0}$	–	0.5–3.0	*In vitro* only	Initial near-diploid Gaussian chromosome-count SD.
$\mu_{4N,0}$	–	83.667–91.667	*In vitro* only	Initial near-tetraploid Gaussian chromosome-count mean.
$\sigma_{4N,0}$	–	1.5–6.0	*In vitro* only	Initial near-tetraploid Gaussian chromosome-count SD.

Supplementary Table 13: Fixed numerical settings used by the reported resource-stress supply–demand model and calibration workflow. Values are reported on the natural scale; settings for inactive optional modules are identified explicitly.

Parameter	Value	Unit	Description	Reference
τ_{O_2}	0.1	day	Oxygen relaxation timescale.	Fixed numerical relaxation setting; empirical workflow choice rather than a literature-derived parameter.
$O_{2,\mathrm{cap}}$	5.0	% O_2	Maximum oxygen level used to cap the supply-demand oxygen trajectory, inherited from the S_0 upper bound.	Bertout et al. (2008);[30] McKeown (2014).[31]
$O_{2,\mathrm{min}}$	0	% O_2	Lower oxygen floor used in the logarithmic oxygen target relation.	Physical/numerical anoxia floor used to bound oxygen values; model constraint rather than literature-estimated threshold.
N_{ref}	10^6	cells	Reference viable burden scale used in the oxygen supply-demand relation.	Oxygen-feedback normalization constant; empirical model scaling value rather than a measured biological quantity.
N_{unit}	22	autosomal chromosomes	Modeled autosomal chromosome count corresponding to ploidy 1.	Model convention defining one haploid autosome set as ploidy 1; definition rather than literature-estimated parameter.
N_{min}	22	autosomal chromosomes	Minimum autosomal chromosome-count state retained in the discrete state space.	Discrete state-space lower bound anchored to the model ploidy convention; grid-design choice.
N_{max}	154	autosomal chromosomes	Maximum autosomal chromosome-count state retained in the discrete state space.	Bielski et al. (2018);[1] Zack et al. (2013).[32]
Δt	0.05	day	Simulation time step.	Simulation time step selected for numerical resolution and stability; computational setting rather than biological parameter.
K	10^{12}	cells	Carrying-capacity setting for the optional logistic crowding term; crowding was disabled in the reported fits.	Waclaw et al. (2015).[33]
$N(0)$	10^6	cells	Initial total tumor population size.	Initial-condition scale for the simulation/calibration workflow; empirical setup choice.
σ_{ploidy}	0.12	chromosomes	Fixed endpoint autosomal chromosome-number observation-noise scale in the harvest likelihood.	Fixed likelihood observation-noise term; empirical measurement-error setting in the calibration workflow.
λ_{nec}	1	unitless	Relative weight of the terminal necrosis loss in the *in vivo* objective (`lambda_necrosis` in the run configuration).	Reported calibration setting.
σ_{nec}	0.75	logit-fraction units	Scale used to standardize terminal necrosis discrepancies (`sigma_necrosis_logit` in the run configuration).	Reported calibration setting.
ϵ_{nec}	10^{-4}	fraction	Clipping constant for observed and predicted necrosis fractions before the logit transform (`necrosis_fraction_eps` in the run configuration).	Numerical safeguard in the reported calibration.
λ_{prior}	0.03	unitless	Relative weight of the soft-prior contribution to the *in vivo* objective.	Reported calibration setting.

Parameter	Value	Unit	Description	Reference
$\varepsilon_{\log B}$	10^{-12}	mm^3	Volume-domain floor used inside $\log\{\max(V, \varepsilon_{\log B})\}$ for burden comparisons.	Numerical floor preventing log(0); computational safeguard.
$N_{\mathrm{min,pop}}$	10^{-12}	cells	Minimum population floor used for numerical stability.	Population floor used to avoid numerical underflow or zero states; computational safeguard.

Supplementary Table 14: Function level contrasts reconstructed from the three selected joint fit winners displayed in Figure 5. Ranges span the C01, C02, and C03 winner values. For survival summaries, the third and fourth columns report the *in vivo* and *in vitro* values. For complete effective missegregation probability, the third column reports the *in vivo*/*in vitro* ratio range and the fourth column is not applicable. The three optimizer selected fits are neither biological replicates nor confidence interval samples.

Derived quantity	Reference state	*In vivo* value or ratio range	*In vitro* reference value	Direction across winners
Per-copy post-missegregation survival	$N = 44$	0.728–0.819	0.204	higher *in vivo*, 3/3
Per-copy post-missegregation survival	$N = 88$	0.904–0.914	0.837	higher *in vivo*, 3/3
Survival gradient, $s_{88} - s_{44}$	$44 \to 88$	0.0849–0.186	0.633	larger *in vitro*, 3/3
Complete effective missegregation-probability ratio	0% O_2, $N = 44$	11.133–20.715	not applicable	> 1, 3/3
Complete effective missegregation-probability ratio	0% O_2, $N = 88$	11.178–20.611	not applicable	> 1, 3/3
Complete effective missegregation-probability ratio	1% O_2, $N = 44$	7.421–17.037	not applicable	> 1, 3/3
Complete effective missegregation-probability ratio	1% O_2, $N = 88$	10.354–19.624	not applicable	> 1, 3/3
Complete effective missegregation-probability ratio	5% O_2, $N = 44$	0.738–5.448	not applicable	> 1, 2/3
Complete effective missegregation-probability ratio	5% O_2, $N = 88$	3.210–5.697	not applicable	> 1, 3/3

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